

PSYCHOLOGY

SENIOR
5

Arts & Humanities
Pathway



STUDENT'S BOOK



Psychology

ARTS AND HUMANITIES PATHWAY

FOR

Senior



STUDENT'S BOOK

EXPERIMENTAL VERSION

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FOREWORD

Dear Student,

The Rwanda Basic Education Board (REB) is pleased to present to you the Senior Five Psychology Student's Book for the Arts and Humanities Pathway. This book is designed to support competence-based teaching and learning, ensuring consistency and coherence in the study of human behaviour and mental processes.

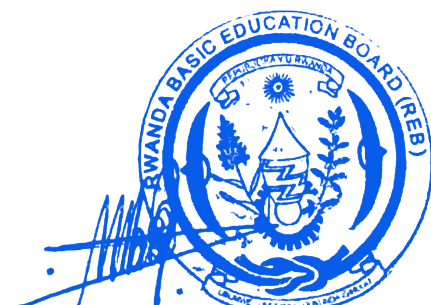
Rwanda's education philosophy emphasises helping every learner achieve him/her full potential by acquiring relevant knowledge, skills, values, and attitudes that prepare him/her to contribute positively to society and seize future opportunities. In line with this vision, the Government of Rwanda places strong importance on aligning teaching and learning materials with the national curriculum to make the learning process effective and meaningful.

This book introduces psychological concepts through activities, real-life situations, and scenarios that encourage you to construct knowledge actively, develop critical thinking, and acquire positive attitudes and values. Each unit begins with a key competence and an introductory activity to spark your interest, followed by content linked to everyday experiences and supported by practical learning tasks. Units conclude with end-of-unit assessments to help you evaluate your progress.

To benefit fully from this textbook, you are encouraged to:

- Engage actively in the activities to develop analytical and practical skills.
- Share ideas with peers through discussions, presentations, role plays, case studies, and research.
- Take responsibility for your own learning and personal growth.
- Apply Psychological theories to real-life situations and reflect on their relevance.

This book is self-explanatory and designed to be learner-friendly, enabling you to use it independently while also benefiting from teacher guidance. By exploring it thoroughly, you will gain different competences needed to understand human behaviour, manage personal challenges, and contribute to the well-being of your community.



Dr. MBARUSHIMANA Nelson

Director General, REB

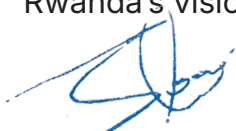
ACKNOWLEDGEMENT

I would like to extend my sincere gratitude to all those who contributed to the development of this Senior Five Psychology Student's Book for the Arts and Humanities Pathway. This achievement reflects the collective effort of dedicated individuals and institutions committed to advancing quality education in Rwanda.

Special recognition is given to the Government of Rwanda for its leadership in promoting competence-based education, and to our education development partners whose technical and financial support was instrumental in curriculum alignment and the production of teaching and learning materials.

I am particularly thankful to the subject experts, university lecturers, and secondary school teachers whose professional insights shaped the content, ensuring that concepts of human behaviour and mental processes are presented accurately and effectively. I also acknowledge the contribution of illustrators and designers who enhanced the book with clear visuals and an accessible layout.

Finally, I recognise the commitment of the Rwanda Basic Education Board staff, especially the Curriculum, Teaching, and Learning Resources Department (CTLRD), for their dedication throughout the development process. Their efforts guarantee that this textbook serves as a practical tool for students and supports Rwanda's vision of quality education for all.



Murungi Joan

Head of Curriculum, Teaching and Learning Resources Department, REB

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UNIT 1

NATURE OF PERSONALITY



Key Unit Competence:

Demonstrate an understanding of how behaviour, emotion, motivation, and thought patterns define an individual.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Define key concepts related to personality.
- Describe the main characteristics of personality.
- Describe individuals' behaviour based on different thoughts in typologies.
- Enumerate the basis that makes people different.
- Explain the meaning and importance of studying personality.
- Explain how the environment and family are factors in human personality development.

Skills

2

- Use concepts of personality when reading, discussing, and talking about personality.
- Justify the role of biological and environmental factors in shaping personality.
- Connect personality typologies to how individuals live and behave in real-life situations.
- Use understanding of personality characteristics to manage or respond to individual differences effectively.
- Apply key concepts of personality when reading, discussing, and communicating about personality -related topics.

Attitudes and values

3

- Recognise individual differences by referring to the main characteristics and dimensions of personality.
- Develop an acceptance of working, dealing, and living with all people in harmony.
- Appreciate that theories of typologies can be proven incorrect or correct by later research.
- Accept individual differences in society.
- Explain and justify the influence of biological and environmental factors in shaping our personality.

Introduction of the unit

Have you ever noticed how two students can face the same situation but react in very different ways? For example, when a teacher asks a difficult question, one student may feel confident and eager to answer, while another may feel shy or nervous. You may also see a classmate who stays calm during problems, while another quickly becomes angry or emotional. These differences show that every person has a unique personality.

In this unit, you will explore the nature of personality and understand how personality shapes the way individuals think, feel, behave, and interact with others. You will learn key concepts such as personality, temperament, character, traits, and behaviour patterns, and examine the factors that influence personality development, including nature (biological influences) and nurture (environment and experiences). This knowledge will help you appreciate individual differences, better understand yourself and others, and recognise how personality influences everyday actions and relationships. Understanding these ideas will help you explain individual differences and improve relationships at school, in workplaces, and in the community.

Introductory activity



The teacher has just finished marking an exam and begins to return the papers. The class becomes quiet, and everyone is curious to see their results. Suddenly, different reactions appear around the room:

- **Kaneza** looks at her paper (15/50) and immediately tears it in frustration.
- **Jane** sees her mark (10/50) and starts crying quietly.
- **Moses** checks his score (14/50) and loudly complains that the teacher must have made a mistake.
- **Toto** receives 5/50 but surprisingly stays calm and says she is okay with the result.
- **Linda** gets 23/50 and angrily insults the teacher, saying she will not return to class again.

The teacher asks the class to pause and think about what just happened.

Discussion questions:

1. Why do you think these students reacted differently, even though they all performed poorly in the exam?
2. In everyday life, what factors make people behave or react differently in similar situations?

1.1. Key Concepts of personality: Personality, personality development, and nature

Learning activity 1.1



Football fans react differently to the scoring of a goal because people have different personalities, emotions, attitudes, and ways of thinking. Even though they are watching the same event, each person interprets and expresses their feelings in their own way. Some fans are very expressive, showing excitement by jumping or running, while others are calmer and may simply smile or remain seated. In some cases, strong emotions or personal attitudes can even make a person react aggressively, such as fighting.

Discussion questions

1. Think about a time when your favourite team scored a goal while you were watching with friends or family. How did you react, and how did others around you react differently?
2. In your class or group of friends, how do students usually react when something exciting happens, such as winning a game or receiving good marks? Are the reactions the same or different? Explain.
3. Have you ever noticed that some students stay calm in situations where others become very emotional or excited? Why do you think people behave differently in the same situation?
4. What personal qualities or characteristics do you think influence how someone reacts to events such as sports victories, exam results, or competitions?
5. How does understanding that people react differently help us respect and better understand others in our community?

Meaning and concept of personality

One of the earliest thinkers who tried to explain differences in human behaviour was the Greek physician Hippocrates (460–377 BC). Today, Psychologists study personality to understand why people think, feel, and behave differently in similar situations. Personality is a psychosocial phenomenon, meaning it involves both Psychological processes and social interactions. It influences how individuals behave in their families, schools, communities, and workplaces.

Psychologists explain personality using two important ideas: **individuality** (which means that every person is unique; people differ in how they think, feel, and act) and **consistency** (which means that a person's behaviour tends to remain relatively stable over time and across different situations). According to **Kluckhohn and Murray (1953)**, every person is: like all other people in some ways, like some other people in certain ways, and like no other person in other ways.

Meaning of personality

Personality refers to the pattern of thoughts, emotions, and behaviours that makes each person unique and relatively consistent over time. Psychologist Gordon Allport (1937) described personality as the dynamic organisation of Psychological and physical systems within a person that influence how the individual adapts to the environment.

In the workplace, for example, in a bank, one employee may politely welcome clients and explain services clearly, while another employee may speak very little and focus more on completing transactions. In a hospital, one nurse may comfort patients and talk to them warmly, while another may focus mainly on completing medical tasks quickly.

In a school, some teachers actively interact with students and encourage discussion, while others prefer strict discipline and quiet classrooms.

These differences show how personality influences behaviour at work. Similarly, during a football match, when a team scores a goal, some fans may jump and celebrate loudly, others may clap quietly, and some may remain calm. These reactions reflect different personalities.

Personality Development: This means to the process through which a person's patterns of thinking, feeling, and behaving develop and change throughout life. It begins in early childhood and continues as the individual grows and interacts with the environment. Personality development is influenced by factors such as heredity, family relationships, education, culture, social experiences, and life events.

Nature of Personality: It refers to the main characteristics and qualities of personality. Personality is unique to each individual, relatively stable over time, but it can also change and develop through experiences and learning. It is influenced by both biological factors and environmental factors, and it affects how individuals think, feel, and behave in different situations.

Application activity 1.1



Think about a friend, classmate, or family member who is feeling overwhelmed by schoolwork, family responsibilities, or personal problems.

1. What advice would you give them to help them manage stress and remain calm?
2. In your school or community, some people eat too much unhealthy food or eat without control when they feel stressed or bored. Why is it important for people to control their eating habits for their health and well-being?

1.2. Key concept of personality: Temperament

Learning activity 1.2



Scenario

Four students, Bebe, Mado, Shaba, and Kodo, were playing a football match. During the game, Mado accidentally stepped on Bebe's foot, and Bebe cried but did not react immediately. After about five minutes, another incident happened: Shaba hit Kodo, and Kodo immediately avenged herself by slapping Shaba. The next day at school, Bebe refused to greet Mado, saying she was still upset because Mado stepped on her during the match the previous day.

Discussion Questions

1. Between Bebe and Kodo, which person would you prefer to live with, study with, or work within a team? Explain your reason.
2. What do the reactions of Bebe and Kodo tell us about different personalities and ways of handling emotions?
3. In your school or community, how do people usually react when someone hurts them during games or disagreements?
4. What are some better ways to manage anger or conflicts among friends, classmates, or colleagues?

Temperament is the natural, inborn way a person responds emotionally and behaves in different situations. It appears early in life and usually stays stable throughout growth. Temperament is influenced by biological and genetic factors, and it forms the foundation for a person's personality, shaping how they interact with others and the world. For example, a cheerful and calm child may stay happy and composed even when faced with challenges in school, showing an "easy-going temperament." A child who gets upset quickly or cries often when things do not go their way demonstrates a "sensitive or difficult temperament." And in the workplace, a manager who remains calm under pressure reflects a stable temperament, while an employee who reacts angrily to small mistakes shows a more reactive temperament.

These examples show that temperament influences how people behave, manage emotions, and relate to others.

1.2.1. Dimensions of temperament

People respond differently to the same experience: some react quickly and strongly, others slowly; some reactions last long, others fade quickly. These differences are what we call temperament. There are four classical dimensions of temperament: choleric, sanguine, melancholic, and phlegmatic.



1. **Choleric temperament** is characterised by quick and strong emotional reactions that last a long time, ambition, energetic and enthusiastic, aspires to success, leadership, or spiritual achievements. Its negative traits may include pride and overconfidence, being quick to anger and resentment, and sometimes showing deceitful behaviour or a lack of sympathy toward others. For example, a businessperson who passionately drives a startup to success may get frustrated or impatient with employees who do not meet their standards. Or a community leader who takes initiative in social projects but struggles with criticism.
2. **A sanguine temperament** is characterized by quick emotional reactions, but they fade quickly. Sanguine individuals are extroverted, friendly, talkative, and entertaining, optimistic, cheerful, compassionate, and sociable. Its dark sides are vanity, jealousy, and a tendency to flirt, love of pleasure, and sometimes avoiding responsibilities. For example: a tour guide who easily engages with tourists and makes them feel welcome but may forget minor details of work due to distraction. Or a teacher who makes lessons lively and interactive but sometimes struggles with routine paperwork.
3. **Melancholic temperament** is characterised by slow emotional reactions that are deep and long-lasting. Individuals are thoughtful, reflective, serious, and often prefer solitude. They tend to be highly committed to tasks once engaged. Examples: a researcher who carefully analyses data and produces high-quality work but may become overly concerned with small details or doubts. Its dark sides are a tendency toward sadness or pessimism, overthinking and worry, being overly critical of oneself and others, and difficulty adapting to change or letting go of past experiences.
4. **Phlegmatic temperament** is characterised by slow and weak reactions; emotions are mild and fade quickly, calm, composed, practical, and patient, and work steadily, avoiding conflicts, and do not demand much. Its dark sides are lazy, unambitious, avoid challenges, and can be indifferent or disengaged. Example: a factory worker who performs routine tasks steadily and reliably but avoids taking on leadership roles. Or a civil servant who maintains composure in stressful situations but prefers routine work over innovative projects.

Summary table of the four temperaments:

Temperament	Reaction speed	Emotional Depth	Example	Personality type
Choleric	Quick	Deep and lasting	Ambitious entrepreneur	Extrovert
Sanguine	Quick	Shallow and fleeting	Tour guide or lively teacher	Extrovert
Melancholic	Slow	Deep and lasting	Researcher or artist	Introvert
Phlegmatic	Slow	Shallow and fleeting	Factory worker or calm civil servant	Introvert

1.2.2. How to determine your temperament

To identify your temperament, consider the following three questions: Do you react immediately and strongly or slowly and gently to events? Do you tend to act right away or wait calmly? Or does your reaction last long or fade quickly? And if you observe your daily behaviour, for example, how do you react if someone criticises you? How do you handle exciting news or achievements? How do you calm down after an argument or stressful event? For example, a student who becomes frustrated very quickly during exams may have choleric tendencies. In contrast, a student who stays calm but takes more time to think and respond may have melancholic tendencies.

1.2.3. Importance of learning about temperament

Understanding temperament helps us recognise why people behave differently in similar situations. It allows us to predict reactions, improve relationships, and work effectively in teams. By knowing our own temperament and that of others, we can manage emotions, reduce conflicts, and make better decisions in both personal and professional settings. In summary, the knowledge about temperament will be useful in:

- Improved communication: Knowing someone's temperament helps tailor how we talk or give instructions.
- Better teamwork: Teams can assign roles according to each member's natural strengths.
- Conflict management: Awareness of temperament allows us to handle disagreements calmly and fairly.
- Personal growth: Understanding our temperament helps us manage stress, set realistic goals, and develop self-awareness.

Practical examples:

- In a **bank**, a choleric branch manager may drive ambitious projects, while a phlegmatic accountant ensures steady, accurate reporting. Knowing this helps the team coordinate effectively.
- In a **tourism company**, a sanguine tour guide easily engages tourists, while a melancholic logistics officer carefully plans itineraries. Together, they provide a smooth and enjoyable experience.

- In a **local cooperative or community project**, understanding that some members are quick-tempered (choleric) and others calm and reflective (phlegmatic) helps prevent misunderstandings and ensures fair participation.

By studying temperament, individuals can adapt to different personalities, enhance workplace productivity, and build harmonious relationships in schools, offices, and communities.

Application activity 1.2



1. Name the four temperaments and give examples of each from everyday life, such as at school, in the community, or at work.
2. Describe a choleric person. Think of a leader, teacher, or family member who shows choleric traits?
3. The sanguine person is often cheerful and fun to be around. But which habits or behaviours of a sanguine friend, classmate, or colleague might annoy you if you were working or living with them?
4. Why is it important to understand different temperaments, especially when working, studying, or living with others?
5. Give examples from **school, family, or community life** where knowing someone's temperament could help you communicate better, avoid conflicts, or work more effectively as a team.

1.3. Key concepts of personality: Characters, Traits, and Patterns

Learning activity 1.3



In a Senior 5 class, students are preparing for end-of-term exams. Aline studies every day, completes her homework on time, and helps her classmates understand difficult topics. Eric is intelligent, but studies only when exams are very close. Sometimes he performs well, but other times he fails. And Claudine often comes late, does not complete assignments, and distracts others during lessons.

Questions

1. Looking at Aline, Eric, and Claudine, what specific behaviours does each one show during exam preparation?
2. From the behaviours of Aline, Eric, and Claudine, which actions do you see happening regularly (repeated over time)?
3. Based on how Aline, Eric, and Claudine behave, what traits can you identify for each of them?
4. Among Aline, Eric, and Claudine, who demonstrates good character? Explain your answer using their behaviours.

1.3.1. Character

Character is the core of a person's identity and shapes how they are viewed, trusted, and respected in society. Just as a company is known for its work culture, a person is recognised by consistent behaviours, attitudes, and moral values. The concept of character has historical roots in early performances where individuals were portrayed through distinct "characters."

Good character develops gradually through daily actions, decisions, responsibilities, and social interactions. It is not inborn; rather, it is shaped by influences from family, school, community, and workplaces. Because people grow up in different environments and experiences, they develop different levels and qualities of character. Cultivating strong character helps individuals avoid harmful behaviours, make responsible choices, and succeed in life.

1.3.2. Dimensions of character

- **Moral Character:** Following laws, being honest, sincere, and courteous. *Example:* A shopkeeper honestly returns extra change to a customer.
- **Spiritual character:** Belief in higher powers, respect for religions, and ethical living. *Example:* A village leader encourages unity and respect for all religious practices in community meetings.
- **Physical character:** Using the body responsibly, avoiding harmful actions, and showing respect for others. *Example:* health workers in a hospital maintain proper hygiene and care for patients respectfully.
- **Materialistic/professional character:** Ethical conduct in work and business. *Example:* a cooperative member ensures fair distribution of produce without cheating or taking shortcuts.

1.3.3. Traits

Personality traits describe consistent patterns of thinking and feeling. Research has identified five major personality traits, known as the "**Big Five**":

- **Extraversion:** Sociable, talkative, energetic, assertive. *Example:* a teacher actively engages students in discussions and group activities.
- **Agreeableness:** Kind, cooperative, trustworthy. *Example:* a community health volunteer patiently helps villagers understand hygiene practices.
- **Conscientiousness:** Organised, responsible, and disciplined. *Example:* a cooperative accountant carefully tracks all financial records.
- **Neuroticism/emotional stability:** Calm, resilient, or prone to anxiety and mood swings. *Example:* a nurse calmly handles emergencies, while another may get easily stressed.
- **Openness to experience:** Curious, imaginative, and willing to learn new ideas. *Example:* a young entrepreneur experiments with innovative farming methods.

1.3.4. Patterns

Patterns refer to the Regular and **consistent ways in which an individual behaves, thinks, or responds to different situations over time**. When a person acts in the same way repeatedly, it forms a pattern that helps others understand their personality and character.

Patterns are important because they show **how stable a person's behaviour is**. For example, if someone is always punctual, responsible, and hardworking, these repeated actions form a **positive behaviour pattern**. On the other hand, if a person often lies, avoids responsibilities, or arrives late at work, this forms a **negative behaviour pattern**.

Examples:

- An employee who consistently submits reports on time and helps colleagues shows a pattern of responsibility and cooperation.
- In a village cooperative, a member who regularly attends meetings and contributes ideas demonstrates a positive participation pattern.
- In school, a student who always completes assignments and respects teachers shows a pattern of discipline and good character.
- Therefore, patterns help people predict behaviour, understand personality, and build trust in relationships at school, workplaces, and in the community.

Application activity 1.3



1. In your school, students show different types of character. Using examples from places such as school or a village, discuss the different dimensions of character (moral, spiritual, physical, and professional/material character).
2. People behave differently in various social and professional contexts. Describe the "Big Five" personality traits and give practical examples of how each trait may appear in a school or a local business.

1.4. Factors of personality

Learning activity 1.4



Two brothers, Kimenyi and Rangira, grew up in the same family. They attended the same school, lived in the same house, and were raised by the same parents. However, people in the village noticed that they behaved very differently.

Kimenyi is very calm, friendly, and enjoys working with others in the family farming cooperative. He likes helping neighbours and easily makes friends.

Rangira, on the other hand, prefers to work alone. He becomes angry quickly when people criticise him, and does not like participating in community activities.

Many people in the village wonder why the two brothers behave so differently, even though they grew up in the same family and environment.

Questions

1. Why do you think Kimenyi and Rangira behave differently even though they grew up in the same family?
2. Do you think a person's behaviour comes from what they are born with or from the environment where they grow up?
3. Can you think of other examples in your school, family, or community where people behave differently because of their background or natural abilities?

1.4.1. Concepts of nature versus nurture

In Psychology, nature and nurture are two important factors that influence personality and human development. Nature refers to the genetic and biological characteristics a person inherits from their parents. These include physical appearance, height, body strength, health, and some natural abilities or talents. For example, a young person may inherit a strong physical body from their parents and later become successful in sports such as football or athletics.

Nurture, on the other hand, refers to the environment and experiences that shape a person's behaviour and personality. These include family upbringing, school experiences, culture, friendships, and the community where a person grows up. For instance, a child raised in a family that values discipline, respect, and hard work may develop these qualities and later become a responsible employee or community leader.

There are several environmental factors influencing personality:

- Physical environment: Where people live affects their behaviour and lifestyle. For example, in rural areas, community activities like farming and Umuganda promote teamwork and cooperation.
- Social environment: People learn behaviours from others around them. For example, an employee may learn professionalism, punctuality, and teamwork.
- Family environment: Parents and relatives play a key role in shaping children's values and attitudes. For instance, children raised in families that encourage education and responsibility are more likely to become hardworking students and professionals.
- Cultural environment: Cultural traditions and values influence personality. In Rwanda, cultural practices such as respect for elders, community cooperation, umuganda, and participation in community meetings (**Inteko z'Abaturage**) help shape responsible citizens.
- School environment: Schools help develop discipline, social skills, and knowledge. For example, students who participate in school clubs, debates, or group projects learn leadership and cooperation that later help them in the workplace.

1.4.2. Interaction between nature and nurture

Most Psychologists agree that nature and nurture work together to shape personality. Inherited traits provide natural potential, while the environment determines how fully these abilities develop. For example, a child may be born with musical talent, but without practice or exposure to music activities, the talent may not grow. Likewise, a person may have genes for tall height, but without proper nutrition, they may not reach it.

This interaction also appears in real-life settings. A bank employee may naturally communicate well (nature) but develop strong customer service skills through training and experience (nurture). In a community cooperative, someone may inherit physical strength but build leadership abilities through teamwork and participation.

Overall, personality and development result from the continuous influence of both inherited characteristics and environmental experiences. This explains why individuals behave and develop differently, even when raised in similar surroundings.

Application activity 1.4



1. People grow up in different environments, such as families, schools, workplaces, and communities. Using examples from Rwandan families, villages, schools, or workplaces (for example, a cooperative, hospital, office, or market), discuss the different aspects of nurture that influence an individual's personality.
2. In real life, a person's personality is shaped by both the natural abilities they are born with and the environment where they grow up. Using examples from your society, such as in schools or sports, explain why the interaction between nature and nurture is important in shaping an individual's personality.

1.5. END UNIT ASSESSMENT

End of unit assessment questions



Knowledge and understanding:

1. Explain how personality, character, and temperament differ, giving one example of each.
2. Describe how nature and nurture interact to shape personality, using examples from daily life.
3. Define the Big Five personality traits and explain how each trait may appear in school or community situations.

Skills:

4. Think of a real situation where someone showed a consistent behaviour pattern. Identify the traits shown and explain how they influence relationships or performance.

5. Imagine leading a team where members have different temperaments. Describe how you would assign tasks and manage teamwork effectively.
6. Choose one Big Five trait and analyse how high or low levels of this trait influence problem-solving or conflict handling.

Attitudes and Values

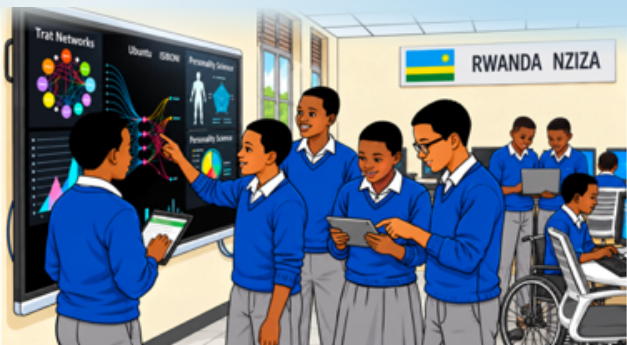
7. In some families, parents raise their children with very strict rules and punishments, expecting total obedience. If you were a teacher, counselor, or community leader, what advice would you give these parents about the possible effects of authoritarian parenting on their children's future personality and behaviour? Which type of parenting would you recommend instead, and how might it help children develop positive behaviours at school, in the workplace, and in the community?
8. Imagine you are working in an office, cooperative, or community project with a person who has a choleric temperament. What kinds of challenging behaviours might you experience in your relationship or teamwork with that person? How might you deal with such situations constructively?
9. People in schools, workplaces, and communities often show different personality traits. Describe the Big Five personality traits by explaining the two extremes of each trait. Give examples of how these traits might appear among people working in places such as a school, hospital, business, or local cooperative.

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UNIT 2

MODERN TRENDS OF PERSONALITY



Key Unit Competence:

Explain modern trends in personality and their application in understanding individual differences.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Describe what is meant by modern approaches to personality in today's Psychology.
- Identify major modern approaches used in the study of personality.
- Enumerate the contributions of contemporary Psychologists to modern personality research.
- Explain the key modern trends in personality and their relevance to understanding human behaviour.
- Describe key features that distinguish modern trends from traditional personality theories.

2 Skills

- Justify with relevant examples why modern trends of personality are emerging in psychology.
- Connect with researchers in finding more information and knowledge about human behaviours
- Summarise how modern trends influence the understanding of individual differences in real-life contexts.

3 Attitudes and values

- Recognise the importance of studying modern trends in understanding human behaviour.
- Value emerging trends in personality as important contributions to psychological practice.

- Express one's position and perspective on modern trends in personality, demonstrating openness, critical thinking, and respect for diverse viewpoints.

Introduction

Have you ever noticed that in the same classroom or workplace, people react very differently to the same situation? For example, when a group task is given, one student may quickly take the lead and guide others, another may quietly observe before contributing ideas, while someone else may feel nervous about speaking in front of the group. In a football match, some fans jump and shout with excitement, others remain calm, and a few may even become aggressive when their team loses. These everyday situations show that people have different personalities and ways of reacting to events.

Modern Psychology studies personality to understand why people think, feel, and behave differently. In this unit, you will explore **modern trends in personality**, including factors such as emotional stability, dominance, liveliness, social boldness, sensitivity, and openness to change. You will also examine the causes of **aggression**, how social situations and culture influence it, and how **genetic factors** like activity, passivity, and responsiveness may shape personality. Understanding these ideas will help you explain **individual differences** and improve relationships at school, in workplaces, and in the community.

Introductory activity



Discussion question:

In today's world, many people spend time using digital technologies such as smartphones, television, radio, and social media platforms. In your opinion, how do the mass media and digital technology influence the personality of men and women in different dimensions, such as liveliness, social boldness, emotional stability, sensitivity, and openness to change? Give examples from schools, workplaces, or communities.

2.1. Factors or dimensions of personality (Sex differences, Warmth, Emotional stability, Dominance, Liveliness, Social boldness, Sensitivity, and Openness to change)

Learning activity 2.1



It is well known that men and women have different physical characteristics. In your opinion, how can these physical differences influence their personality traits and behaviours? Give examples from everyday situations in Rwanda, such as in schools, workplaces, sports activities, or community leadership, to explain why personality differences may be observed between men and women.

2.1.1. Introduction to personality dimensions

Psychologists have tried to understand why people behave differently. One Psychologist, Raymond Cattell, studied many personality traits and grouped them into 16 personality factors. He also created a personality test called the 16 Personality Factors (16PF). Cattell explained that personality traits are not simply present or absent. Instead, everyone has these traits, but at different levels. For example, one person may be very friendly while another may be less friendly.

Some important personality traits include:

- Warmth: being friendly and supportive to others.
- Reasoning: the ability to think and solve problems.
- Emotional stability: staying calm in stressful situations.
- Dominance: the ability to lead or influence others.
- Liveliness: being energetic and enthusiastic.
- Rule-consciousness: respecting rules and responsibilities.
- Social boldness: confidence when interacting with others.
- Sensitivity: being caring and understanding.
- Openness to change: willingness to accept new ideas.

Examples: In class, a teacher who shows warmth and sensitivity listens to students' problems and supports them in learning. A nurse in a health centre: emotional stability helps a nurse remain calm when treating many patients during emergencies. In the local government leader (Village Leader), Dominance and social boldness help a leader guide community meetings and resolve conflicts. In a bank employee, rule-consciousness is important because the employee must follow financial regulations and procedures carefully. These examples show that personality traits influence how people perform their duties and interact with others.

2.1.2. Sex differences in personality

Men and women have physical differences, but Psychologists have long debated whether there are also personality differences between them. Research shows small general trends, although individuals can still be very different.

For example:

- Girls often develop language skills earlier and may perform well in communication activities.
- Boys may show stronger skills in spatial reasoning or physical activities.
- Boys are sometimes more physically active and competitive.
- Girls often show higher emotional sensitivity and cooperation.

However, these differences are influenced by culture, education, and social expectations. Examples: in classroom discussions, girls may participate actively in debates and storytelling, while boys may show strong interest in mathematics or technical subjects. In construction sites, men often dominate heavy physical work, while women may work in project planning, administration, or engineering roles. In

the Health sector, many nurses show strong empathy and patience with patients, which reflects sensitivity and agreeableness. In entrepreneurship, both men and women run businesses in markets or shops, but they may manage them with different communication styles and leadership approaches. These differences are general patterns, not strict rules. Personality is shaped by experience, environment, and individual choices.

2.1.3. Gender differences in the “Big Five” personality traits

Another way Psychologists study personality is through the Big Five personality traits.

a) Neuroticism (Emotional stability)

This trait refers to how people respond to stress and negative emotions. Example: A customer service worker in a telecom company must stay calm when customers complain. A person with high emotional stability can handle such pressure effectively. Research suggests women sometimes report higher emotional sensitivity, but this varies among individuals.

b) Agreeableness

Agreeableness refers to kindness, cooperation, and empathy. Example: A community health worker visiting families in villages must show care, patience, and understanding when advising parents about children's health. Women often score slightly higher in these traits, but many men also show strong agreeableness.

c) Conscientiousness

This trait includes responsibility, organisation, and discipline. Example: An accountant in a company must carefully record financial information and respect deadlines. High conscientiousness helps them avoid mistakes and manage responsibilities well.

d) Extraversion

Extraversion describes how sociable, confident, and energetic someone is. Example: A tour guide in the tourism sector needs strong extraversion to interact with visitors, explain attractions, and create a friendly atmosphere. Women often show more warmth and friendliness, while men may show more assertiveness.

e) Openness to experience

This trait includes creativity, curiosity, and a willingness to learn new things. Example: A young entrepreneur developing digital services or mobile applications needs openness to learn new technologies and adapt to market changes. Research usually finds very small differences between men and women in this trait.

2.1.4. Temperament and biological influences

Psychologists Hans and Sybil Eysenck believed that personality is partly influenced by genetics (biology). They described two main personality dimensions:

Extraversion versus introversion

Extroverts are outgoing and enjoy social interaction. Introverts prefer quiet environments and reflection. Example: In a company meeting, an extroverted employee may confidently present ideas, while an introverted employee may prefer to analyse information carefully before speaking.

Neuroticism versus emotional stability

Some people react strongly to stress, while others remain calm. Example: . A police officer managing traffic during busy hours requires emotional stability to remain calm and make sound decisions.

Understanding personality helps us work better with others, respect individual differences, and improve cooperation in society.

Application activity 2.1



Based on what you observe in schools, workplaces, and community life, in which personality traits do women often show stronger tendencies, and in which traits do men often show stronger tendencies? Explain why these differences may appear, giving examples from real-life situations such as hospitals, schools, businesses, or community leadership.

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2.2. Aggression: definition, biological and emotional causes of aggression

Learning activity 2.2



Observe the image and answer the questions

1. Look carefully at the picture. What do you think is happening between the two people? Based on their body language and facial expressions, do you think they are having a disagreement or argument? Explain what signs in the image make you think so.

2. In your opinion, what other aggressive behaviours can people show in daily life? Give examples you have observed in schools, families, or community situations (for example, shouting at someone, threatening others, fighting, or insulting someone).

2.2.1. Aggression

Aggression is one of the oldest human behaviours. It has been used by both humans and animals, especially in the past when people had to fight to survive. In primitive societies, people sometimes used aggression or hostility to protect themselves or their families. Today, society has changed because of technological development. People no longer rely only on physical force. However, aggression still exists and can even be more dangerous because modern weapons are easily available and can cause serious harm.

Aggression is any behaviour that is intended to harm another person who does not want to be harmed. Aggression can appear in different forms, for example:

- Verbal aggression: insulting someone, threatening, or calling bad names.
- Physical aggression: pushing, hitting, or fighting.
- Severe aggression (violence): actions that can seriously injure or kill someone.

For example, A student insulting a classmate during a disagreement. A driver shouting at another driver during traffic in the city. A worker pushing a colleague during an argument.

Aggression can also appear in everyday social interactions through anger, hostility, or violent behaviour. Because of this, social psychologists study aggression to understand how aggressive behaviour develops, what factors influence it, and how it can be reduced in society. Example: Aggression may appear in workplaces when employees insult each other, when a supervisor shouts at workers, or when conflicts between motorcycle taxi drivers lead to physical fights.

In aggression, we can find violence, which is a very serious form of aggression that can cause severe injury or death. Examples of violence include stabbing, shooting, and killing someone. This means that all violence is aggression, but not all aggression is violence. Example: Two children pushing each other over a toy is aggression, but not violence. Attacking someone with a weapon is both aggression and violence.

2.2.2. Causes of aggression

Aggression can arise from both biological factors and emotional factors.

a) Biological causes of aggression

Some aggressive behaviours are influenced by biological factors such as genetics, hormones, and the functioning of the brain and nervous system.

❖ Genetic aspects

Research suggests that genetics may influence aggressive behaviour. Some people may inherit biological tendencies that make them more emotionally reactive or

impulsive. Studies involving twins and adopted children show that aggression can sometimes run in families, suggesting that genes may partly contribute to aggressive behaviour.

However, genetics alone does not determine aggression. Environmental factors such as parenting, education, peer groups, and life experiences also strongly influence how people behave. A person may inherit a tendency toward aggression, but supportive family and social environments can help them develop self-control and peaceful behaviour.

Scientists also study how genes interact with brain chemicals and hormones to influence aggression. For example, some genetic factors may affect serotonin regulation, emotional control, and impulsive behaviour.

❖ The brain

A part of the brain called the amygdala helps control fear and aggression. When a person feels threatened or angry, this part of the brain becomes active. Another part of the brain called the prefrontal cortex helps people control their anger and aggressive behaviour. Example: A police officer controlling traffic in a city may face angry drivers. Their self-control (prefrontal cortex) helps them remain calm instead of reacting aggressively.

❖ Hormones and aggression

Hormones in the body can also influence aggression.

- **Testosterone**

Testosterone is a hormone that is often linked to higher levels of aggression, especially in men. Example: In some cases, young men involved in violent fights in bars or streets may have higher levels of testosterone and physical strength that support aggressive behaviour.

- **Serotonin**

Serotonin helps people control impulses and emotions. Low serotonin levels may increase impulsive aggression.

❖ Alcohol and aggression

In psychology, alcohol is usually considered a biological factor because it directly affects the brain, nervous system, judgment, and self-control. Alcohol consumption can increase aggression because it reduces self-control and judgment. Examples: Fights happen in bars or nightclubs after excessive drinking. A person becomes aggressive in a community gathering after drinking alcohol.

Alcohol makes people less aware of consequences, which increases the risk of aggressive behaviour.

b) Emotional causes of aggression

Aggression often happens when people experience negative emotions. These include anger, frustration, stress, pain, and disappointment.

Frustration and aggression

Frustration occurs when someone cannot achieve an important goal. Examples: A motorcycle taxi driver (moto rider) becomes angry when traffic delays his work. A student becomes aggressive after failing an important exam. A worker becomes angry after being paid less than colleagues.

Frustration can sometimes lead to aggressive reactions.

c) Other situations that increase aggression

Aggression may increase when people experience pain, extreme heat, and stressful conditions. For example, Workers on construction sites during very hot weather may become easily irritated and argue more frequently.

Displaced aggression

Sometimes people cannot express anger toward the person who caused the problem. Instead, they direct their anger toward another person. This is called displaced aggression. Example: A worker who is scolded by their boss at work may go home and shout at family members. A student who is disciplined by a teacher may express frustration or anger by fighting with classmates.

Catharsis and Aggression

Some people believe that expressing anger aggressively will reduce aggression. This idea is called catharsis. For example: hitting objects, shouting, playing violent games.

However, research shows that these actions usually increase aggression instead of reducing it. A better solution is to control emotions and calm down. Examples of healthier solutions: talking to someone, taking a break, exercising, solving the problem peacefully. For example, instead of fighting during a community disagreement, local leaders often encourage dialogue and mediation to solve the conflict peacefully.

Application activity 2.2



1. Using examples from schools or community life, explain some biological factors (such as the brain or hormones) that can influence aggressive behaviour in people.
2. Sometimes people cannot express anger toward the person who caused the problem and instead direct it to someone else. Using examples from family life, the school explains how displaced aggression can happen.

2.3. Aggression: The influence of social situation on aggression

Learning activity 2.3



1. In your school, why do you think some students bully others? Explain using examples you have seen or heard about.
2. As a student, what actions can you take to help stop bullying and create a safe and respectful school environment?

2.3.1. Social learning approaches to aggression

According to the social learning theory, aggression is a learned behaviour. People learn it the same way they learn other social behaviours. Psychologist Albert Bandura (1997) explained that aggression can be learned through:

a) Direct experience

People repeat aggressive behaviour if it brings them rewards. *Example:* If a worker threatens others and always gets what he wants, he may continue using aggression.

b) Observation (vicarious learning)

People can learn aggression by observing others. This can happen through family members, leaders, colleagues, and media such as movies and television. *Examples:* People can learn aggression by observing others. This can happen through family members, leaders, colleagues, and the media, such as movies and television. For example, young workers may copy aggressive communication styles used by senior staff, and children may imitate violent actions seen in movies or online videos.

2.3.2. Social Influences on Aggression

a) Exposure to mass media and aggression

Studies show that many television programs and films contain violent scenes. When people frequently watch violent content, they may become more likely to behave aggressively. Bandura's famous experiment showed that children who watched aggressive models later behaved aggressively themselves. *Examples:* Youth imitating violent behaviour seen in action movies. Some young people copy aggressive language from music or social media.

b) Social rejection

Human beings need social relationships. When people feel rejected or excluded, they may develop anger and aggression. Social rejection can damage a person's self-esteem and lead them to interpret others' actions as hostile. *Examples:* A new employee ignored by colleagues may become hostile. A worker excluded from group decisions may respond aggressively. In Rwanda, this may happen in offices, factories, cooperatives, and community groups.

c) Influence and need fulfilment

Sometimes people use aggression to control others or get what they want. Aggression may occur when someone believes that aggression will help them succeed or that the other person will not fight back. *Examples:* A supervisor shouting at employees to force them to work faster. A trader threatens another trader to take a selling space in the market.

d) Peer pressure and crowd behaviour

Peer pressure can also encourage aggression. Young people may behave aggressively to gain acceptance from friends or social groups. For example, some students may join fights or bully others to get approval from their peers. In crowds or large groups, people may also behave more aggressively because they feel less personally responsible for their actions.

2.3.3. Reducing aggression

Aggression results from interactions between personal, social, and environmental factors. Social psychologists suggest several ways to reduce it.

a) Social learning: punishment and modelling

Since aggression is learned, it can also be reduced through learning processes.

- **Punishment**

Punishment can discourage aggressive behaviour if it is given immediately, is fair, and is explained clearly. *Examples:* A worker who insults colleagues receives a warning letter. A student who fights at school is disciplined. However, punishment must be used carefully.

- **Positive modelling**

People can learn about peaceful behaviour by observing good role models. *Examples:* Managers who solve conflicts calmly. Community leaders promote dialogue instead of violence. Exposure to pro-social behaviour encourages cooperation and respect.

b) Catharsis

Catharsis means expressing anger in a harmless way to reduce aggressive feelings. Examples include playing sports, shouting alone, and exercising. *Examples:* A worker playing football after work to release stress. A person is taking a walk to calm down after a conflict.

However, research shows that these effects are temporary, and aggression may return if the real problem is not solved.

c) Cognitive strategies

Cognitive strategies focus on changing how people think to control aggression.

- **Apology**

A sincere apology can reduce anger and conflict. *Example:* A colleague apologising for making a mistake during teamwork.

- **Self-control and self-regulation**

People should learn to control their emotions and think before reacting. However, when someone is very angry, it becomes difficult to think clearly.

- **Avoiding rumination**

Repeatedly thinking about past conflicts can increase anger. Instead, people can: read books, watch comedy films, talk to friends, and engage in hobbies. *Example:* A worker choosing to relax with family or attend community activities instead of thinking repeatedly about workplace conflicts.

2.3.4. Bullying behaviour

Bullying includes behaviours such as insulting someone, teasing, spreading rumours, threatening, and excluding someone from a group. Bullying is similar to aggression. According to Olweus (1999), bullying is when a person repeatedly harms someone who has less power to defend themselves.

Bullying can occur in many forms: verbal, physical, Psychological, social, and online (cyberbullying)

Although bullying is common in schools, it also occurs in workplaces, homes, prisons, and social groups. *Examples:* Workers mock a colleague because of their rural background. Senior staff humiliating interns. Students spreading rumours about classmates on social media.

a) Motives behind bullying

Researchers suggest two main reasons why people bully others:

- Desire for power: Bullies want to control others.
- Desire to belong to a powerful group: Some people bully others to gain acceptance from certain groups.

Research also shows that some individuals bully others because they feel unhappy or insecure, and bullying temporarily makes them feel stronger. *Example:* A worker humiliates new employees to appear powerful in the office.

b) Steps to reduce bullying behaviour

Bullying can have serious effects on victims, including low self-esteem, anxiety, depression, and social isolation. In extreme cases, victims may even attempt suicide. To reduce bullying, several actions should be taken:

- Recognise that bullying is a serious problem.
- Authorities such as teachers, managers, parents, and supervisors must address it seriously.
- Institutions should establish clear rules against bullying.
- Victims should be informed about support systems available.
- Immediate disciplinary action should be taken against bullies.

For example, schools implement anti-bullying policies. Companies establish workplace anti-harassment policies. Community leaders promote respectful communication.

Application activity 2.3



1. From your observations in your school, family, or community, explain how people learn aggressive behaviour. Give examples of situations where someone may have learned aggression from others.
2. In your opinion, how can mass media (such as television, films, social media, music, or online videos) influence people to behave aggressively? Use examples from your daily life.

2.4. Aggression: Personal and cultural influences on aggression

Learning activity 2.4



1. In your community, what situations or factors do you think can cause people to behave aggressively? Discuss different causes and give real-life examples.
2. In your opinion, why might some people from poor socio-economic backgrounds become aggressive? Explain using examples from your community or society.

Aggression results from the interaction between personal factors and situational influences. A situation may encourage aggression, but people do not react in the same way. For example, some people watch violent movies or play violent video games but do not become aggressive, while others may become aggressive after watching such content.

Just like some people smoke for many years without getting lung cancer, while others become sick, we cannot always predict who will be affected by violent influences.

Therefore, psychologists study personality factors that influence aggression and how these factors interact with social situations.

2.4.1. Individual differences in aggression

People differ in how aggressive they are. Some individuals may display aggressive behaviour depending on their experiences and environment. Women and girls often use non-physical aggression, such as shouting, insulting, spreading rumours, or excluding others. Men and boys are more likely to display physical forms of aggression, such as hitting, pushing, or kicking. People also differ in their beliefs about violence. Some believe aggression is an acceptable way to solve problems. These people are usually more aggressive.

Social environments also influence attitudes toward aggression. For example, members of youth gangs may see violence as normal because their group values strength and fear. Interestingly, people with high self-esteem can sometimes be more aggressive when their self-image is threatened. For example, a bully may become angry when criticised because they want to protect their reputation. Research also shows that people who feel socially or economically disadvantaged may show more aggression, especially if they feel unfairly treated.

2.4.2. Gender differences in aggression

Research across many cultures shows that males are more likely to display physical forms of aggression than females. Females may more often express indirect forms of aggression, such as spreading rumours, social exclusion, or verbal insults, while males may more frequently display direct physical aggression, such as hitting or pushing. However, aggression is influenced not only by biological factors but also by socialisation, cultural expectations, family environment, and individual experiences.

It is important to avoid gender stereotypes when discussing aggression. Not all males are aggressive, and not all females are non-aggressive. Individuals differ greatly

in personality, emotional control, upbringing, and life experiences. Both males and females are capable of peaceful conflict resolution, cooperation, empathy, and self-control.

Biological factors such as testosterone may contribute to aggressive tendencies, but environmental influences such as parenting style, peer groups, media exposure, education, and cultural values also play an important role in shaping behaviour.

Another reason is evolution and social roles. In the past, men were often responsible for hunting, fighting, and defending their communities. These roles encouraged competitive and aggressive behaviours. Studies also show different reactions to stress: males often react with fight or flight. Females often respond by seeking social support and caring for others. In many societies, girls are also socialised to control direct aggression and express it in indirect ways.

2.4.3. Age and aggression

Many people believe that adults are more aggressive than children, but the relationship between age and aggression is complex.

Young children and adolescents often use physical aggression because they have not yet developed strong social skills to solve conflicts.

As children grow older, socialisation teaches them better ways to manage conflicts, such as communication and cooperation. As a result, they gradually learn to control aggressive behaviour.

2.4.4. Cultural influences on aggression

Culture also influences aggression. Research comparing different societies shows that aggression can increase in complex societies with strong hierarchies, such as kingdoms or states. However, societies can change over time. Some societies become more peaceful by promoting values of cooperation and respect.

Many cultural factors, including social values, economic conditions, social organisation, gender roles, and environmental conditions influence aggression. Studies of peaceful societies suggest that aggression can be reduced by promoting values of peace and cooperation, teaching children non-violent behaviour, and creating institutions that encourage conflict resolution. These strategies help build more peaceful communities.

Cultural beliefs and traditions influence how people express anger and aggression. In some societies, aggressive behaviour may be accepted as a sign of strength or authority, while in others, peaceful conflict resolution is encouraged. Family upbringing, religious teachings, community values, and exposure to violence in society can shape attitudes toward aggression. For example, children raised in violent homes may learn that aggression is a normal way to solve problems.

2.4.5. Socio-Economic Stress

Poverty, unemployment, overcrowding, and social inequality may increase frustration and stress, which can contribute to aggressive behaviour in communities. However, not all people experiencing hardship become aggressive because individual self-control and social support also play important roles.

Application activity 2.4



In your school, what actions can people or institutions take to reduce aggression and promote peaceful relationships among different social groups? Give examples.

2.5. Genetic aspects: Activity, Passivity, Fussiness, Cuddliness, and Responsiveness

Learning activity 2.5



One evening, the family was preparing dinner together. The brother was very active and quickly helped with different tasks, while his young sister preferred to sit quietly and observe. When their mother asked them to help, the brother responded immediately, while the sister took some time before responding. Suddenly, the father changed the usual dinner arrangements, the sister became upset, while the brother remained calm. From that time, the mother started sitting close to her children and showing affection, while the father and the children enjoyed the way the mother of the house was treating them.

Question:

1. What differences in behaviour can you identify among the four family members? Why do you think they behave differently even though they live in the same family?
2. Can a child be naturally quiet or less active without being sick or unhappy? Why? community or society.

2.5.1. Meaning of Genetic Aspects of Personality

Genetic aspects of personality refer to characteristics or tendencies that are influenced by heredity and biological factors. Some differences in the way people behave, react to situations, and relate to others may be partly influenced by characteristics inherited from their parents. However, personality does not develop from heredity alone. A person's family, education, culture, relationships, experiences and environment also influence personality development. Some genetic aspects that may be observed in behaviour include activity, passivity, fussiness, cuddliness, and responsiveness.

2.5.2. Activity

Activity refers to a tendency to be physically energetic, restless, or constantly engaged in movement and activities. An active child may enjoy playing, moving around, exploring the environment and participating in different activities. For example, a learner who enjoys participating in sports, moving around during break time and actively taking part in classroom activities may show a high level of activity.

2.5.3. Passivity

Passivity refers to a tendency to remain quiet, less physically active, or less likely to initiate activities. A passive child may prefer observing others before joining an activity and may need encouragement to participate. For example, a group member may remain seated and observe classmates during a group activity before deciding to participate.

Passivity should not automatically be considered a negative characteristic. A quiet person may still be attentive, thoughtful and capable of participating effectively when given appropriate opportunities.

2.5.4. Fussiness

Fussiness refers to a tendency to become easily dissatisfied, uncomfortable, irritated or difficult to please, particularly when needs or expectations are not met. A person who is easily upset by changes in routine or becomes dissatisfied with ordinary situations may show fussiness. For example, a young child may become upset whenever the usual daily routine is changed or may repeatedly complain about small changes in their environment.

2.5.5. Cuddliness

Cuddliness refers to a tendency to seek physical closeness, comfort, affection and contact with familiar or trusted people. It may be observed especially among young children. For example, a young child may enjoy being held by a parent or caregiver, stay close to familiar people, or seek comfort from them when feeling tired or uncomfortable.

2.5.6. Responsiveness

Responsiveness refers to the tendency to react or respond to people, situations or stimuli in the environment. A responsive child may quickly react when spoken to, called, encouraged or exposed to a new situation. For example, a learner who listens and responds quickly when a teacher asks a question may demonstrate responsiveness.

Application activity 2.5



Two children, Mwiza and Gakuru, are raised in the same family. From an early age, Mwiza is very active, enjoys moving around and quickly joins games with other children. Gakuru is usually quiet, prefers sitting and observing others, and takes more time to respond when spoken to. Another child in their neighbourhood often becomes upset when routines change, while another enjoys being held, staying close to familiar people, and quickly responds when someone calls or talks to them.

Discuss the following questions:

1. What differences in behaviour can you identify among the Mwiza, Gakuru, and their neighbouring children?
2. Which child appears to be active and which one appears to be passive? Give reasons for your answers.

3. Which child shows fussiness, and what behaviour suggests this?
4. What behaviour can show cuddliness in a child?
5. What does it mean for a child to be responsive?

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2.6. End unit assessment



In a certain community, teachers and local leaders have noticed increasing aggressive behaviour among people in schools, workplaces, and neighbourhoods. Some students insult or fight with their classmates, while in some workplaces, employees shout at colleagues or refuse to cooperate when disagreements arise. Community members also observe that younger people sometimes solve conflicts through physical fights, while older people tend to control their anger and use discussion to solve problems.

They also notice that boys and girls may react differently when they are angry. Because of these situations, the community leaders want to understand how aggression can be reduced among people and how factors such as age and gender may influence aggressive behaviour. They also want to examine common beliefs about differences between men and women and other factors that may contribute to aggressive behaviour in society.

Based on this situation, discuss strategies that can help reduce aggression in the community, explain how age and gender can influence aggressive behaviour, and evaluate the truthfulness of different statements related to personality and aggression.

Questions

A. Knowledge, Skills, Attitudes, and Values

1. Based on this situation, what strategies do you think people in your school or community can use to reduce aggressive behaviour? Give examples.

B. Knowledge, Skills

2. From your observation, how do age and gender influence the way people express aggression in your community?

C: Knowledge, Skills, Attitudes, and Values

3. Think about behaviours you have seen among students, or community members. Which actions in the situation above can be considered aggressive, and why?
4. What attitude should you develop towards people who have different behavioural tendencies from yours, and why?

UNIT 3

THEORIES OF PERSONALITY



Key Unit Competence:

Use knowledge acquired in personality theory to explain individual differences

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Explain the main personality theories and their key concepts in psychology.
- Identify different perspectives on personality according to various schools of thought.
- Describe the contributions of key theorists in the development of personality Psychology.
- Outline the main differences and similarities among various personality theories.

Skills

2

- Integrate multiple thoughts and theories about personality when defining and analysing it in various situations.
- Use thoughts and views embedded in theories of personality to understand human behaviour
- Analyse the similarities and differences between major personality theories and their approaches to explaining human behaviour.
- Create a visual diagram or concept map integrating key concepts from different personality theories to show their relationships.
- Evaluate the relevance and practical application of a chosen personality theory in explaining individual behaviour.

Attitudes and values

3

- Value different thoughts and views as cited in theories of personality.
- Recognise and value the contributions of various theorists in the development of personality studies.

- Respect individual differences in personality as highlighted by various theories.

Introduction of the Unit

Have you ever noticed how two classmates can face the same situation but react in completely different ways? One learner may enjoy speaking in front of others, while another prefers to stay quiet and observe. Some people are very organised and careful with their work, while others are more spontaneous and flexible. These differences show that every person has a unique personality that influences how they think, feel, and behave.

This unit introduces the theories of Personality and explores how Psychologists explain individual differences among people. You will study major perspectives such as the Psychoanalytic theory, Neo-Freudian approaches, humanistic theories, and the cognitive approach, which help explain how personality develops and shapes behaviour. The unit also examines personality typologies, including classifications by Heymans and Wiersma, the trait approach (extraversion and introversion), and Myers-Briggs personality preferences.

In addition, you will learn about important personality characteristics such as openness, conscientiousness, extraversion, agreeableness, and neuroticism. Through this unit, you will develop the ability to use knowledge from personality theories to explain why individuals differ in their behaviour, attitudes, and ways of interacting with others. Understanding these concepts will help you interpret human behaviour more accurately and appreciate the diversity of personalities found in everyday life.

Introductory activity



Introductory activity:

Gwiza, Karabo, and Gisa are three students in the same secondary school who show different personality traits. Gwiza is quiet, disciplined, and very religious; he avoids cheating and relationships because of his parents' advice, but he sometimes spends so much time praying that he forgets to review his school lessons. Karabo, on the other hand, is impulsive and careless; she dislikes working hard, accepts average results, cheats in exams, and focuses mainly on satisfying her personal desires without thinking about consequences. Gisa is sociable, responsible, and balanced; he manages both his studies and religious activities well, works hard to succeed honestly, and decides to delay having a romantic relationship so she can focus on her education.

Questions:

1. From this scenario and from a personality point of view, how can you explain the different behaviours of Gwiza, Karabo, and Gisa?
2. Which student's personality do you think is closest to your own? Explain why.

3.1. Psychoanalytic theories of personality

Learning activity 3.1



In a family, Sangwa and Hirwa are two young children aged 8 and 4. Sangwa is calm, polite, and comfortable sleeping alone in his room because he believes he is mature enough. In contrast, Hirwa is very active, noisy, and refuses to sleep alone; he insists on sleeping with his mother. When their parents try to stop him, especially when the father uses punishment, Hirwa becomes upset and even develops negative feelings toward his father because he thinks his father prevents him from staying with his mother. At night, Hirwa often cries and asks to sleep in his mother's room, but he feels happy when his father is away because he then gets the chance to stay with her. Meanwhile, Sangwa laughs at Hirwa's behaviour and does not understand why his younger brother acts that way.

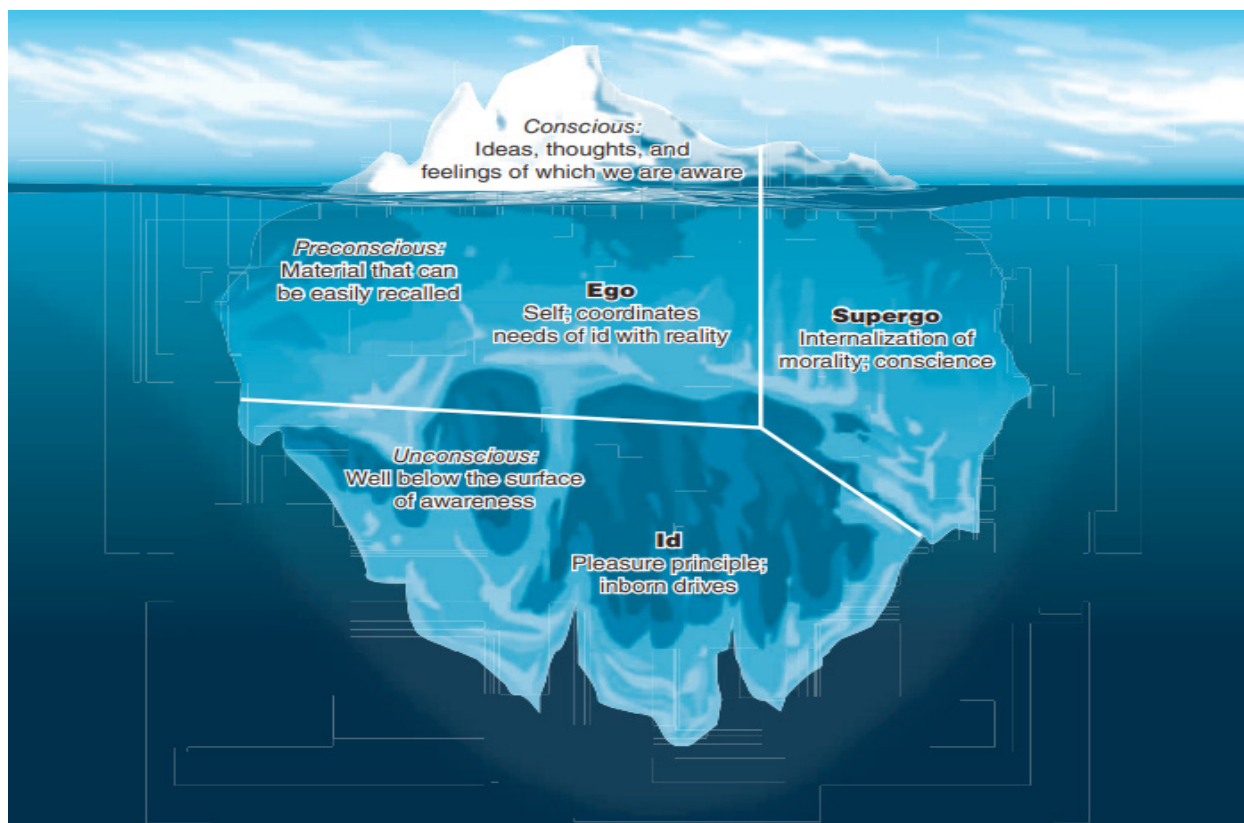
Questions

1. From the scenario, explain the behaviours of Sangwa and Hirwa and describe how they are different from each other.
2. In your opinion, does Hirwa exaggerate in his behaviour, and why do you think he develops negative feelings toward his father? Explain your answer.
3. Have you ever experienced a situation like Sangwa or Hirwa when you were younger? Briefly explain.

Sigmund Freud (1856–1939) was known as the founder of Psychoanalytic theory, and his early students Carl Gustav Jung and Alfred Adler are among the main figures in this approach. According to this view, much of human development takes place beyond conscious awareness because children are often unaware of the motives underlying their behaviour and mental activities. They also emphasised that early childhood experiences with parents play a significant role in shaping personality development. Sigmund Freud argued that personality development is grounded in the principle of psychological determinism, which holds that human behaviour is largely influenced by unconscious psychological processes.

3.1.1. Freud and the Mind

According to Freud, the mind has three levels: the conscious, preconscious, and unconscious. The conscious mind includes thoughts, perceptions, memories, and feelings that a person is currently aware of, and it represents only a small part of the mind, like the tip of an iceberg.



Source: Cocioppo & Freberg (2013, p. 575)

Below the conscious level is the preconscious, which contains memories and information that are not currently in awareness but can easily be recalled into consciousness. Freud believed that the conscious and preconscious together form only a small portion of mental life.

The unconscious mind forms the largest part. It contains hidden drives, instincts, memories, and emotions, especially those related to traumatic experiences. Although people are not aware of these processes, they strongly influence behaviour and personality. Freud's Psychoanalytic theory, therefore, emphasizes the powerful role of the unconscious mind in determining human behaviour (Sibi, 2020).

3.1.2. Freud's Model of the Structure of Personality

Freud proposed that personality is made up of three interacting structures: the id, the ego, and the superego. These are not physical parts of the brain, but Psychological forces or concepts used to explain human behaviour. Their existence is inferred from the way people act. When these three parts work together in balance, they contribute to a well-adjusted personality.

- **The Id**

The Id is present at birth and is the most primitive and instinctive part of personality. It deals with basic biological drives, instincts, and reflexes, and operates completely in the unconscious mind (KIE, 2008). The id follows the pleasure principle, seeking immediate satisfaction of needs and the reduction of pain, tension, or anxiety. It is the reservoir of psychic energy, containing urges related to food, water, warmth, and sexual satisfaction. The id behaves like a demanding and impulsive child, seeking pleasure without considering consequences.

Freud believed the id includes two main instincts: Eros and Thanatos. Eros, the life instinct, supports survival and directs life-sustaining activities such as breathing, eating, and reproduction. It produces energy called libido. Thanatos, the death instinct, represents destructive tendencies that may appear as aggression or violence. Freud believed that the life instinct (Eros) is generally stronger and helps maintain human survival.

- **The Ego**

The ego represents the individual's sense of self and begins to develop during the first year of life. Its role is to satisfy the desires of the id in realistic and socially acceptable ways. The ego operates in both the conscious and unconscious levels and follows the reality principle. It evaluates the demands of the id and considers the expectations of the superego and the realities of the external world. The ego, therefore, mediates between the id, the superego, and reality, using logical thinking, time awareness, and the ability to distinguish between what is real and what is not.

- **The Superego**

The superego develops between the ages of five and seven years and represents the moral aspect of personality. The superego develops through the process of socialisation, as children internalise the values, norms, and ideals of society from their parents and other authority figures. It enables individuals to distinguish between right and wrong and functions as the moral conscience that guides their behaviour. It encourages behaviour that follows moral standards and social expectations. The superego functions through reinforcement processes:

- a) For good behaviour: it provides rewards, leading to feelings of pride and self-esteem.
- b) For bad behaviour: it applies punishment, leading to feelings of guilt or inferiority.

Freud described the superego as a strict and demanding moral authority that strives for perfection and may ignore individual abilities or environmental circumstances.

3.1.3. Anxiety and defence mechanisms

Inner-psychic conflicts are inevitable in every human being, often arising from sexual and aggressive impulses and internal tensions. In many cases, these conflicts resolve themselves after a short time; however, sometimes they may persist and cause Psychological distress. When such internal struggles continue for a long time, they may weaken an individual and affect daily functioning. Feelings of fear or anxiety may appear and should not be underestimated. When the ego cannot successfully balance the conflicting demands of the id, the superego, and reality, anxiety is produced. Anxiety is a feeling of tension that interferes with normal functioning and signals to the ego that action is needed to protect the personality. To reduce this anxiety, the ego uses defence mechanisms, which are unconscious mental strategies that protect the individual from overwhelming fear, feelings of guilt, or other unpleasant experiences.

Freud identified nine primary defence mechanisms: repression, sublimation, rationalisation, displacement, reaction formation, projection, denial, regression, and identification. These mechanisms help individuals redirect or manage distressing thoughts and impulses in more acceptable ways. However, Freud also noted that excessive reliance on a particular defence mechanism can be harmful.

Other mechanisms, such as compensation and withdrawal, are also commonly discussed in Psychology.

Freud's Defence Mechanisms

Defense mechanism	Definition	Example
Repression	Keeping distressing thoughts or feelings buried in the unconscious mind.	A person forgets a traumatic childhood event; a student forgets an embarrassing moment in class.
Sublimation	Redirecting unacceptable urges into socially acceptable activities.	An aggressive youth practices martial arts; a person channels anger into sports or hard work.
Rationalization	Creating logical excuses to justify unacceptable behaviour.	A learner who cheats on a test says, "Everyone else was doing it."
Displacement	Redirecting emotional feelings from the real source to a safer target.	A student scolded by a teacher goes home and argues with a younger sibling.
Reaction formation	Behaving in a way opposite to one's true feelings.	A person who dislikes a colleague acts extremely friendly toward them.
Projection	Attributing one's own unacceptable thoughts or feelings to others.	A person who lies often accuses others of being dishonest.
Denial	Refusing to accept reality or facts that cause anxiety.	A person with a serious illness refuses to believe the medical diagnosis.
Regression	Returning to childish behaviour to cope with stress.	A teenager throws a tantrum or cries when facing difficult situations.
Identification	Reducing anxiety by adopting the characteristics of another person.	A child imitates a respected teacher or parent to feel confident.
Compensation	Covering weaknesses by emphasising strengths in another area.	A learner who struggles academically excels in sports or music.
Withdrawal	Avoiding situations that create stress or anxiety.	A student who fears criticism avoids group discussions or social activities.

Source: Adapted from information by John T. Cacioppo and Laura A. Freberg (2013), Grohol (2016), and Rwanda Education Board (2020).

3.1.4. Psychosexual stages and personality development

Freud's most controversial argument concerns personality formation. Freud proposed that personality develops through a series of Psychosexual stages, a theory later discussed and expanded by some of his followers, including Anna Freud. He believed that individuals pass through five stages of Psychosexual development: oral, anal, phallic, latency, and genital. At each stage, a specific part of the body becomes the main source of pleasure, called an erogenous zone (such as the mouth, anus, or genitals). Freud argued that adult personality is shaped by how conflicts between instinctual drives and social expectations (such as weaning, toilet training, or social rules) are resolved during these stages.

Fixation: In Freud's theory, fixation occurs when a person fails to successfully resolve the conflicts of a particular Psychosexual stage. As a result, part of the individual's Psychological energy remains attached to that stage, which may influence behaviour in adulthood. When fixation occurs, the person may show immature behaviours or difficulties in forming healthy social and emotional relationships later in life.

The five stages of Psychosexual development proposed by Sigmund Freud are described below (Hansell, Ehrlich, Katz & Minter, 2008).

1. The Oral Stage (Birth–18 months)

This is the first stage of development occurring during the infant's first 18 months. The **erogenous zone is the mouth**. Infants gain pleasure through activities such as sucking, chewing, biting, and licking. These actions provide both nourishment and comfort. For example, a baby may suck a thumb or pacifier even when not hungry because the action provides pleasure and helps reduce tension.

Fixation at the oral stage may occur if feeding is inconsistent or if the child becomes overly dependent on oral comfort.

Consequences:

- An adult may become overly dependent on others.
- Behaviours such as excessive eating, smoking, or nail-biting may develop.
- A child may regress to thumb-sucking when feeling insecure, for example, after the birth of a sibling.

2. The Anal Stage (18 months–3 years)

This stage occurs when children begin to gain voluntary control over bowel movements. The erogenous zone is the anus, and pleasure is associated with controlling or releasing bodily waste. During this stage, toilet training becomes important.

Freud believed that toilet training creates the first major conflict between a child's natural impulses and society's expectations. If parents are too strict or too lenient during toilet training, fixation may occur.

Consequences:

- Excessive neatness, orderliness, and perfectionism in adulthood.
- Alternatively, some individuals may become messy, careless, or disorganised.

3. The Phallic Stage (3–6 years)

During this stage, the erogenous zone is the genitals. Children become more aware of their bodies and may explore them. Freud suggested that children also develop emotional attachments to the opposite-sex parent and rivalry toward the same-sex parent.

In boys, this conflict is known as the Oedipus complex, while in girls, Freud described a similar idea called the Electra complex. For example, a boy may want his mother's attention and feel jealous of his father.

Resolution:

Children eventually resolve this conflict by identifying with the same-sex parent, adopting their behaviours, attitudes, and moral values. This process helps form the superego.

4. The Latency Stage (6 years–puberty)

This stage occurs between 6 years and puberty. Sexual impulses become less active and are largely repressed. Children focus more on schoolwork, friendships, hobbies, and learning social skills.

Although sexual energy is dormant during this period, children may still experience emotional challenges. For example, jealousy may occur when a new sibling is born. However, most energy during this stage is directed toward intellectual development and social relationships.

5. The Genital Stage (Puberty–Adulthood)

The genital stage begins at puberty and continues into adulthood. During this stage, sexual energy reappears and is directed toward mature relationships outside the family. The genitals become the main erogenous zone, and individuals begin forming romantic and emotional relationships.

Freud believed that an important developmental task during this stage is gaining independence from parents. Young people gradually separate emotionally from their parents and build their own identities, relationships, and adult responsibilities.

Summary of Freud's Psychosexual Stages of Development

Stage	Approximate ages	Main characteristics
Oral stage	Birth to 1 year	Pleasure obtained through sucking and oral activities
Anal stage	1 to 3 years	Conflict between bowel control and toilet training expectations
Phallic stage	3 to 6 years	Oedipus/Electra conflicts and identification with same-sex parent
Latency stage	6 to 12 years	Focus on learning, social skills, and internalising social rules
Genital stage	12 years and above	Development of mature sexual interests and relationships

Source: Adapted from John T. Cacioppo & Laura A. Freberg (2013).

Application activity 3.1



1. Based on the ideas of Sigmund Freud, summarise the main view of the Psychoanalytic theory of personality.
2. Using examples from everyday life, explain the concepts of the Oedipus complex and fixation in personality development.
3. Identify the Psychosexual stages of personality development proposed by Sigmund Freud and indicate the approximate age for each stage.
4. Using the behaviour of a young child in a family, explain the concepts of Oedipus complex and fixation as described in the Psychoanalytic theory of Sigmund Freud.
5. Imagine you are explaining personality development to a classmate. Enumerate the Psychosexual stages of personality development proposed by Sigmund Freud and indicate the approximate age for each stage.

3.2. Neo-Freudian-Psychoanalytic theory

Learning activity 3.2



1. Think of some Rwandan proverbs that show how people's behaviour is acquired, shaped, or changed through interaction with society.
2. Using your family as an example, explain how your behaviour has been influenced by the actions, attitudes, and guidance of your family members.
3. Using Rwandan culture, explain at least 2 cultural factors that shape our personality.

Sigmund Freud attracted many scholars who followed his Psychoanalytic ideas. However, some of them later disagreed with him on several points. These scholars became known as Neo-Freudians. Freud believed that personality is mainly formed during the first seven years of life through the Psychosexual stages. Neo-Freudians disagreed with this idea. They argued that social and cultural experiences also shape personality throughout life, not only childhood instincts.

Because of these disagreements, some of Freud's followers developed their own theories. The most important Neo-Freudians were Carl Gustav Jung, Alfred Adler, and Karen Horney.

3.2.1. Carl Gustav Jung (1875-1961)

Carl Gustav Jung was a Swiss Psychiatrist who created Analytical Psychology. His ideas influenced Psychology, culture, religion, and Philosophy.

1. Differences between Jung and Freud

Jung disagreed with Freud on several important points:

- a) **Meaning of Libido:** Freud believed libido mainly meant sexual energy. Jung believed libido is a general Psychological energy that includes motivation, creativity, ambition, and other driving forces behind human behaviour. Example: A teacher in secondary school may work hard to prepare lessons not because of sexual drives, but because of the motivation to help students succeed.
- b) **Past and Future Influences:** Freud focused mainly on past childhood experiences. Jung believed people are influenced by both past experiences and future goals.
- c) **Example:** A young nurse in a health centre may work hard because of childhood experiences and because she dreams of becoming a medical doctor in the future.
- d) **Role of the Unconscious:** Jung emphasised the unconscious mind even more than Freud. He introduced the idea of the collective unconscious, which includes inherited experiences shared by all humans.

2. The Collective Unconscious

Jung believed that all humans share common mental images called **archetypes**. These are symbols that appear in stories, traditions, and cultures around the world. Examples of archetypes: Mother, Father, Hero, God, and Sun or Moon. *Example:* In many Rwandan stories and cultural narratives, the **hero who protects the community** reflects the archetype of the hero.

3. Anima and Animus

Jung explained that both men and women have Psychological traits of the opposite sex.

- **Anima:** the feminine side of a man.
- **Animus:** the masculine side of a woman.

Example: A male school headmaster may demonstrate gentleness and care when supporting students (anima), while a female bank manager may exhibit strong leadership and assertiveness (animus), showing that both men and women can express a balance of traditionally feminine and masculine traits in their personalities

4. Jung's Psychological Types

Jung also introduced the concepts of introversion and extraversion.

- **Introverts:** Quiet, prefer working alone, do not easily share their feelings. Example: A laboratory technician who prefers working quietly and carefully.
- **Extroverts:** Sociable, Talkative, enjoy interacting with others. Example: A market seller in Kigali who interacts with many customers daily.

3.2.2. Alfred Adler's Individual Psychology (1870-1937)

Alfred Adler introduced the idea of the inferiority complex. An inferiority complex is a strong feeling of being less capable or less important than others. These feelings may come from real or imagined weaknesses. Example: A young employee in a company may feel inferior because colleagues have higher education.

Sometimes people overcompensate to hide these feelings. Example: Someone may buy an expensive car or phone to appear more successful than others.

Adler also emphasised the importance of social relationships and cooperation. He believed that people develop through social interactions, not only through biological drives.

Birth Order

Adler suggested that a person's **position in the family** can influence personality.

- First-born children: may become responsible and hardworking.
- Middle children: may learn to cooperate and negotiate.
- Youngest children: may receive more attention and sometimes become dependent. Example: In a farming family in rural Rwanda, the firstborn child may take leadership in helping parents manage the farm.

Three Concepts in Adler's Individual Psychology

1. Striving for Superiority

All humans try to improve themselves and succeed in life. This does not necessarily mean dominating others, but becoming better and achieving goals. Example: A teacher studying for a master's degree to improve professional skills. However, some people may misuse this motivation by seeking power or prestige.

2. Inferiority

People naturally try to overcome weaknesses. Types of inferiority:

- Physical inferiority: weakness in the body.
- Psychological inferiority: low self-confidence.
- Social inferiority: feeling less important in society.

Example: A student who feels weak in mathematics may study harder to improve. Adler emphasised the importance of realistic self-understanding to achieve success.

3. Style of Life

A style of life is the unique way a person deals with problems and goals. Adler believed this style begins around age five and becomes part of personality. Example: One worker may solve problems through teamwork, while another prefers working independently.

3.2.3. Karen Horney (1885-1952)

Karen Horney criticised Freud's ideas, especially his views about women.

She believed that social and cultural factors, not biological differences, explain many differences between men and women.

Horney also argued that personality development is strongly shaped by a person's relationships with parents and the wider social environment. Example: Children who grow up in supportive families and communities often develop confidence and emotional stability. She also introduced the idea of womb envy, suggesting that some men may feel envy toward women's ability to give birth.

a) The Childhood need for safety

Horney believed that children need security, love, and protection. If parents behave negatively, children may develop anxiety. Examples of harmful parental behaviours: Favouring one child over another, unfair punishment, breaking promises, and humiliating the child. Example: A child who is constantly compared with a more successful sibling may feel insecure and anxious.

b) Neurotic Needs

Horney identified 10 neurotic needs, which are unhealthy ways people try to deal with anxiety. Examples include the need for affection and approval, power, admiration, perfection, and independence. These needs serve as ways for individuals to cope with feelings of fear and insecurity.

c) Three Personality Types by Karen Horney

1. Compliant Personality

These people move toward others. They strongly seek love, approval, and protection. Example: An employee who always agrees with the manager and avoids disagreement to be liked.

2. The Aggressive Personality

These people move against others. They believe the world is competitive and try to dominate others. Example: A business manager who constantly competes with colleagues and seeks power and recognition.

3. Detached Personality

These people move away from others. They prefer independence and avoid close relationships. Example: A computer programmer who prefers working.

Application activity 3.2



1. During your Psychology studies, you learned about some scholars who expanded or challenged the ideas of Sigmund Freud. Describe the three Neo-Freudian Psychologists studied in this unit and explain how their ideas contributed to understanding personality.

2. Imagine you are teaching your classmates about Neo-Freudian theories. Explain how the ideas of Carl Gustav Jung, Alfred Adler, and Karen Horney differ from those of Sigmund Freud.
3. In your own words and using examples from Rwandan society or workplaces, explain the concept of the collective unconscious as proposed by Carl Gustav Jung.
4. Consider a situation where a student or worker feels less capable than others and tries to prove their worth. Explain the concept of an inferiority complex according to Alfred Adler and how it may influence behaviour.
5. Explain the concept of womb envy introduced by Karen Horney and discuss why she used it to challenge Freud's ideas about gender.
6. In everyday life, people respond differently to relationships and anxiety. Explain the three personality types proposed by Karen Horney and give a real-life example for each type.

3.3. Humanistic theory of personality



Learning activity 3.3



Taking your family as an example, describe your life by answering these questions:

1. Do you always struggle to find basic needs in your family? If yes, explain how. If not, explain.
2. Do you have enough security at home and on the way to and from school? What are the elements of insecurity in your location?
3. Can you describe the kind of love you experience at home and at school? What are your aspirations for the future?

Psychoanalysis, developed by Sigmund Freud, viewed human beings as largely driven by unconscious instincts and biological drives. In contrast, humanistic Psychology views human beings as basically good, capable of growth, and striving to reach their full potential.

Humanistic Psychologists criticized Freud and other Psychoanalysts because they mainly studied people with mental problems. They argued that if Psychologists only study neuroses and psychoses, they cannot fully understand positive human qualities such as creativity, kindness, happiness, and success. Instead, humanistic Psychologists suggested studying healthy and successful people to understand how individuals grow and develop.

The humanistic approach to personality is mainly represented by the work of Abraham Maslow and Carl Rogers. Their theories emphasise human strengths and abilities, free will, personal growth, and self-realisation. Humanistic Psychology presents an optimistic view of human nature, describing people as active, creative, and motivated to grow. Example: A teacher who works hard to improve students' performance, or a nurse who cares deeply for patients, demonstrates the human potential for growth and helping others.

3.3.1. Abraham Maslow: Needs-hierarchy theory

Abraham Maslow is considered one of the founders of humanistic Psychology. He criticised both Psychoanalysis and behaviourism for focusing mainly on problems rather than positive human qualities such as happiness and personal fulfilment. Maslow believed Psychologists should study healthy and successful individuals to understand human potential. His theory suggests that all people are born with natural needs that guide their behaviour and development.

Personality development: The Hierarchy of Needs

Maslow proposed that human needs are arranged in a hierarchy of five levels: physiological needs, safety needs, belongingness and love needs, esteem needs, and self-actualisation needs

Lower needs must usually be satisfied before higher needs become important. Example: A worker who has no food or shelter will focus first on survival, not on gaining recognition or promotion at work. Similarly, a farmer who is concerned about providing food for the family will focus more on harvesting crops than on gaining social status.

However, when basic needs are satisfied, people begin to pursue respect, achievement, and personal development.

Abraham Maslow also noted that the order of needs may sometimes change, especially during difficult or unusual situations. Example: If an employee loses a job during an economic crisis, safety and financial needs may again become more important than prestige or promotion.

Characteristics of needs

Maslow described several important characteristics of human needs:

- Lower needs are stronger and more urgent than higher needs.
- Higher needs develop later in life. For example, children first need food and safety, while self-development becomes important later.
- Failure to satisfy lower needs causes serious problems, while higher needs can sometimes be postponed.
- Higher needs support personal growth and happiness.
- Meeting higher needs often requires better social and economic conditions.
- A need does not need to be completely satisfied before the next one begins. Example: A teacher with a stable salary may start focusing on professional development or leadership roles.

1. Physiological needs

These are the basic survival needs, such as food, water, air, and rest. When these needs are not met, they dominate behaviour. Example: A worker who has not eaten all day will focus on finding food, rather than thinking about workplace recognition or status. In communities where poverty exists, physiological needs remain strong motivators.

2. Safety needs

These include security, stability, and protection from danger. Adults often try to maintain safety through stable employment, savings, or insurance. Example: A civil servant may choose to keep a secure government job rather than take the risk of starting a new business.

3. Belongingness and love need

Once basic needs are satisfied, people seek relationships, friendship, and acceptance. These needs may be fulfilled through family, friendships, community groups, and religious communities. Example: Teachers may form professional groups or churches where they feel accepted and supported.

4. Esteem needs

Esteem involves self-respect, recognition from others, achievement, and success.

When esteem needs are satisfied, people feel confident and capable. Example: A nurse who receives recognition for good service may feel motivated to work even harder. However, when esteem is lacking, people may feel inferior or discouraged.

5. The self-actualisation need

Self-actualisation is the highest level of Maslow's hierarchy. It refers to achieving one's full potential and personal growth.

People at this level use their talents and abilities fully.

Maslow explained that individuals feel satisfied when they express their abilities. Examples: A teacher who enjoys helping students succeed, a farmer who improves agricultural practices, an artist who creates traditional Rwandan artwork.

Each person reaches self-actualisation in different ways depending on their talents. For self-actualisation to occur, several conditions are helpful: freedom to express ideas, secure relationships, realistic understanding of oneself.

However, some people may sacrifice basic needs for strong beliefs or values. Example: Some religious leaders or activists may sacrifice comfort for service to the community.

The study of self-actualizers

Maslow believed that only a small number of people reach full self-actualisation. He identified several characteristics of self-actualised individuals: realistic understanding of life, acceptance of themselves and others, creativity, independence, strong relationships, and interest in helping society. Example: A community leader who works to improve education or health services demonstrates concern for the wider community.

3.3.2. Carl Rogers

Carl Rogers also developed a humanistic theory of personality. Unlike Freud, Rogers did not emphasise unconscious drives or childhood conflicts. Instead, he believed that present experiences and personal perceptions are most important in shaping personality. According to Rogers, personality can only be understood by looking at how individuals view themselves and their experiences.

Person-Centered Perspective

Rogers developed person-centred therapy, a Psychological approach that focuses on helping individuals grow through supportive relationships. Rogers believed people are naturally good and capable of growth, like plants that grow when given the right conditions. Growth occurs when three conditions are present:

1. Genuineness

Being honest and open with others. Example: A manager who openly discusses challenges with employees builds trust.

2. Acceptance (unconditional positive regard)

Accepting people without judging them, even when they make mistakes. Example: A teacher who supports a struggling student instead of criticising them.

3. Empathy

Understanding another person's feelings and experiences. Example: A nurse who listens carefully to a patient's concerns. Rogers believed these qualities help people grow in many relationships: teacher and student, manager and staff, parent and child, and friends.

The self and the tendency toward actualisation

Rogers believed that people naturally try to develop, improve, and maintain themselves. He studied how environmental factors influence personality, including family environment, health, education, culture, economic conditions, and social relationships. However, he also emphasised the importance of self-understanding. A person who understands themselves can make responsible choices and develop positively.

Congruence and incongruence

Rogers explained that sometimes people's self-image does not match reality.

- Congruence: when self-concept matches reality.
- Incongruence: when there is a mismatch.

Example: An employee may believe he is very honest, but frequently lies about being late for work. Rogers believed that conditional love from parents may create incongruence. If parents only love children when they behave in certain ways, children may hide their true feelings. However, unconditional love helps children develop confidence and honesty.

Characteristics of fully functioning people

Rogers described a fully functioning person as someone who continues to grow throughout life.

Characteristics include openness to experiences, living fully in the present, trusting their own decisions, creativity, and ability to adapt to change. Example: An innovative entrepreneur who adapts to new business opportunities shows these qualities. Rogers used the term "actualising" rather than "actualised" because personal growth never truly ends.

Critics of humanistic theories

Some Psychologists have criticised humanistic theories.

Common criticisms include concepts that are sometimes too vague and difficult to test scientifically. Theories may rely on personal opinions rather than strict research, and critics argue that humanistic Psychology may ignore human capacity for harmful behaviour.

Despite these criticisms, humanistic Psychology remains important because it highlights human potential, dignity, and personal growth.

Application activity 3.3



1. During a Psychology discussion in class, students compared the ideas of humanistic psychologists with those of Sigmund Freud. Explain how humanistic Psychologists criticized the Psychoanalytic view of personality, using examples from everyday life.

2. Think about people in your community or workplace who demonstrate positive qualities such as cooperation, creativity, or leadership. Explain why humanistic Psychologists suggested studying human strengths and virtues instead of focusing only on mental disorders.
3. Imagine you are explaining personality development to new students. Describe the central idea of the theory proposed by Abraham Maslow and illustrate it with an example from real life.
4. In relationships such as teacher–student, manager–employee, or parent–child, people can help others grow and develop. Explain the three ways people nurture growth according to Carl Rogers and give examples from daily life.
5. Some Psychologists have criticised humanistic theories of personality. Discuss at least two criticisms of the humanistic approach and explain why some Psychologists question these theories.

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3.4. Cognitive theories of personality

Learning activity 3.4



1. Imagine you are observing classmates, friends, or colleagues at work. Explain the theory of social learning as proposed by Albert Bandura, using examples from school or workplace settings in Rwanda.
2. Think about the Rwandan proverb “Uwiba ahetse aba abwiriza uwo mu mugongo.” Explain how this saying relates to Bandura’s social learning theory, giving an example from your community or school where someone imitates another person’s actions.
3. In classrooms or training sessions, students often pick up habits from teachers. Describe how students learn by observing and copying a teacher’s behaviour, and give real examples you have seen in Rwandan schools or colleges.

Cognitive personality theories focus on thoughts and perceptions rather than biological or genetic factors. They explain that a person’s self-image, values, and way of thinking shape emotions and influence behaviour. For instance, if a worker in Kigali believes they are honest and hardworking, they will behave responsibly at their job.

Modern Psychology embraced cognitive approaches in the 1960s when Albert Bandura introduced Social Learning Theory, which later evolved into Social Cognitive Theory by the 1980s. Today, many cognitive approaches explain personality development.

3.4.1. Social cognitive theory

Social Cognitive Theory suggests people learn by observing others, and that thinking is central to personality. Behaviour is shaped by the interaction between environmental factors, personal characteristics, and cognitive processes. For example, a young apprentice in a carpentry workshop may learn polite customer interaction by observing the master carpenter.

Albert Bandura (1925-2021)

Bandura developed the concept of reciprocal determinism, showing that behaviour, thoughts, and the environment influence one another. For example, an employee who is praised for good performance is likely to repeat the behaviour, which can also contribute positively to the workplace culture. Similarly, an adolescent who is bullied may respond with aggression, showing the two-way interaction between behaviour and environment.

The social-cognitive perspective emphasises:

- **Choosing environments:** A student choosing to attend a high-performing school in Kigali interacts with a learning-focused environment that influences personality.
- **Interpreting events:** An anxious worker may perceive constructive criticism as a threat, affecting their responses.
- **Creating situations:** A teacher's supportive attitude may encourage students to engage positively; a negative attitude may cause disengagement.

Bandura's model highlights interactions among personal factors, behaviour, and context, as well as gene-environment influences. For example, someone naturally quick-tempered may face conflicts if raised in a stressful workplace, increasing aggressive behaviour.

3.4.2. Other Cognitive Theories

❖ Rotter's Locus of Control

Rotter expanded Bandura's ideas, suggesting that people differ in their beliefs about control over life outcomes:

- **Internal locus of control:** A student in Huye who studies hard and attributes their high grades to their effort.
- **External locus of control:** Another student blames low grades on a teacher or test quality, feeling powerless.
- **Walter Mischel**

Mischel emphasised that behaviour depends on both the situation and personal perception. He identified five personal variables affecting behaviour:

1. **Competencies:** skills and knowledge, e.g., a nurse's ability to follow medical procedures.
2. **Encoding strategies:** How people interpret events, e.g., seeing a supervisor's feedback as helpful or critical.

- 3. Expectancies:** Anticipated outcomes of actions, e.g., believing extra effort will earn promotion.
- 4. Subjective values:** Importance of potential outcomes, e.g., valuing teamwork recognition over salary.
- 5. Self-regulation:** The ability to manage behaviour, e.g., a market vendor resisting gossip to focus on customers.

Criticism of social cognitive theory

- It does not fully explain how social cognition, behaviour, environment, and personality connect.
- It is not a unified theory; some aspects, like observational learning and self-efficacy, are not linked clearly.
- Not all social learning is observable, making measurement difficult.
- It does not fully account for development across the lifespan; children and adults learn differently from observation.

Comparing major personality theories

Personality theory	Key proponents	Assumptions	View of personality	Assessment methods
Psychoanalytic	Freud	Emotional problems come from unconscious conflicts and childhood fixation	Personality = id, ego, superego	Association, dream analysis
Psychodynamic	Adler, Horney, Jung	Childhood experiences and defence mechanisms shape personality	Interaction of conscious & unconscious motives	Projective tests, therapy
Humanistic	Rogers, Maslow	Focus on healthy people and self-realisation	Meeting basic needs allows growth toward self-actualisation	Questionnaires, therapy
Trait	Allport, Eysenck, McCrae, Costa	Stable traits influenced by genetics	Big Five traits shape personality	Personality inventories
Social Cognitive	Bandura	Personality develops through interaction with the environment	Cognition, behaviour, and environment interact to shape behaviour	Past behaviour in similar situations predicts future behaviour

Application activity 3.4



1. Explain in your own words how social cognitive theory describes the development of personality and the role of thoughts, behaviour, and environment.
2. How does Albert Bandura's concept of reciprocal determinism help us understand the interaction between a person and their environment in shaping personality?

Describe the following cognitive concepts: self-efficacy, internal locus of control, self-regulatory system, and cognitive processes, and give an example of each from everyday life in Rwanda, such as in schools, workplaces, or communities.

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3.5. End unit assessment



Question 1: Knowledge and Skills

Explain Freud's view of personality development and how people use defence mechanisms to cope with anxiety. Give examples from everyday life in Rwanda.

Question 2: Knowledge and understanding

How did humanistic Psychologists like Maslow and Rogers view personality, and what was their main goal in studying it?

Question 3: Knowledge, Skills, Attitudes, and Values

Describe Maslow's hierarchy of needs, including physiological needs, and explain how these needs influence behaviour in schools, workplaces, or communities.

Question 4: Knowledge and Skills

How does the social-cognitive perspective explain the mutual influence of personal traits, behaviour, and environment in shaping personality? Provide a real-life example.

Question 5: Knowledge, Skills, and Values

Discuss the contributions and limitations of social-cognitive theory in understanding personality, including concepts like self-efficacy, locus of control, and self-regulation.

UNIT 4

PERSONALITY TYPOLOGIES



Key Unit Competence:

Analyse personality typologies to understand individual behaviour.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Identify the main personality typologies and their key characteristics.
- Illustrate the key concepts associated with typology.
- Explain how different personality typologies influence individual differences.

Skills

2

- Compare and contrast theories of typologies.
- Evaluate different views embedded in typologies.
- Connect typologies to individuals' lifestyles and behaviour in real life.
- Apply personality typologies to interpret individuals' behaviours in real-life situations.

Attitudes and values

3

- Adopt the spirit of being aware of different views and thoughts behind typologies.
- Have ethical respect and openness when considering various personality typologies and when interpreting human behaviour in different contexts.
- Accept one's own typology.

Introduction

Have you ever noticed that some students in your class enjoy speaking and sharing ideas freely, while others prefer to listen quietly and think before responding? Or wondered why one person likes planning everything carefully while another prefers to act spontaneously? These differences show that people have different personalities, which influence how they think, feel, and behave. Psychologists study these differences through personality typologies, which classify individuals according to common patterns of behaviour, attitudes, and emotional responses.

In this unit, you will learn the concept of personality typologies and explore different ways psychologists classify personality. You will study the typologies proposed by Heymans and Wiersma, the trait approach focusing on extroversion and introversion, and the personality preferences described by the Myers–Briggs model. You will also examine the major personality characteristics such as openness, conscientiousness, extraversion, agreeableness, and neuroticism. By the end of the unit, you will be able to classify personality typologies and better understand individual differences in everyday life.

Introductory activity



Imagine you are introducing yourself to someone you have never met before. Write a short **letter to a stranger** describing who you are. Try to create a clear picture of yourself using words. Start with simple facts about your life, such as the school you attend or the activities you participate in, and then describe some of your personal characteristics. For example, you may mention whether you are **outgoing or shy, sensitive or thick-skinned, aggressive or timid, or emotional or calm**. You can also talk about groups or clubs you belong to, such as a school club or community activity.

After writing the letter, reflect on your description. Do you think the **traits you described will still represent you in ten years**? Explain your answer with clear reasons. Also, think about whether **people's thoughts and behaviours change over time**. Why do you think such changes happen?

4.1. Concepts of personality typologies

Learning activity 4.1



Look at the picture showing a job interview panel assessing a candidate. In many recruitment tests, candidates are asked to select adjectives that describe their personality to determine whether they are suitable for the job. Examples include confident, sociable, shy, imaginative, courageous, cooperative, optimistic, introvert, extrovert, responsible, and sensitive.

Question:

- a) Choose seven adjectives from the list that best describe your personality and briefly explain why.
- b) Choose seven adjectives that describe a person with weak or unsuitable personality traits for a job and describe such a person.

4.1.1. Understanding typology

Typology means classifying people or things based on similar characteristics. It is a system used in fields such as Psychology and social sciences to organise individuals into a limited number of categories.

In everyday life, people often talk about “types.” For example, someone might say, “I know that type of person” or “He is not my type.” This means they are describing a certain kind of person. Typology works in a similar way by grouping people with similar behaviours. For instance, drivers may be classified as cautious, confident, or careless based on their driving behaviour and attitudes on the road.

In Psychology, personality types refer to categories used to group people based on their behaviour patterns and Psychological characteristics. For example, some people are introverts (quiet and reflective), while others are extroverts (sociable and talkative). Personality types help Psychologists understand people better. They also help explain behaviour and sometimes predict how individuals may act in different situations, such as in school, at work, or in relationships.

4.1.2. Typological method

By observing many people, psychologists identified different ways of classifying personality:

- **Morphological typologies:** These classify people according to their physical body structure or appearance and relate it to behaviour. Example: Some theories suggest that body shape may be linked to certain behaviour patterns.
- **Physiological typologies:** These are based on how the nervous system and hormones (endocrine glands) function in the body. Example: Differences in brain activity or hormones may influence how calm, active, or emotional a person is.
- **Psychological typologies:** These focus on observing behaviour, emotions, and attitudes to classify people. Example: Classifying people as introverts or extroverts based on their social behaviour.
- **Mixed typologies:** These combine physical, biological, and psychological factors to explain personality.

What is a personality trait?

A personality trait is a stable and lasting characteristic that influences how a person thinks, feels, and behaves in different situations. Traits help explain the differences between people.

For example:

- **Extraversion:** People who are talkative, energetic, and sociable.
- **Introversion:** People who are quiet, reserved, and prefer being alone.

People do not fall completely into one category. Instead, they may fall somewhere between the two, forming a continuum. Personality traits help Psychologists describe, understand, and predict behaviour.

Three assumptions about personality traits

1. Traits are relatively stable over time. This means personality traits do not change easily. Example: A person who is shy in their youth may still be shy many years later.
2. Traits are consistent across situations. People tend to behave in similar ways in different places. Example: A person who is dominant at work may also behave dominantly at home or in social situations.
3. People differ in the strength and combination of traits. Everyone has traits, but they vary in degree. Example: Many people are sociable, but some are very sociable, while others are less sociable. Most people fall somewhere between these extremes.

Application activity 4.1



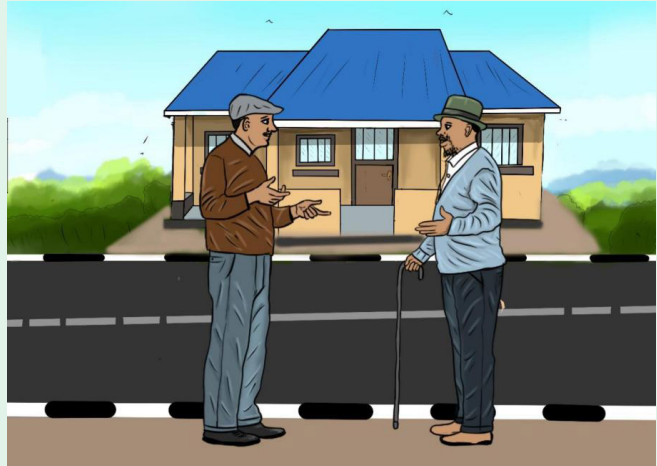
1. In a health centre, nurses and other staff members often have different ways of working. Some communicate easily with patients, while others prefer focusing quietly on technical tasks. Explain at least three reasons why understanding personality types can help workers cooperate better and improve service to patients.
2. In a local company, the manager learns that Psychologists classify personality in different ways. Using clear examples from the workplace or community, differentiate between morphological typologies and physiological typologies.
3. During a community meeting, some members actively share ideas and lead discussions, while others prefer to listen carefully before speaking. Using such a situation, differentiate between introversion and extroversion.
4. In a cooperative, market, or village community, people behave differently; some are very sociable and energetic, while others are calmer and more reserved. Using examples from community or workplace situations, explain why Psychologists say that individual differences result from differences in the strength, number, and combination of personality traits.

4.2. Typologies: Classification of personality by Heymans and Wiersma

Learning activity 4.2



One afternoon in a Rwandan neighbourhood, two neighbours, Mr Kanani and Mr Mutabazi, were discussing how people respond during conversations and disagreements. Mr. Kanani explained that his partner usually expresses her opinions immediately, especially when she disagrees with something, while Mr. Mutabazi said that his partner prefers to take time to think before responding.



Sometimes Kanani feels uncomfortable when someone reacts too quickly, while Mutabazi prefers quick responses so that problems can be solved without delay. As they talked, they realised that people have different personality traits: some express their feelings quickly and directly, while others think carefully before speaking. They concluded that understanding these differences helps people communicate better and respect each other's personalities in families, communities, and workplaces.

Discuss

1. Based on the story and the picture, what personality traits can you identify in the two women described by Mr. Kanani and Mr. Mutabazi? Write at least four traits for each person.
2. In your school or community in Rwanda, people respond to situations differently. Give two real-life examples of people who respond quickly and two examples of people who take time to think before responding.
3. Which type of personality do you prefer to work or interact with: a person who responds immediately or a person who takes time to think before responding? Explain your answer with one practical example from daily life.

The Dutch Psychologist Gerardus Heymans used different research methods to study personality differences among people. Instead of experiments, he mainly used biographical studies and questionnaires. In one study, he examined the biographies of 110 well-known historical figures (94 men and 16 women). He recorded information such as their name, occupation, nationality, and century, and analysed them using 88 personality characteristics.

Heymans focused on three main personality traits: emotionality, activity, and primary or secondary function (which is like today's idea of introversion and extraversion). Emotionality refers to how easily and strongly a person expresses emotions, while activity refers to the level of energy shown in meaningful actions, not just physical movement.

The primary function describes people who react mainly to present situations, while the secondary function refers to people who are more influenced by past experiences.

To explain his theory, Gerardus Heymans used a cube model based on three dimensions: activity, emotionality, and secondarity (long-term influence of experiences). Each dimension has two opposite poles, positive (+) and negative (-). By combining these dimensions, Heymans identified eight personality or temperament types, including neurotic, sentimental, sanguine, phlegmatic, choleric, passionate, amorphous, and apathetic. These types are represented in a model called Heymans' cube, which illustrates how different combinations of personality traits form different temperaments.

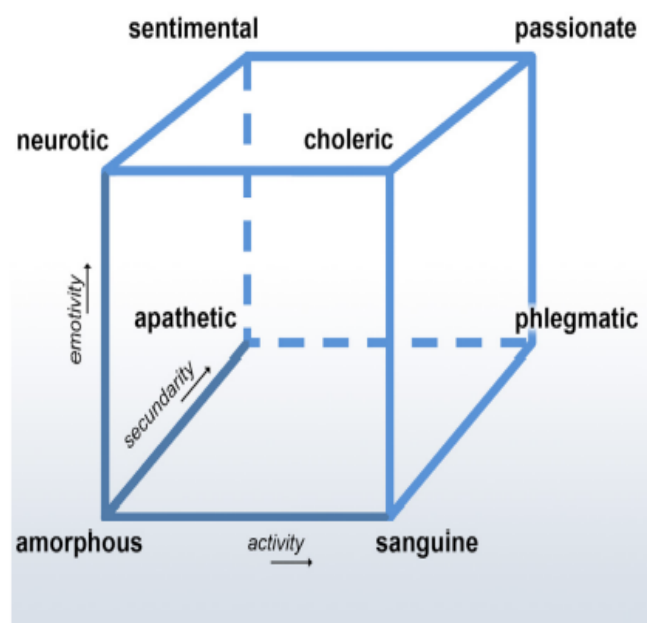
With every dimension having two poles: + and -; Heymans constructed the following eight types (temperaments) by combination of the three variables in binary form:

Table: Heymans' eight types

Type	Emotionality	Activity	Primary and Secondary Function
<i>Neurotic (E+A-S-)</i>	+	-	-
<i>Sentimental (E+ A- S+)</i>	+	-	+
<i>Sanguine (E- A+ S-)</i>	-	+	-
<i>Phlegmatic (E- A+ S+)</i>	-	+	+
<i>Choleric (E+ A+ S-)</i>	+	+	-
<i>Passionate (E+ A+ S+)</i>	+	+	+
<i>Amorphous (E-A- S+)</i>	-	-	-
<i>Apathetic (E- A- S+)</i>	-	-	+

Source: Adapted from Heiser (2008) and Vandenberg (1966)

The eight types were presented using a cube known as Heymans' cube:



Cube of Heymans

On the apexes the eight temperaments

Application activity 4.2



1. During your Psychology lesson, you learned about the work of the Dutch Psychologist Gerardus Heymans. In your own words, briefly describe the work and contributions of Heymans in the study of personality.
2. Heymans identified several temperament types based on combinations of personality traits. Write the formulae (E, A, S signs) for the following temperament types:
 - Neurotic
 - Sentimental
 - Choleric
 - Phlegmatic
 - Amorphous
3. In Heyman's personality theory, the cube model is used to explain different personality types. Explain the three main axes used in Heymans' Cube and what each axis represents in human personality.

4.3. Trait approach

Learning activity 4.3



Last month, Sangwa celebrated her 20th birthday. Her friends organised a surprise party and took her into a decorated room where a birthday cake was waiting on the table. When Sangwa saw the surprise, she became very emotional and cried with happiness. After the celebration, her friends discussed her reaction. Mugeni said that expressing emotions like that is normal, while Rugwiro felt that people should control their emotions in public.



Kagoyire, who was quiet and reserved, listened to the conversation without giving her opinion. This situation shows that people express their feelings and react to events differently depending on their personality traits, such as being expressive, talkative, or reserved.

1. Imagine you were one of Sangwa's friends at the birthday party. Based on the story, who among Sangwa, Mugeni, Rugwiro, and Kagoyire would you describe as introverted and who would you describe as extroverted? Explain why.
2. Think about your classmates, friends, or colleagues. Some people easily express their emotions like Sangwa, while others prefer to stay quiet like Kagoyire. What other personality traits can you identify in Sangwa, Mugeni, Rugwiro, and Kagoyire?

3. In your daily life at school, at home, or in the community, which type of person do you feel more comfortable interacting with: someone expressive and talkative or someone quiet and reserved? Give one example from your experience.

Trait theories propose that personality consists of a set of stable characteristics, or traits, that influence how individuals think, feel, and behave across different situations. Everyone has these traits, like being friendly or shy, but on different levels, which makes each person unique. These ideas were developed by Psychologists such as Gordon Allport, Raymond Cattell, and Hans Eysenck.

4.3.1. Allport's trait approach

Gordon W. Allport studied words used to describe people and grouped personality traits into three types. Cardinal traits are very strong traits that guide most of a person's behaviour, but they are rare (for example, strong kindness or selflessness). Central traits are the main qualities that describe a person, such as being kind, friendly, or honest, and everyone has them in different amounts. Secondary traits are less important and only appear in certain situations, like preferences or habits. For example, someone who is usually calm may feel nervous when speaking in public. Trait theory is helpful because it clearly explains personality and helps Psychologists measure it. However, this approach also has limitations: it may oversimplify individuals by placing them into fixed categories and does not fully account for the influence of culture and environmental factors on behaviour.

4.3.2. Cattell's personality traits

Raymond Cattell explained personality using 16 main traits. He first reduced a long list of personality descriptors originally compiled by Gordon Allport and then used a statistical method called factor analysis to identify the most important personality traits. Cattell said that every person has all 16 traits, but at different levels. For example, one person can be more outgoing while another is more reserved, or one can be more practical while another is more imaginative. So, personality is like a scale; people fall at different points on each trait.

Cattell also grouped traits into four types. Common traits are shared by many people (like honesty), while unique traits belong to one person only. Surface traits are easy to see in behaviour (like being talkative), while source traits are deeper traits that cause behaviour (like emotional strength). His theory helps explain personality clearly, but it can be complex because it includes many traits.

4.3.3. Eysenck's dimensions of personality

Hans Jürgen Eysenck said that personality is influenced by both biology (like genes) and the environment. He believed that every person has a unique temperament, which explains how they think, feel, and behave in daily life. People with different temperaments act differently, and Eysenck emphasised that genetics plays a big role in these differences.

His theory is explained using the PEN model, which has three main traits. Psychoticism shows how caring or aggressive a person is (high = less empathy, low = kinder and more caring). Extraversion–Introversion explains whether a person is outgoing and social (extravert) or quiet and reserved (introvert). Neuroticism–Emotional stability shows how a person handles emotions (high = anxious and easily stressed, low = calm and stable). These traits help describe and understand personality simply.

4.3.4. Typology of Carl Gustav Jung: Introvert and extrovert

Carl Jung explained that people have two main attitudes: introversion and extroversion. **Extroverts** are outgoing, talkative, and active, while introverts are quiet, reserved, and prefer to be alone. He said that people fall somewhere between these two, and this affects how they behave in different situations.

Introverts are usually shy, careful, thoughtful, and like to be alone. They think before speaking, focus easily, and may avoid attention or social situations. Extroverts, on the other hand, are social, active, and enjoy being with others. They speak quickly, make fast decisions, seek attention, and often act before thinking.

Application activity 4.3



1. How did Raymond Cattell describe the 16 primary traits of personality, and what are the opposite tendencies for traits such as reserved, humble, shy, trusting, practical, conservative, group-dependent, and relaxed?
2. How can you explain the personality theory developed by Hans Jürgen Eysenck in your own words?
3. What are some examples of common traits according to Cattell's classification of personality?
4. What are the three main dimensions in Eysenck's PEN model of personality, and what do each of them describe?

4.4. Characteristics of personality: the Big Five theory (openness, conscientiousness, extraversion, agreeableness, and Neuroticism)

Learning activity 4.4



Which of the following characteristics best describe you: worried, emotional, nervous, sociable, fun-loving, active, imaginative, artistic, creative, sympathetic, antipathic, reliable, hardworking, and punctual? After selecting them, how would you describe and summarise your personality in your own words?

The Big Five theory of personality explains that a person's behaviour can be understood using five main traits. These traits are Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism (often remembered as OCEAN). Each trait includes

many smaller qualities. For example, agreeableness includes being kind and friendly, while the opposite includes being rude or aggressive. Everyone has these five traits, but at different levels.

- Openness means being curious and willing to try new things (high = creative and imaginative, low = practical and prefers routine).
- **Conscientiousness** means being organized and responsible (high = hardworking and reliable, low = careless or lazy).
- **Extraversion** shows how social a person is (high = outgoing and energetic, low = quiet and reserved).
- Agreeableness is about how kind and cooperative a person is (high = friendly and helpful, low = rude or uncooperative).
- **Neuroticism** shows how a person handles emotions (high = anxious and worried, low = calm and confident. The following memory: Picturing an OCEAN/CANOE will help you.

Low score	Traits	High score
Practical, prefers routine, conforming	Openness	Imaginative, prefers variety, independent
Disorganised, careless, impulsive	Conscientiousness	Organised, careful, disciplined
Retiring, sober, reserved	Extraversion	Sociable, fun-loving, affectionate
Ruthless, suspicious, uncooperative	Agreeableness	Soft-hearted, trusting, helpful
Calm, secure, self-satisfied	Neuroticism (Emotional stability vs instability)	Anxious, insecure, self-pitying

Source: Information from McCrae & Costa (2008)

The Big Five personality factors have some strengths and weaknesses. One strength is that they can fairly predict how a person may behave over time. They are also widely used by researchers and have contributed greatly to the study of areas such as learning, gender differences, culture, personality disorders, and career success.

However, there are also weaknesses. The traits are not separate; for example, a friendly person may also be outgoing. The model also overlooks certain important traits, such as a sense of humour and honesty.

Additionally, its results may not always be accurate, as they often rely on self-reported data, where individuals may provide biased or untruthful responses.

Application activity 4.4



1. What does the abbreviation OCEAN stand for, and how does each factor help describe a person's personality?
2. In your opinion, what are one strength and one weakness of the Big Five personality model, and why do you think so?
3. How would you describe someone with high agreeableness? Can you give examples of how they might act toward friends, family, or classmates?

4.5. Personality preferences of Myers-Briggs

Learning activity 4.5



Read each question carefully and choose the answer that best describes how you usually feel or act. After answering all the questions, reflect on your choices and write a short paragraph describing your personality.

1. When doing tasks or going out, do you prefer to plan carefully or be spontaneous? Give an example.
2. Are you more social and outgoing, or quiet and reserved? How does this show in your daily life?
3. Do you usually follow your heart or your head when making decisions? Explain why.
4. Do you prefer having many friends or a few close friends, and do you admire people who follow the rules or those who are more original?

The Myers-Briggs Type Indicator (MBTI) is a questionnaire that helps people understand their personality, strengths, and preferences. It was created by Isabel Myers and her mother, Katharine Mary Briggs, based on Carl Jung's theory of personality types. The goal of the Myers-Briggs Type Indicator (MBTI) is to help individuals understand themselves better, including identifying careers and activities that may best suit their personality. There are no "better" or "worse" personality types; each type has value.

The MBTI measures personality using four main pairs of traits:

1. **Extraversion (E) – Introversion (I):** Extraverts are outgoing, enjoy social interaction, and get energy from being with others. Introverts are quiet, reserved, and recharge by spending time alone.
2. **Sensing (S) – Intuition (N):** Sensing people focus on facts, details, and reality. Intuitive people focus on ideas, patterns, and possibilities.
3. **Thinking (T) – Feeling (F):** Thinking types make decisions based on logic and facts, while Feeling types consider people's emotions and values.

- 4. Judging (J) – Perceiving (P):** Judging types prefer order, plans, and structure. Perceiving types are flexible, open to change, and like to keep options open.

By combining these four pairs, the MBTI identifies 16 personality types, each with its own characteristics. For example:

- **ISTJ (The Inspector):** Reserved, practical, loyal, and organised.
- **ENFP (The Champion):** Energetic, creative, and outgoing.
- **INTJ (The Architect):** Logical, analytical, and creative.
- **ESFP (The Performer):** Fun-loving, social, and spontaneous.

The MBTI is useful because it helps people understand themselves and others, which is helpful in school, work, and group projects. It helps assign tasks based on strengths and encourages better teamwork.

Difference from other personality tools: MBTI is not a “test” with right or wrong answers. It focuses on understanding personality, not diagnosing mental health. Other tools, like the Big Five or Cattell’s 16 Personality Factors, measure traits differently, often with more detail.

Strengths of MBTI:

- Simple and easy to understand.
- Describes personality in a positive, neutral way.
- Helps people see their strengths and work better with others.

Weaknesses of MBTI:

- Some people think they are experts after a little study.
- People may ignore real problems by focusing only on the positive descriptions.
- Its simplicity can make people think it’s less accurate or too basic.

Application activity 4.5



1. In your own words, what is the Myers-Briggs Personality Type Indicator (MBTI), and how can it help you understand yourself?
2. Which psychologist’s ideas influenced Myers and Briggs, and how do you think these ideas shaped their work?
3. The MBTI looks at four main traits that make up a personality. Can you name these traits and explain what each one means to you personally?
4. Choose two MBTI types, ENFP and ESFP, and describe what kind of person you think each type represents. Can you think of someone you know who fits these types?

4.6. Accept One's Typology

Learning activity 4.6



Each family member has their own way of communicating, spending time with others, making decisions, and responding to different situations. What differences in the way people think, feel, and behave can you identify, and how can you appreciate these differences?

4.6.1. Meaning

Accepting one's typology means recognising, understanding, and appreciating one's own personality characteristics while acknowledging that people have different personality patterns. People differ in the way they think, feel, communicate, make decisions, relate to others, and respond to situations. These differences contribute to individual uniqueness.

Accepting one's typology does not mean believing that every behaviour is good or that a person cannot change. It means being able to recognise one's characteristics without feeling inferior or ashamed, while being willing to improve behaviours that need development. Example, a group member who is naturally quiet can accept this characteristic while also developing confidence and communication skills when participating in group activities.

4.6.2. Importance of accepting one's typology

Accepting one's typology helps a person understand oneself better, recognise personal strengths and areas for improvement, develop self-confidence, set realistic personal goals, avoid unnecessary comparison with others, improve relationships with other people, appreciate individual differences, and make better use of personal strengths. For example, a student who recognises that he/she is good at organising activities can use this strength in group work while also learning to listen to the ideas of others.

4.6.3. Accepting and Respecting Different Personality Typologies

Every person has characteristics that can support their development and others that they may need to improve. Accepting one's typology involves recognising both. Accepting one's typology means recognising personal strengths and areas for improvement while understanding that people have different personality characteristics. A person should appreciate their strengths, work on behaviours that need improvement, and avoid using personality as an excuse for inappropriate behaviour. At the same time, people should respect, tolerate, and appreciate individual differences, since different personality characteristics can complement one another and support effective cooperation.

Application activity 4.6



Complete the following table by filling in the possible strength and area for development for each personality characteristic. The first row is given as an example

Personality characteristic	Possible strength	Area for development
<i>Sociable</i>	<i>Communicate easily with others</i>	<i>Avoiding dominating conversations</i>
Quiet		
Assertive		
Patient		
Independent		

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4.7. End unit assessment



Question 1: Knowledge and understanding

How do Psychologists use personality traits to describe how people think, feel, and behave?

Question 2: Knowledge, understanding and Skills

What are the main differences between extroverts and introverts, and how do these differences show in everyday life?

Question 3: Skills, Attitudes and Values

Think about your own personality traits. How would you classify them according to Gordon Allport's three levels: cardinal, central, and secondary traits?

Question 4: Attitude and Values

Which of your traits do you think gives the most useful information about who you are as a person?

Question 5: Skills and Attitudes

What are some criticisms of the Big Five personality traits, and do you think they apply to you or people you know?

Question 6: Skills, Attitude & Values

Where would you place yourself on the Big Five traits: conscientiousness, agreeableness, neuroticism (emotional stability), openness, and extraversion? How do you think your friends would describe you on these traits?

Question 7: Knowledge and Skills

Based on Eysenck's PEN model, which category would you place a student like Eddy, who is aggressive, rude, lacks empathy, struggles to make friends, and shows antisocial behaviour? Why do you think this category fits him?

UNIT 5

HISTORY OF LEARNING PSYCHOLOGY



Key Unit Competence:

Examine the history of learning Psychology.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Describe early thoughts on learning.
- Enumerate psychologists' and learning theorists' contributions to learning.
- Explain the major schools of thought that shaped the study of learning.
- Identify the key milestones in the historical development of learning Psychology.
- Describe how modern neuroscience and technology are integrated to enhance understanding of learning processes.

Skills

2

- Refer to Psychologists and learning theorists to explain how people acquire or change their behaviours.
- Analyse the contribution of early thoughts to learning psychology.
- Apply historical insights from learning Psychology to explain current approaches to teaching and learning.
- Assess the contributions of early approaches, Psychologists, and learning theorists to the understanding of learning.

Attitudes and values

3

- Appreciate the contribution of early and modern Psychologists in shaping our understanding of learning and human behaviour.
- Exhibit open-mindedness and critical reflection when examining the contributions of key Psychologists and theorists to learning.

- Adopt a spirit of curiosity and respect for diverse perspectives in the study of learning Psychology, recognising their relevance in education and everyday life.

Introduction to the unit

Have you ever wondered how you learned to speak your first words, solve a problem in class, or even control your emotions when you are upset? Or noticed how some students quickly understand new ideas while others take more time but still succeed? These everyday experiences show that learning is not just about memorising; it is a process that has been studied, explained, and improved over time by different thinkers and Psychologists.

This unit introduces the History of Learning Psychology, helping you understand how ideas about learning have developed from early philosophers such as Socrates and Plato to modern psychologists such as Edward Thorndike, John B. Watson, Hermann Ebbinghaus, B. F. Skinner, and Jean Piaget. You will explore how learning involves emotional and behavioural self-regulation, cognitive skills like thinking and problem-solving, and qualities such as curiosity and creativity.

By studying this unit, you will be able to examine how different theories explain the way people learn, grow, and adapt in school and daily life. You will also develop skills to better understand your own learning process, improve your study habits, and become a more active, curious, and independent learner.

Introductory activity



Think about the first time you learned to ride a bicycle, solve a tricky puzzle, or even manage your feelings when frustrated. Did you know that Psychologists have been trying to understand how people learn these skills for hundreds of years?

5.1. Understanding learning

Learning activity 5.1



Have you ever wondered why some things are easy to remember while others are really hard? Or why you quickly learn a new game or skill, but forget it if you don't practice?

5.1.1. Definition of learning.

Different Psychologists and theorists define learning in various ways. The following are some of the meanings they provide:

- **Behavioural theorists:** Behaviourists define learning as a relatively permanent change in behaviour resulting from experience. They believe that learning has taken place only when a behaviour change can be observed. They also explain that learning is a lifelong process through which individuals acquire knowledge, skills, positive behaviours, and cultural values from one generation to another.

- **Cognitivists:** Cognitivists define learning as a semi-permanent change in mental processes or associations. They focus on internal processes such as thinking, problem solving, language, concept formation, and information processing. For them, learning occurs when a person understands the concept.
- **Constructivists:** Constructivists view learning as a process in which the learner actively builds new ideas or concepts based on prior knowledge and experiences. Learners use their thinking processes to construct understanding. According to this view, the learner plays a central role in learning. Knowledge is created from experience, and outcomes may differ from one learner to another.
- **Social constructivists:** Social constructivists emphasise that learning occurs through social interaction. Learners construct knowledge together by sharing ideas and experiences. They believe that children are naturally curious and actively involved in their own learning. Much important learning happens through interaction with more knowledgeable people, such as teachers or parents, who provide guidance and support. As learners gain competence, they gradually become more independent.
- **Social learning theorists:** Social learning theorists argue that people learn by observing others and imitating their behaviour. They emphasise that learning can occur by watching models such as parents, teachers, and peers. They also note that individuals may learn by seeing others being rewarded or punished, without directly experiencing it themselves.

In summary, learning is a lifelong process of acquiring knowledge, skills, positive behaviours, and cultural values through experience and interaction.

5.1.2. Types of Learning

- **Rote learning:** It is the process of memorising information without understanding its meaning, so that it can be recalled exactly as it was learned.
- **Meaningful learning:** This is learning in which the learner fully understands the information and can connect it with other knowledge.
- **Imitation learning (observational, modelling, or social learning):** This is a type of learning in which people acquire behaviours by observing and imitating others. For children, common models include parents, caregivers, teachers, and peers. For example, students may imitate how their teachers motivate them.
- **E-learning (Electronic learning):** E-learning refers to the use of information and communication technologies to access and support teaching and learning resources online.

Application activity 5.1



In your class and daily life, you learn in different ways. Think about your own learning experiences and answer the following:

- Describe how you have learned in the following situations:
 - When you change your behaviour after experience (for example, after being corrected or rewarded).

- When you understand something by thinking deeply or solving a problem.
- When you build new knowledge based on what you already know.
- When you learned through interaction with teachers, parents, or classmates.
- When you learned by observing and imitating others.

b) For each situation above, identify the group of theorists that best explains your learning (Behavioural theorists, Cognitivists, Constructivists, Social constructivists, and social learning theorists).

5.2. Philosophical roots of learning

Learning activity 5.2



1. What do you know about Socrates and Plato?
2. How do you think people learned skills and behaviours before formal schools existed?

Before psychology became a scientific discipline, philosophers discussed how people acquire knowledge and how learning takes place. Their ideas, developed over many centuries, provided important foundations for the later scientific study of learning.

Idea of Plato (Allegory of the Cave)

Plato used the Allegory of the Cave, presented in *The Republic* around 380 BCE, to explain how people move from limited understanding to deeper knowledge. He suggested that learning involves moving beyond appearances and developing a better understanding of reality through education, reflection, and reasoning.

Socrates with Socratic Dialogue

Socrates, who lived around 470–399 BCE, believed that people could develop knowledge through questioning and dialogue. His Socratic method encouraged learners to examine their ideas, identify errors, and develop understanding through critical thinking and guided questioning.

Aristotle (Empiricism)

Around 350 BCE, Aristotle emphasised the importance of experience, observation, and the senses in acquiring knowledge. He believed that people develop knowledge by interacting with and observing the world around them. His ideas contributed to the development of empiricism, which later became important in the study of learning.

René Descartes (Dualism and role of reason in learning)

In 1641, René Descartes developed ideas associated with dualism, distinguishing between the mind and the body. He also emphasised the importance of reason and rational thinking in acquiring knowledge, contributing to later discussions about the role of mental processes in learning. According to René Descartes, reason, rational thinking, and reflection are important because knowledge should not depend only on what we see or experience through the senses. The mind can use reasoning to examine information, question assumptions, and arrive at clearer and more certain conclusions.

John Locke (Tabula Rasa)

In 1690, John Locke presented his ideas about Tabula Rasa, meaning a “blank slate.” He argued that the mind begins without fully developed ideas and that knowledge develops through experience and interaction with the environment. This idea emphasised the important role of experience in learning.

Associationism (David Hume)

In 1739–1740, David Hume developed ideas associated with associationism, explaining how the mind connects ideas through experience. He identified principles such as similarity, contiguity, and cause and effect, helping explain how one experience or idea can become connected with another.

Emergence of Psychology as a Scientific Discipline (Wilhelm Wundt)

In 1879, Wilhelm Wundt established a psychological laboratory in Leipzig, Germany, an event widely regarded as an important milestone in the emergence of psychology as an independent scientific discipline. Wundt used introspection, in which trained participants reported their conscious experiences under controlled conditions, to study mental processes systematically. His work helped establish psychology as a scientific field and created a foundation for the later scientific study of learning.

Application activity 5.1



1. Explain how the ideas of Socrates, Plato, and Aristotle contributed to understanding how people acquire knowledge.
2. Compare the views of René Descartes and John Locke on how knowledge is acquired.
3. Explain how the philosophical ideas of David Hume and Wilhelm Wundt contributed to the development of the scientific study of learning.

5.3. Early approaches to learning: Emotional and behavioural Self-Regulation

Learning activity 5.3



Think about a time when you were trying to learn something difficult in class, but you felt distracted, frustrated, or too excited.

- a) A) How did your emotions or behaviour affect your ability to learn?
- b) B) What did you do (or what could you do) to control your emotions and behaviour so that you can learn better?

The early approaches to learning focus on how children learn. They refer to the skills and behaviours that children use to actively engage in learning. For example, how a child controls emotions, pays attention, or behaves in class affects how well they learn.

5.3.1. Emotional self-regulation

Self-regulation is the ability to control your thoughts, emotions, and behaviours to make good decisions for your future. It also means calming yourself when you are upset and encouraging yourself when you feel sad. Even adults sometimes fail to control themselves. For example, a person may know that eating too much sugar is unhealthy, but still does it. This shows that self-regulation is a skill that develops over time.

a) Meaning of emotional self-regulation

Emotional self-regulation is the ability to understand and manage your emotions positively and acceptably. It involves controlling strong feelings and choosing how to respond to them. For example, instead of shouting when angry, a student may choose to stay calm or talk to the teacher.

In simple terms, emotional self-regulation means managing your emotions, reactions, and impulses so that you can behave appropriately in different situations.

b) Importance of emotional self-regulation

Emotional self-regulation is important for learning and personal growth in many ways:

- It helps people understand themselves better and manage stress.
- It supports creativity and allows people to express their ideas.
- It helps people create new knowledge and think independently.
- It brings joy and meaning to learning.
- It encourages people to develop new and innovative ideas.
- It gives people freedom to explore and learn from their environment.
- It improves problem-solving and critical thinking skills.
- It motivates people to learn, especially when they feel confident and interested.

For example, a person who can control frustration during a difficult task is more likely to continue trying and succeed.

c) Strategies that can help to manage emotions in a healthy and helpful way

People can use the following strategies to regulate their emotions:

- **Goalsetting:** Setting clear and achievable targets.
- **Self-monitoring:** Checking one's own behaviour and progress.
- **Effective use of self:** Using personal strengths wisely.
- **Self-instruction (self-talk):** Talking to oneself positively (e.g., "I can do this").
- **Self-reinforcement:** Rewarding oneself after success.
- Thinking about the story you tell yourself (avoiding negative thoughts).
- Choosing how to respond instead of reacting immediately.
- Focusing on positive emotions.

d) Healthy activities that help you regulate your emotions

Some activities help people manage their emotions better, such as:

- Talking to a friend.
- Exercising.
- Writing in a journal.
- Practising meditation or relaxation.
- Seeking help from a counsellor or therapist.
- Getting enough sleep.
- Taking care of your health.
- Identifying and managing negative thoughts.

For example, a student who feels stressed before exams can talk to a friend or take a short walk to relax.

e) How to teach emotional regulation

Teachers and parents can help children regulate their emotions by:

- Encouraging them to stay calm and think before acting.
- Teaching them to respond instead of reacting impulsively.
- Being patient and supportive.
- Giving positive feedback.

With time and guidance, children learn to manage their emotions independently.

5.3.2. Behavioural self-regulation

Behavioural self-regulation is the ability to control your actions and avoid harmful behaviours when dealing with emotions. Some people try to manage emotions through unhealthy behaviours like overeating or substance use, but these do not solve the

problem. Behavioural self-regulation means choosing actions that support your long-term goals. For example, waking up early to study or exercise, even when you feel tired, shows good self-regulation

a) Why is it important to have behavioural self-regulation skills?

Behavioural self-regulation is important because it helps individuals develop healthy ways of managing emotions. Without it, people may depend on unhealthy habits and even develop addictions. For children, behavioural self-regulation helps them:

- Stay calm.
- Adapt to different situations.
- Perform well at school.
- Build good relationships.
- Feel confident about themselves.
- For example, a student who can control distractions can concentrate better and perform well academically.

b) How can teachers support the development of behavioural self-regulation in children?

Teachers can support children by:

- Managing their own stress and acting as role models.
- Focusing on teaching skills, not just correcting behaviour.
- Setting realistic expectations based on children's abilities.
- Staying calm and showing empathy when children are upset.
- Encouraging and supporting children positively.
- Ensuring children have enough sleep, food, and exercise.
- Reducing unnecessary pressure and workload.
- Providing clear routines and structure.
- Making learning about self-regulation fun and engaging.
- Teaching children to express and understand their feelings.
- Helping children become aware of their emotions and behaviours.
- Giving opportunities to practice self-regulation skills.
- Providing immediate and constructive feedback.
- Using praise and rewards to encourage positive behaviour.

For example, a teacher can help a child calm down by speaking gently, guiding them to take deep breaths, and praising them when they manage their emotions well.

Application activity 3.2



Think about your own experience as a learner in school.

1. Describe how your ability to control your emotions (such as anger, excitement, or stress) has affected your learning.
2. Explain how your behaviour (such as paying attention, avoiding distractions, or completing tasks) has influenced your academic performance.
3. Based on your experience, assess the importance of emotional self-regulation and behavioural self-regulation in helping you learn effectively.

5.4. Early approaches to learning 'Cognitive Self-Regulation (Executive Functioning)

Learning activity 5.4



Imagine you oversee organising a 10-minute class presentation about your favourite topic. Write down three steps you will take to prepare for the presentation. And think about what you need, how long each step will take, and the order of tasks.

Reflection questions:

- What worked well in your planning?
- Did you face any distractions? How did you handle them?
- If you did it again, what would you do differently?

5.4.1. Meaning of Cognitive Self-Regulation

Cognitive self-regulation is the ability to control your thinking and attention, stay focused, and resist distractions. It is also called executive functioning. These skills help you:

- Maintain attention.
- Control impulses.
- Think flexibly.
- Use working memory (holding information in your mind to complete tasks).

These skills begin to appear in infants and toddlers and grow stronger during the preschool years. Cognitive self-regulation involves planning, adjusting, and using your thinking skills to reach personal goals in different situations.

a) Examples of cognitive self-regulation:

- Paying attention and staying focused on a task.
- Avoiding distractions.
- Staying engaged in an activity until it is finished.

- Thinking carefully before acting or making decisions.

b) Activities that help develop cognitive self-regulation:

- Reading books or solving challenging puzzles.
- Learning a new language or musical instrument.
- Trying a new hobby.
- Teaching or tutoring others.
- Listening to podcasts or watching educational videos.
- Paying attention to your thoughts about yourself and others.

5.4.2. Importance of Cognitive Self-Regulation

Cognitive self-regulation is important because it helps:

- Manage reactions to negative emotions like anger, frustration, or embarrassment.
- Calm down when excited or upset.
- Focus on tasks and filter out distractions.
- Control impulses.
- Think carefully and plan before acting.
- Stay engaged in activities.
- Improve memory and recall information.
- Work well in school.
- Set and pursue learning goals that are achievable and meaningful.
- Interact well with others.
- Achieve personal goals and expectations.

Example: A worker in the office who can focus on an assigned task without being distracted by a phone is more likely to complete it effectively and remember what they have done.

How to Improve Cognitive Self-Regulation

Teachers and parents can help children develop these skills through:

- Active learning activities: Moving, dancing, writing, doing physical exercises, art, music, singing, storytelling, and games.
- Creating a focused learning environment: Keep the classroom inviting, comfortable, and free from unnecessary distractions (proper lighting, temperature, noise levels, and minimal clutter).
- Family support: Encourage children to explore, research, and practice skills at home; provide resources and opportunities for play and learning.
- Students who develop emotional, cognitive, and behavioural self-regulation:
 - Think critically before reacting.
 - Control impulses effectively.

- Plan their learning carefully.
- Achieve better learning outcomes.

Application activity 5.4



Think about a time when you were learning something challenging, like solving a difficult math problem, reading a new story, or practising a skill.

- a) How did being able to focus, plan your steps, and control distractions help you learn better?
- b) How do you think these skills, like paying attention, thinking before acting, and staying engaged, can help other children succeed in school?

5.5. Initiative, Curiosity, and Creativity

Learning activity 5.5



Imagine you are given a box of random materials (paper, string, bottle caps, markers) and asked to do something useful or fun in 10 minutes.

1. Which of these materials do you think could be used in a new or unexpected way, and why did that idea come to your mind?
2. How did you decide which idea to start with, and what made you act instead of waiting for someone to tell you what to do?
3. What unique feature did you add to your creation to make it different or more interesting than what others might make?

5.5.1. Initiative

a) Meaning of initiative

Initiative is the ability to act and get things done without waiting to be told. It is the first step in a series of actions and shows a person's willingness to take responsibility. People who show initiative think for themselves, act when needed, and use their ideas to achieve goals. Example: A student notices the classroom is messy and decides to organise it before being asked. In school, student initiatives can include projects in culture, education, sports, environment, health, or civic engagement. In the workplace, initiative is seen when someone acts on an idea and solves problems without waiting for instructions.

b) Ways teachers can promote initiative in students

- Be a role model by showing initiative yourself.
- Link lessons to clear goals, encouraging students to act.
- Use group activities and let students work independently.
- Encourage sports, arts, music, and creative activities.

- Help students set goals and take responsibility for learning.
- Praise effort more than results and motivate them to try new challenges.
- Include homework, projects, and presentations to stimulate initiative.
- Let students think critically, join discussions, clubs, or competitions.

c) Importance of initiative in learning

- Students act without being told, solve unseen problems, and seize learning opportunities.
- Builds resourcefulness, resilience, and determination.
- Encourages independent thinking and drives success in school and life.

5.5.2. Curiosity

a) Meaning of curiosity

Curiosity is a natural desire to learn and explore the world. Curious students ask questions, investigate, and have a strong desire to discover new things. Culture and environment influence how curiosity develops.

b) Importance of curiosity in learning

- Curiosity leads to knowledge and helps make connections between ideas.
- Prepares the brain for enjoyable and effective learning.
- Drives students to explore the unknown, take risks, and find solutions.
- Prevents boredom by encouraging active learning and play.
- Enhances problem-solving, decision-making, and creativity.

Example: A student wonders why plants grow differently and decides to experiment with sunlight and water to see the effect.

c) Tips to promote curiosity in learning

- Use activities that involve predicting, guessing, solving problems, or exploring big questions.
- Encourage questions and provide resources for exploration.
- Allow trial-and-error learning and open-ended play.
- Let students explore materials and ideas that interest them.
- Reward curiosity and model it yourself by showing interest in new topics.

5.5.3. Creativity

a) Meaning of creativity

Creativity is the ability to produce new ideas, solutions, or products by combining knowledge, skills, and experiences. It involves both thinking and doing, often through problem-solving.

b) Importance of creativity in learning

- Allows students to express thoughts and feelings.
- Helps explore their inner world, cope with stress, and gain self-awareness.
- Encourages new ideas and innovative solutions.
- Promotes critical thinking, problem-solving, and making connections between subjects.
- Increases motivation by letting students tie learning to personal interests.

Example: A student creates a colourful poster combining art and science to explain climate change.

c) Common characteristics of a creative person

- **Flexible:** Willing to try new ideas and change plans if needed.
- **Playful:** Enjoys experimenting without fear of mistakes.
- **Open-minded:** Listens to and explores others' ideas.
- **Independent:** Makes decisions and works without constant guidance.
- **Intuitive:** Follows feelings and instincts.
- **Risk-taker:** Willing to try new approaches.
- **Curious:** Loves learning and exploring new things.

d) Ways to encourage creativity in the classroom

- Create flexible and collaborative classroom layouts.
- Set up a classroom library and involve students in organising it.
- Encourage students to visualise ideas and goals.
- Use reflective activities and outdoor projects like gardens.
- Introduce multimedia and hands-on learning materials.
- Explore different cultures and perspectives.
- Incorporate humour and colour to make learning fun.
- Use team-building exercises and reward creative efforts.

Initiative, curiosity, and creativity are essential for learning. Students who are curious and creative perform better academically, think critically, solve problems, communicate effectively, and develop skills for lifelong learning. Teachers can foster these skills by creating supportive, stimulating, and engaging learning environments, helping students become responsible, curious, and creative citizens.

Application activity 5.5



Explain how your own initiative, curiosity, and creativity help you learn better, solve problems, and achieve your goals in school and everyday life.

5.6. Psychologists and learning theories: Socrates and Plato

Learning activity 5.6



Imagine you are sitting in a classroom over 2,000 years ago, listening to a teacher asking questions that make you think deeply about the world around you. How would you learn if you were encouraged to question everything instead of just memorising facts?

5.6.1. Socrates (469-399 BC)

a) Contributions of Socrates to learning

Socrates is considered one of the founders of Western philosophy. He created the Socratic method, a way of learning through persistent questioning. He believed that “the unexamined life is not worth living,” meaning that thinking deeply about life is essential.

Some key questions Socrates asked about learning:

- What is knowledge? He separated knowledge into trivial (useless) and important (helpful for living well and making good choices).
- Why learn? He believed goodness, truth, and morals are in everyone, but learning brings them out.
- How do we learn? Learning comes from questioning, observing, and reflecting on the wisdom of others, as well as recognising one’s own mistakes.
- Who do we learn from? No one has all the answers; we learn by sharing experiences and knowledge with others.
- Where do we learn? Learning isn’t just in school; it can happen anywhere people meet and share ideas.
- When do we learn? Learning happens whenever people have meaningful conversations and are open to understanding themselves.

b) Socratic teaching approach

Socrates encouraged teachers to ask leading questions that guide students to discover answers themselves. This interaction between teacher and student is central to learning.

c) Theories of Socrates

- **Theory of Learning:** Learning is the pursuit of truth by questioning, analysing, and observing others. People learn collaboratively.
- **Theory of Transmission:** Knowledge, skills, and attitudes are shared through open dialogue between teachers and students.
- **Theory of Society:** Society teaches knowledge, skills, and attitudes through its various roles, like builders, farmers, and clothiers.

- **Theory of Opportunity:** Practical knowledge of experts may not help others unless shared. Socrates challenged those who thought they knew everything to improve intellectually and morally.

5.6.2. Plato (427-327 BC)

a) Plato's Allegory of the Cave

Plato, a student of Socrates, wrote about morality and truth. He believed our senses show only shadows of reality, while true knowledge is inside us. He explained this with the Allegory of the Cave:

1. Imagine being shackled in a cave, seeing only shadows on the wall. That's all you know.
2. If released, you start seeing things as they really are and question what is real.
3. Eventually, you leave the cave and understand the full reality.
4. Returning to the cave to teach others, you might be rejected, but perseverance reveals the truth.

Lesson from the allegory: The cave represents our limited view of reality. Plato teaches that questioning assumptions, thinking critically, and discovering your own solutions are essential skills. Teachers should encourage abstract thinking and discussion about the real world.

b) Other contributions of Plato

- A just society gives the best education to all according to ability.
- He influenced fields like science, mathematics, ethics, music, drama, poetry, and government.
- Education for all: Both boys and girls should receive education.
- State education: Education should be organised by the state, not just parents.

Curriculum and stages:

- **Elementary (7-18 years):** Co-education; study math, poetry, music, literature.
- **Military training:** Physical education for two years; select the best for higher education.
- **Higher education (20-35 years):** Study math, literature, philosophy; later study dialectics for leadership roles.

Women's education: Plato emphasised equal education for women. Strong women should learn the same skills as men to contribute to society. In his ideal republic, both men and women could be rulers if they were knowledgeable and capable.

Teaching methods:

- Play-based learning at the elementary level. Children learn by doing and exploring.
- Higher education focuses on reasoning, abstract thinking, and motivation.
- Plato opposed force in education and encouraged joyful learning through engaging, hands-on activities.

Application activity 5.6



Reflecting on what you've learned, summarise how Socrates and Plato each contributed to education and teaching.

5.7. Psychologists and learning theories: Thorndike, Watson, and Ebbinghaus

Learning activity 5.7



In a classroom where some students quickly grasp new concepts while others require more time and repeated practice, you observe distinct learning needs among your learners. Alice, for example, performs best when she receives immediate feedback after answering questions, whether her responses are correct or incorrect. In contrast, Brian finds it difficult to retain information unless he reviews it several times over the course of a week. Meanwhile, Carlos shows strong motivation and improved performance when he is given rewards or praise after completing a task.

Questions:

1. Why do you think Alice, Brian, and Carlos learn in different ways even though they are in the same classroom? What does this tell us about how learning happens?
2. If you were the teacher, how would you change your teaching strategies to help each student learn better, considering their unique learning styles?
3. How do you think Thorndike's, Watson's, or Ebbinghaus' ideas about learning could explain the differences in how Alice, Brian, and Carlos learn?

5.7.1. Edward Lee Thorndike

a) Trial and Error Learning

Edward Lee Thorndike (1874–1949) developed the trial-and-error theory of learning. He studied animals such as cats, rats, and chickens to understand how learning happens. According to Thorndike, learning occurs by trying different actions until the right response works. When a person or animal makes a mistake, they learn from it and improve with practice.

b) Cat Experiment

Thorndike placed a hungry cat in a puzzle box with a door that could be opened by pulling a loop. The cat first tried random actions like biting the bars or hitting the box with its head. After several attempts, it finally pulled the loop and got the food. When the experiment was repeated, the cat learned faster each time. For example, the first success took 160 seconds, but later attempts only took a few seconds.

c) Principles of Thorndike's Work

- Teachers should reward successful attempts to encourage learning.
- Actions that are repeated and rewarded become stronger.
- Intelligence grows by making connections and practising them.
- Learning can be transferred to new situations because of prior experiences.

5.7.2. John Watson

a) Biography

John Broadus Watson (1878–1958) was an American psychologist and the founder of behaviourism. He believed psychology should focus only on behaviours that can be observed. Watson studied how the environment shapes behaviour, not inherited traits.

b) Little Albert Experiment

Watson taught an infant, Little Albert, to fear a white rat by pairing it with a loud noise. After repeated trials, Albert cried at the sight of the rat even without the noise and feared similar furry objects. This showed that behaviours, including fear, can be learned from the environment. While famous, the experiment is considered unethical today.

c) Legacy in Education

- Watson's behaviourism focused on measurable behaviour and environmental influences.
- Teachers can help students learn desired behaviours by providing positive reinforcement. For example, praising a student for finishing homework encourages them to do it again.
- Three key assumptions:
 1. Only observable behaviours are important.
 2. Environment shapes behaviour.
 3. Learning happens through association (contiguity) and reinforcement.

5.7.3. Herman Ebbinghaus

a) Biography and Forgetting Curve

Herman Ebbinghaus (1850–1909) studied memory and discovered the forgetting curve, which shows how quickly people forget new information over time if it is not reviewed.

Examples of Forgetting Curve Theory

- Strength of memory: People remember important or vivid experiences better than dull information.
- Spaced practice: Reviewing material several times with breaks (e.g., once a day) improves retention.

- Cramming: Studying all at once may help short-term memory, but leads to fast forgetting.

b) Strategies to Improve Memory

- Ask questions: Encourage students to understand and question the material.
- Use rhymes and songs: Creating songs or rhymes can help remember facts.
- Active learning: Discuss topics regularly to retain information longer.
- Visual materials: Pictures, videos, and flashcards make learning more engaging.
- Real-life examples: Connecting lessons to personal experiences helps memory.
- Multisensory experiences: Reading aloud, listening, watching, and doing activities enhance memory.
- Listing keywords: Making lists of important terms helps build associations.

Note: Students with strong memory skills tend to perform better academically.

Application activity 5.7



4. How do different ways of learning, like trial-and-error or focusing on key points, help you remember and understand new information? Can you share an example from your own experience?
5. How can teachers use rewards, active engagement, or other strategies to make learning more effective and enjoyable for students like you?
6. Why do Psychologists like John Watson and Edward Thorndike think our environment and experiences shape the way we learn? How have you noticed this in your own learning?

5.8. Psychologists and learning theories: Skinner, Piaget, Pavlov, Jerome Bruner, Vygotsky, and Bandura

Learning activity 5.8



In a Rwandan primary classroom, Mr. Mukunzi helps students learn multiplication by combining rewards and problem-solving. Students earn tokens for correct answers, motivating them through positive reinforcement, while also working in groups to discuss strategies and discover solutions together.

1. How does earning tokens for correct answers motivate students to actively participate and improve their learning of multiplication?
2. How does working in groups help students share ideas and discover different strategies for solving multiplication problems?
3. In your own learning, how can you combine rewards (such as praise or incentives) with problem-solving activities to better understand difficult subjects?

5.8.1. Skinner

a) Bibliography

Burrhus Frederic Skinner (1904–1990) was an American Psychologist famous for his work in behaviourism. His most important contribution is operant conditioning, a method of learning through rewards and punishments. Learning happens when people or animals connect a behaviour with its consequence. Skinner used experiments with rats in a Skinner box, where pressing a lever gave the rats food. Over time, the rats learned to press the lever faster because they expected a reward.

b) Experiment Operant Conditioning

Operant conditioning is learning where voluntary behaviours are shaped by consequences. Reinforcement increases behaviour, while punishment decreases it.

- **Reinforcement:** Encourages behaviour.
 - Positive reinforcement: Adding something good, like praising a student for completing homework.
 - Negative reinforcement: Removing something unpleasant, like fastening a seatbelt to stop the car's beeping.
- **Punishment:** Discourages behaviour.
 - **Positive punishment:** Adding something unpleasant, like extra homework for misbehaviour.
 - **Negative punishment:** Taking away something enjoyable, like TV time, to reduce bad behaviour.

c) Principles of Operant Conditioning

1. **Law of effect:** Behaviours followed by rewards are repeated; those followed by punishment are not.
2. **Law of exercise:** The more you practice a behaviour, the stronger the connection; the less you practice, the weaker it becomes.
3. **Law of readiness:** Learning is effective when the learner is ready; forcing learning leads to frustration.
4. **Law of similarity:** The more similar the situations, the easier it is to respond correctly.
5. **Law of contiguity/immediacy:** Reinforcement should happen immediately after the response.
6. **Law of spacing:** Learning is stronger when practice is spaced over time, not crammed.

Example: A Rwandan teacher might reward students with tokens for correct answers (positive reinforcement) or remove privileges if they disrupt the class (negative punishment). Over time, students learn which behaviours are expected.

5.8.2. Jean Piaget

a) Bibliography

Jean Piaget (1896–1980) was a Swiss Psychologist known for his theory of cognitive development, which studies how children think and learn as they grow. Piaget believed children are like little scientists, actively exploring and trying to understand the world.

b) Stages of Cognitive Development

- 1. Sensorimotor stage (birth–2 years):** Babies learn by seeing, touching, and manipulating objects. Example: A toddler shaking a rattle to hear the sound.
- 2. Preoperational stage (2–7 years):** Children play pretend but struggle with logic and understanding others' perspectives. Example: A child pretending a stick is a sword.
- 3. Concrete operational stage (7–11 years):** Kids begin logical thinking but find abstract ideas challenging. Example: Understanding that $3 + 4 = 7$, but struggling with "what if" questions like "What if there were 3 red balls and 4 blue balls in total?"
- 4. Formal operational stage (12 years and up):** Teenagers and adults can think logically, reason abstractly, and solve hypothetical problems. Example: Solving algebraic equations or debating ethical dilemmas.

c) Advice for Teachers and Parents

When preparing lessons or activities, it's important to match them to the child's cognitive stage. Younger children need hands-on and concrete experiences, while older students can handle abstract ideas and problem-solving tasks.

5.8.3. Other Early Psychologists' contributions

Ivan Pavlov

Ivan Pavlov (1849–1936) contributed to the study of learning through his theory of classical conditioning. He demonstrated that learning can occur when a person or animal associates one stimulus with another. In his famous experiment, dogs learned to associate the sound of a bell with food and eventually responded to the bell alone. Pavlov's work showed that some behaviours can be learned through repeated associations between stimuli.

Jerome Bruner

Jerome Bruner (1915–2016) contributed to learning psychology through his theory of discovery learning. He argued that learners understand concepts better when they actively explore, ask questions, investigate, and discover relationships rather than simply receiving information from the teacher. Bruner also introduced the idea of a spiral curriculum, in which important ideas are introduced at a simple level and revisited at more complex levels. His work emphasised active participation and learner-centred learning.

Lev Vygotsky

Lev Vygotsky (1896–1934) emphasised the importance of social interaction and culture in learning. He argued that learners develop knowledge through interaction with teachers, parents, peers, and other more knowledgeable people. His concept of the Zone of Proximal Development (ZPD) refers to what a learner can achieve with appropriate support compared with what they can do independently. Vygotsky also highlighted the important role of language in developing thinking and learning.

Albert Bandura

Albert Bandura (1925–2021) explained that people can learn by observing and imitating the behaviour of others. His social learning theory emphasised the role of models such as parents, teachers, peers, and other people in the learning process. Through observation, learners can acquire new behaviours without directly experiencing them themselves. Bandura's work showed that learning is influenced by observation, imitation, social interaction, and the consequences of behaviour.

Application activity 5.8



Explain how the ideas of Skinner and Piaget continue to influence the way students learn, and teachers teach in classrooms today, giving examples from Rwandan schools if possible.

5.9. Modern Integration of Neuroscience and Technology Learning

Learning activity 5.9



Using an example, explain how you remember something you have learned and how technology can help you.

Modern psychology increasingly combines knowledge from neuroscience, psychology, and technology to understand how people learn. Neuroscience examines how the brain and nervous system function, while technology provides new ways of supporting learning and studying brain activity. This integration helps explain how attention, memory, emotions, motivation, and practice influence learning.

5.9.1. Role of Neuroscience in Understanding Learning

Neuroscience helps psychologists understand what happens in the brain when a person learns. Learning involves changes in connections between brain cells, and repeated practice can strengthen these connections. Research on the brain also shows the importance of attention, sleep, emotions, and motivation in supporting effective learning. This knowledge can help teachers design learning experiences that are more suitable for learners' cognitive and emotional needs.

Technology and Learning

Technology has introduced new ways of accessing information, practising skills, communicating, and receiving feedback. Digital learning platforms, educational applications, simulations, videos, and interactive activities can make learning more engaging and provide opportunities for people to practise at their own pace. However, technology is most effective when it is used appropriately and supports clear learning objectives.

Neuroscience and Digital Learning

The integration of neuroscience and technology has contributed to the development of digital learning approaches that consider how individuals pay attention, process information, remember content, and respond to feedback. For example, learning technologies can present information in different formats and provide immediate feedback, while knowledge from neuroscience can guide the design of activities that support memory and attention.

5.9.2. Emerging Technologies in Learning

Modern technologies such as artificial intelligence (AI), virtual reality (VR), augmented reality (AR), educational games, and adaptive learning systems are creating new opportunities for learning. These technologies can provide simulations, personalised practice, interactive experiences, and immediate feedback. Their educational value depends on how appropriately they are selected and integrated into the learning process.

5.9.3. Benefits and Challenges

The integration of neuroscience and technology can improve understanding of learning, support personalised learning, increase access to educational resources, and provide interactive learning experiences. However, challenges include excessive screen use, distraction, unequal access to technology, privacy concerns, and the risk of relying on technology without considering learners' social and emotional needs. Therefore, technology should complement rather than replace effective teaching and human interaction.

Application activity 5.9



How can technology such as smartphones, videos, educational apps, or AI help you learn or solve problems in your everyday life? Mention one benefit and one possible challenge.

5.10. End unit assessment



Question 1: Knowledge and understanding

Who developed the Allegory of the Cave, and what key lessons does it teach about learning and knowledge?

Question 2: Knowledge and understanding & Skills

- In what ways can learning happen outside of schools, and why is this important for students' development?
- How can the use of rewards in the classroom influence student behaviour, and what are some effective ways to implement them?

Question 3: Skills & Attitudes and values

How can self-regulation skills influence students' performance in school, and can you give examples from your own learning experience?

Question 4: Skills & attitudes and Values

What strategies would you suggest to help children develop emotional self-regulation and initiative, and how could these strategies be used in a Rwandan classroom?

UNIT 6

THEORIES OF LEARNING



Key Unit Competence:

Analyse different theories of learning and the way they influence the current education system.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Define key terms related to theories of learning.
- Explain the key principles, processes, and theorists associated with each type of learning theory.
- Differentiate different types of learning theories.
- State the implications of different types of learning in an individual's daily life.
- Discuss how various learning theories affect everyday life.

Skills

2

- Compare and contrast theories of learning.
- Categorise the different types of learning theories based on their principles and processes.
- Critique the different types of learning theories in terms of their strengths, limitations, and practical applications.
- Create scenarios that reflect types of learning in the teaching and learning process.

Attitudes and values

3

- Appreciate that types of learning theories can be proven incorrect or limited by later research.
- Appreciate the relevance of different types of learning theories in promoting effective teaching and learning.

- Exhibit confidence in analysing and judging differing theoretical perspectives.

Introduction of the unit

Have you ever wondered how you learned to speak your first words, solve a difficult problem, or even develop habits like greeting others politely or studying before exams? Why do some students learn quickly through practice, while others learn better by observing others or thinking deeply about ideas? These everyday experiences show that learning is not a simple process; it happens in different ways and is explained by different theories.

This unit introduces you to the theories of learning, which explain how people acquire new knowledge, skills, attitudes, and behaviours. You will begin by exploring key concepts such as learning, cognition, stimulus, response, reinforcement, punishment, observation, modelling, and socialisation, which provide a strong foundation for understanding how learning occurs in real-life situations. Building on this, you will examine major learning theories that have shaped modern education, particularly behaviourism. This includes classical conditioning, as demonstrated by Ivan Pavlov, and operant conditioning, as developed by B. F. Skinner. Through these perspectives, you will gain an understanding of how behaviour is influenced by stimuli, rewards, and consequences. You will also explore social learning theory by Bandura, which highlights learning through observation and imitation, as well as cognitive learning theory, guided by Piaget and Bruner, which focuses on how the mind processes and organises information. Furthermore, you will be introduced to insight learning and constructivist approaches that emphasise understanding, problem-solving, and active participation in learning. Throughout the unit, you will relate these theories to everyday experiences in schools, families, and communities, such as how teachers use reinforcement, how individuals learn by observing others, and how thinking processes shape learning. By the end of the unit, you will be able to analyse different theories of learning and explain how they influence the current education system, while developing skills in critical thinking, observation, and the application of knowledge in daily life.

Introductory activity



Think about your own experiences with music and advertising. You may remember that learning happens through experience and can influence how you feel and behave. For example, imagine hearing a song you loved as a child being used in an advertisement; you might start to feel positive about that product without even realising why.

- a) Can you recall an advertisement you have seen or heard (on TV, radio, or social media) that used a song you personally like or remember from your childhood? How did it make you feel about the product?
- b) Now imagine that the same product is advertised using a song you dislike. How do you think this would affect your interest in the product? Explain based on your own feelings and experiences.

6.1. Key concepts related to learning

Learning activity 6.1



A student named Aline comes to school every day. When the bell rings, she quickly enters the classroom. One day, her teacher praises her for completing homework on time, and she feels happy and continues working hard. Another student, Eric, is punished for making noise and later avoids that behaviour. During break time, Aline observes her friend greeting teachers politely and begins to do the same. At home, her parents guide her on how to behave well in the community.

Discuss

1. Identify at least three things Aline or Eric learned from the situation.
2. What caused Aline to enter the classroom when the bell rang?
3. Why did Aline continue doing her homework?
4. Why did Eric stop making noise?
5. How did Aline learn to greet teachers politely?
6. Who influenced Aline's behaviour at home?

Learning is a continuous process that takes place in everyday life, at school, at home, and in different workplaces such as offices, shops, farms, hospitals, and businesses. People learn new skills, improve performance, and adapt their behaviour through experience and interaction with others. To better understand how learning occurs, it is important to study the following key concepts.

6.1.1. Learning

Learning is a relatively permanent change in behaviour, knowledge, or skills that results from experience or practice. Examples: A new employee in an office learns how to use a computer system after several days of practice. A tailor improves sewing skills through repeated work. A farmer learns better planting techniques after attending training. Learning leads to improvement and lasting change.

6.1.2. Cognition

Cognition refers to the mental processes involved in thinking, understanding, remembering, and problem-solving. Examples include a cashier who quickly calculates change using mental skills, a nurse who determines the appropriate treatment after carefully analysing a patient's symptoms, and a student who plans how to answer examination questions. These situations illustrate how cognition enables individuals to make informed decisions and solve problems effectively.

6.1.3. Stimulus

A stimulus is anything in the environment that triggers a response. Examples:

a supervisor's instruction at a workplace. The ringing of a bell in a school or factory. A customer requesting service in a shop. A stimulus initiates behaviour.

6.1.4. Response

A response is the reaction or behaviour that follows a stimulus. Examples: Workers start or stop tasks when a bell rings. A shopkeeper serves a customer after being called. A driver slows down when seeing a traffic sign. Learning often involves connecting stimuli to appropriate responses.

6.1.5. Reinforcement

Reinforcement is anything that increases the likelihood that behaviour will be repeated. There are two main types of reinforcement:

- **Positive reinforcement:** This involves giving a reward, such as praise or a salary bonus, to encourage and strengthen a desired behaviour.
- **Negative reinforcement:** This involves removing something unpleasant, such as reducing supervision when performance improves, to encourage the continuation of a desired behaviour.

Examples: An employee receives praise for completing work on time and continues working hard. A student studies more after receiving good grades. Those are positive reinforcements. A worker avoids mistakes to prevent criticism. Reinforcement strengthens good behaviour.

6.1.6. Punishment

Punishment is a consequence that reduces the likelihood of a behaviour being repeated. Examples: An employee is warned for coming late and becomes more punctual. A worker loses a bonus due to poor performance. A student is disciplined for breaking school rules. Punishment discourages unwanted behaviour.

6.1.7. Observation

Observation is learning by watching others and what happens around them. Examples: A new worker learns how to operate a machine by watching experienced workers. A trainee observes how a manager communicates with clients. A young person learns farming techniques by watching their parents. Learning can occur without direct practice.

6.1.8. Modeling

Modelling is the process of imitating or copying the behaviour of others, especially those seen as role models. Example: A junior employee emulates a supervisor's professionalism. A student imitates a teacher's way of explaining ideas. An apprentice learns a trade by copying a skilled worker. People often copy behaviours they admire.

6.1.9. Socialization

Socialisation is the process through which individuals learn the values, norms, and behaviours of their society or workplace. Examples: New employees learn workplace rules, teamwork, and communication styles. Young people learn respect and responsibility from family and community. Workers adopt professional behaviour expected in their job. Socialisation helps individuals fit into society and workplaces.

These key concepts show that learning is not limited to the classroom. It occurs in real-life situations, especially in workplaces where people interact, observe, practice, and adapt their behaviour. Understanding these ideas helps learners explain how individuals acquire skills, improve performance, and become responsible members of society.

Application activity 6.1



Jean is a new employee working at a busy hotel in Kigali. On his first day at work, he carefully observes how experienced staff members welcome guests and carry out their duties. When a guest arrives, Jean pays close attention to how his senior colleagues greet the guest politely and assist them, and he tries to do the same.

As he continues working, Jean thinks carefully about how to handle different situations. When he performs his tasks well, his supervisor praises him, which encourages him to keep improving. When he makes mistakes, the supervisor gently reminds him to follow the correct procedures. Through daily practice, observation, and feedback, Jean gradually gains confidence, improves his skills, and becomes able to work on his own in a professional manner.

Questions

1. What did Jean do when he first saw a guest arriving at the hotel, and why do you think he behaved that way?
2. How did observing his senior colleagues help Jean improve his work? Give one clear example.
3. In what ways did the supervisor's praise and reminders influence Jean's behaviour and performance?
4. After one month, what changes can you notice in Jean's skills and his ability to work independently?

6.2. Behavioural learning (Behaviourism)

Learning activity 6.2



You usually take your coffee break at 10:00 am at your workplace. What would you do if, on a particular day, the coffee is not ready on time, or instead, your favourite snack is available in the break room? How might these situations affect your mood or behaviour?

6.2.1. Description of behaviourism

Behaviourism is a theory of learning that focuses on how behaviours are acquired, maintained, or changed through experience and interaction with the environment. Learning occurs when a person or animal responds to a stimulus, and these responses can be shaped through practice, reinforcement, or punishment. It focuses on observable behaviour rather than internal thoughts or feelings, demonstrating how experience shapes the way we act in everyday life, school, or work.

Within behaviourism, there are two main types of conditioning. Classical conditioning, demonstrated by Pavlov's experiment with dogs, occurs when a neutral stimulus (like the sound of a bell) is repeatedly paired with a natural stimulus (like the smell of food) until the neutral stimulus alone triggers a response (salivation). Operant conditioning, developed by Skinner, explains how voluntary behaviours are shaped by consequences: behaviours that are reinforced (through praise, rewards, or removing unpleasant situations) are more likely to be repeated, while behaviours that are punished are less likely to occur. Together, these conditioning methods show how both involuntary and voluntary behaviours can be learned and influenced by experience, helping people develop skills, habits, and behaviours in real life.

6.2.2. Introduction to classical conditioning

Classical conditioning is a type of learning in which a neutral stimulus becomes associated with a natural stimulus to produce a new, learned response. A famous example is Pavlov's experiment with dogs: by repeatedly pairing the sound of a bell (a neutral stimulus) with the smell of food (a natural stimulus), the dogs eventually began to salivate in response to the bell alone. This shows how involuntary responses can be learned through association.

6.2.3. Key Concepts in Classical Conditioning

a) Conditioned, Unconditioned, and Neutral Stimulus

- An unconditioned stimulus (US or UCS) is something that naturally triggers a reflex or automatic response without any prior learning. For example, smelling food makes your mouth water. Touching a hot iron makes you pull your hand back. Stepping on something sharp makes you scream. Inhaling dust makes you sneeze.
- A neutral stimulus (NS) is something that does not cause a response at first. For example, in Pavlov's experiment, the bell was neutral because the dogs did not salivate just at its sound.
- A conditioned stimulus (CS) is a neutral stimulus that, after being paired with an unconditioned stimulus, triggers a learned response.

b) Stimulus Generalisation and Discrimination

- Stimulus generalisation happens when a person or animal responds to stimuli that are similar to the original stimulus. Example: If you get sick from eating red berries, you might also avoid purple berries, even though they are not the same.
- Stimulus discrimination is the ability to tell the difference between similar stimuli and respond only to the correct one. Example: Pavlov's dogs learned to salivate to the specific bell tone used in training, but not to other tones.

c) Conditioned and Unconditioned Response

- An unconditioned response (UR) is a natural reaction caused by an unconditioned stimulus. Example: Feeling pain when a needle pricks your skin.
- A conditioned response (CR) is a learned reaction that happens after a neutral stimulus has been paired with an unconditioned stimulus. Example: A child who feels pain from a vaccination (US) may start to feel scared or distressed when seeing the doctor or needle (CS). The fear or distress is the conditioned response.

6.2.4. Process of classical conditioning (Experiment of Pavlov)

Classical conditioning occurs in three main steps, with specific stimuli and responses at each stage. Pavlov's experiment with dogs is a classic example that illustrates how a neutral stimulus can be associated with a natural stimulus to produce a new learned response.

1. Before Conditioning

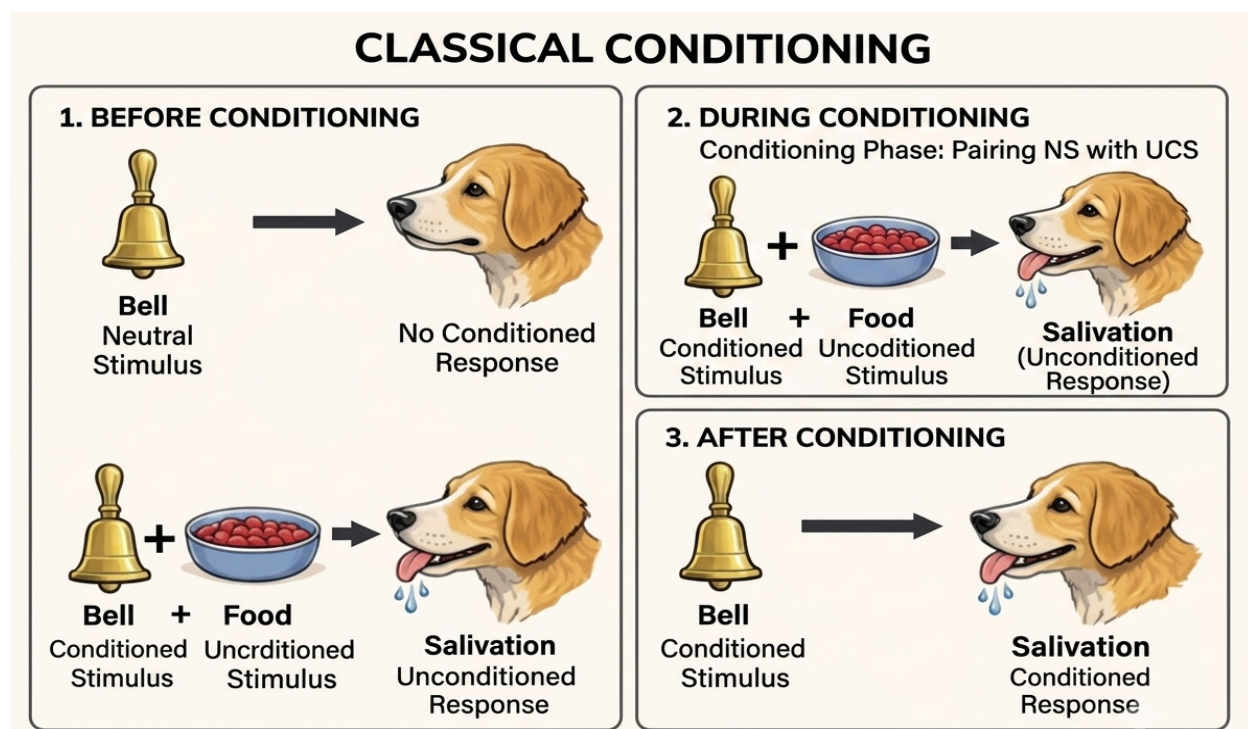
- Natural stimulus: Food → naturally causes salivation in dogs (unconditioned response).
- Neutral stimulus: Bell → does not cause salivation.

2. During Conditioning

- Pavlov repeatedly paired the ringing bell (neutral stimulus) with the presentation of food (natural stimulus).
- Over time, dogs began to associate the bell with the food.

3. After Conditioning

- Conditioned stimulus: Bell → now triggers salivation even without food.
- Conditioned response: Salivation in response to the bell.



Source: <https://www.wattsburg.org/Unit5Learning.aspx>

6.2.5. Key Concepts in Classical Conditioning

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6.2.6. Laws of Classical Conditioning

1. Acquisition

This is the learning phase, when an organism gradually associates a neutral stimulus with an unconditioned stimulus to produce a conditioned response. For example, Pavlov's dogs learned to salivate at the sound of the bell after repeated pairings with food.

2. Generalisation

After conditioning, an organism may respond to stimuli similar to the original conditioned stimulus (CS). Example: If a dog is conditioned to salivate to a bell, it may also salivate to a chime or other similar sounds.

The more similar the new stimulus is to the original CS, the stronger the response.

3. Discrimination

This is the ability to distinguish between the conditioned stimulus and other similar stimuli, responding only to the correct one. For example, Pavlov's dogs salivated only in response to the specific bell tone used in training, not to other tones.

4. Extinction

Extinction happens when the conditioned stimulus is repeatedly presented without the unconditioned stimulus (US). The conditioned response (CR) gradually disappears. For example, if the bell rang repeatedly without presenting food, the dogs eventually stop salivating.

5. Spontaneous Recovery

Even after extinction, a conditioned response can suddenly reappear after some time. Example: Days later, the dogs might salivate again when the bell rings, even without food.

6. Higher-Order Conditioning

A new stimulus can become conditioned by being associated with an already established conditioned stimulus. Example: If a light is flashed before the bell rings, eventually the light alone might trigger salivation in the dogs.

Application activity 6.2



1. In your own words, explain the difference between a conditioned stimulus and an unconditioned stimulus. Can you give an example from your daily life?
2. Alice trained her younger sister to react every time she said the word "pancake" by tickling her afterwards. What kind of stimulus do you think the word "pancake" has become? Explain why.
3. Describe how stimulus generalisation and stimulus discrimination are opposites. Can you give examples from school, home, or work where someone might generalise or discriminate between similar signals or cues?

6.3. Behavioural learning (Behaviourism): Applications and Factors affecting classical conditioning

Learning activity 6.3



At a busy office, every day at 10:00 a.m., a bell rings to signal that coffee and snacks are ready in the break room. At first, the bell doesn't mean much to the employees; they keep working. But over time, every time the bell rings, employees get excited and head to the break room before even seeing the coffee.

One morning, the bell rang, but no coffee was ready. Some employees still walk toward the break room automatically, expecting the coffee. On another day, the bell rings, and a new snack they really like is served. Employees become even more excited and eager to go to the break room when the bell rings in the future.

Questions:

1. What do you think caused the employees to react automatically to the bell after a few days?
2. Why do some reactions fade or change when the bell rings, but no coffee is served?
3. How does the type of snack, timing, or frequency of the bell influence how strong the reaction becomes?
4. Can you think of similar examples from school, home, or other workplaces where people develop automatic responses to signals or events?

6.3.1. Applications for classical conditioning

Classical conditioning is used widely in daily life, education, workplaces, and therapy. Some examples include:

1. Education: Teachers use a pleasant tone or rewards to encourage participation. Over time, students may feel motivated to engage when the teacher signals an activity.
2. Advertising: Brands pair products with enjoyable music or images. Over time, consumers feel positive emotions toward the product.
3. Workplace: A bell or notification signals break time. Employees may feel automatically excited or alert when it rings.
4. Therapy & Behaviour Change: Phobias and anxieties are treated using techniques like systematic desensitisation, which pairs feared stimuli with relaxation to reduce fear.

6.3.2. Factors Affecting Classical Conditioning

The strength and effectiveness of classical conditioning depend on several factors:

1. **Frequency of Pairing:** The more times the neutral stimulus is paired with the unconditioned stimulus, the stronger the learned response. Example: The more consistently the office bell signals coffee, the faster employees respond.
2. **Timing of Stimulus Presentation:** The neutral stimulus should be presented just before or at the same time as the unconditioned stimulus. Example: The bell should ring shortly before coffee is available. If it rings too early or too late, the association may be weaker.
3. **Intensity of Stimuli:** Stronger or more noticeable stimuli create stronger associations. Example: A bright, loud alarm paired with a reward may produce a quicker response than a soft sound.
4. **Generalisation:** A stimulus similar to the original conditioned stimulus may trigger the same response. Example: Employees may respond to a chime that sounds like a coffee bell.
5. **Discrimination:** Learners can distinguish between similar stimuli and respond only to the correct one. Example: Employees may only react to the specific office bell used for coffee, not other alarms.

Classical conditioning shows that behaviours can be learned through association. Its applications are seen in education, advertising, workplace routines, and therapy, while its effectiveness is influenced by frequency, timing, intensity, and the learner's ability to generalise or discriminate stimuli. Understanding these principles helps us explain and predict behaviour in real-life situations.

Application activity 6.3



1. Think of a real-life situation (such as in school, workplace, or advertising) where a signal, sound, or event makes people react automatically. Explain how this response was learned and how classical conditioning is applied in that situation.
2. Describe a situation where a learned response becomes stronger or weaker over time. How do factors like frequency of pairing, timing, and intensity of stimuli influence this change? Give a clear example.
3. Using examples from your daily life, explain how people may respond to similar stimuli or learn to distinguish between different stimuli. How does this show the processes of generalisation and discrimination?

6.4. Operant conditioning

Learning activity 6.4



Think about a time when you were expected to do cleaning at home or help with chores, but did not do it properly. What happened afterwards, and how did the reaction from others influence what you did the next time?

6.4.1. Description, origin, and foundation of Skinner's theory (Work of E. L. Thorndike)

Operant conditioning is a theory of learning that explains how behaviour is influenced by its consequences. It was mainly developed by B.F. Skinner, an American Psychologist, believed that people and animals learn best when their actions are followed by rewards or punishments. According to Skinner, behaviours that are reinforced are more likely to be repeated, while those that are punished are less likely to occur. This type of learning focuses on voluntary behaviours, such as studying, working, or following rules.

The foundation of Skinner's theory comes from the earlier work of Edward L. Thorndike, who introduced the Law of Effect. Thorndike conducted experiments using animals, especially cats, placed in a "puzzle box." The cats learned to escape by pressing a lever or pulling a string. At first, they made many random attempts, but over time, they repeated the actions that helped them escape successfully. From this, Thorndike concluded that behaviours followed by satisfying outcomes are likely to be repeated, while those followed by unpleasant outcomes are less likely to be repeated.

Building on Thorndike's ideas, Skinner developed the concept of operant conditioning,

emphasising that behaviour is shaped and maintained by reinforcement and punishment. He used the famous Skinner Box experiment, where animals such as rats or pigeons learned to press a lever or peck a key to receive food or avoid discomfort. This showed clearly that behaviour can be controlled and modified through consequences.

In everyday life, Skinner's theory can be seen in schools, workplaces, and homes. For example, students study harder when they are praised or rewarded, and employees improve their performance when they receive incentives or recognition. This shows that learning is not only about thinking but also about how the environment shapes behaviour through consequences.

6.4.2. Key concepts of operant conditioning

Operant conditioning is a type of learning in which behaviour is influenced by its consequences. According to this theory, individuals are more likely to repeat behaviours that lead to positive outcomes and avoid those that lead to negative outcomes. The main processes involved in operant conditioning are reinforcement, punishment, and extinction. These processes help explain how behaviour is strengthened, weakened, or eliminated in everyday life.

1. Reinforcement

Reinforcement refers to any consequence that increases the likelihood that behaviour will be repeated. It plays an important role in encouraging desirable behaviours in school, at home, and in workplaces.

There are two main types of reinforcement:

- **Positive reinforcement:** This involves adding a pleasant or rewarding stimulus after a desired behaviour. Example: A teacher praises a student for answering questions correctly, encouraging the student to participate more in class. In a workplace, an employee may receive a bonus for good performance, motivating them to continue working hard.
- **Negative reinforcement:** This involves removing an unpleasant stimulus after a desired behaviour. Example: A student completes assignments on time to avoid being scolded by the teacher. Similarly, a worker may finish tasks early to avoid strict supervision. Reinforcement strengthens behaviour and increases the chances that it will occur again.

2. Punishment

Punishment refers to any consequence that reduces the likelihood that behaviour will be repeated. It is used to discourage undesirable or inappropriate behaviour.

There are two types of punishment:

- **Positive punishment:** This involves adding an unpleasant consequence after undesired behaviour. Example: A student who talks during a lesson may be given extra work. In a workplace, an employee who breaks rules may receive a warning.
- **Negative punishment:** This involves removing something pleasant after undesired behaviour. Example: A student may lose the chance to participate in a school activity due to misconduct. At home, a child may have their phone taken away for not following instructions. Punishment aims to reduce or stop unwanted behaviour.

3. Extinction

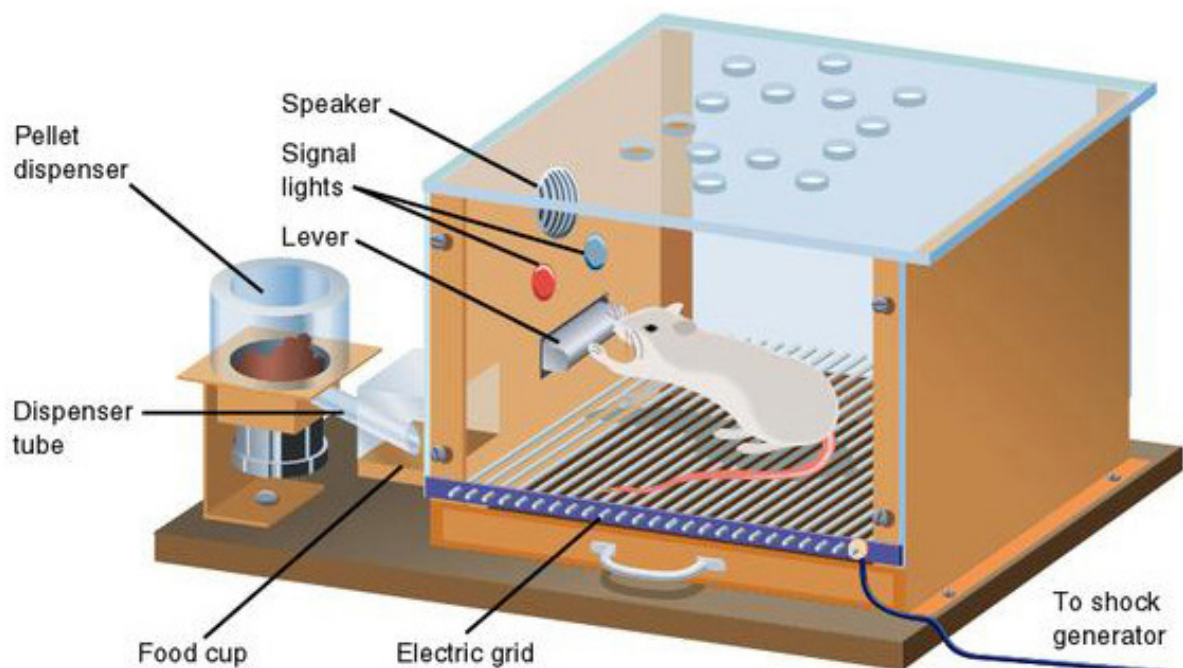
Extinction occurs when behaviour gradually decreases and eventually disappears because it is no longer reinforced. When behaviour is not rewarded or recognised, the individual may stop performing it over time. Examples: A student who stops receiving attention for making noise in class may eventually stop that behaviour. An employee who is not recognised for extra effort may stop going beyond their duties. Without reinforcement, behaviour weakens and may disappear.

Reinforcement, punishment, and extinction are essential concepts in operant conditioning. They explain how behaviour can be encouraged, discouraged, or eliminated through consequences. These principles are widely used in education, parenting, and workplaces to guide behaviour and improve performance. Understanding these concepts helps individuals make better decisions and develop positive habits in their daily lives.

6.4.3. The experiment of F. Skinner (Skinner Box)

B. F. Skinner, an American psychologist, conducted experiments to explain how behaviour is influenced by its consequences. He used a special device called the Skinner Box (also known as an operant conditioning chamber) to study how animals learn through reinforcement and punishment.

The Skinner Box is a controlled environment that contains a lever (for rats) or a key (for pigeons), a food dispenser, and sometimes a light or sound signal. In his experiment, Skinner placed a hungry rat inside the box. At first, the rat moved around randomly. By chance, it pressed the lever, and immediately, food was released. After several attempts, the rat learned that pressing the lever resulted in receiving food. As a result, it began to press the lever more frequently.



Source: <https://www.simplypsychology.org/operant-conditioning.html>

Skinner also showed that behaviour could be shaped using different consequences. For example, if pressing the lever stopped an unpleasant stimulus (such as a mild

electric shock), the rat would learn to press the lever to avoid discomfort. On the other hand, if pressing the lever produced no reward, the behaviour gradually decreased. This demonstrated the processes of reinforcement and extinction.

The Skinner Box experiment proved that behaviour is not random but can be learned, strengthened, or weakened based on its consequences. It also showed that learning can occur step by step, as animals gradually form associations between their actions and outcomes.

In real life, this type of learning can be seen in schools, workplaces, and homes. For example, students study harder when they are rewarded with good grades or praise, and employees improve their performance when they receive incentives or recognition.

B.F. Skinner's experiment demonstrates that behaviour is shaped by its consequences, and that learning occurs through reinforcement, punishment, and extinction.

6.4.4. Types of reinforcement and punishment

In operant conditioning, behaviour is influenced by its consequences. Behaviours that are reinforced become stronger and are likely to be repeated, while behaviours that are punished become weaker and are less likely to occur again. Since operant conditioning deals with voluntary behaviour, individuals choose their actions based on expected outcomes.

There are four main types of consequences:

1. **Positive reinforcement:** Adding a desirable outcome to strengthen behaviour.
2. **Negative reinforcement:** Removing an unpleasant condition to strengthen behaviour.
3. **Positive punishment:** Adding an unpleasant consequence to reduce behaviour.
4. **Negative punishment:** Removing a desirable outcome to reduce behaviour.

Example: A student studies to receive praise (positive reinforcement), completes homework to avoid punishment (negative reinforcement), is given extra work for misbehaviour (positive punishment), or loses privileges such as phone use (negative punishment).

1. Positive reinforcement

Positive reinforcement is a concept developed by B.F. Skinner. It refers to the process of adding a rewarding stimulus after a desired behaviour, making that behaviour more likely to occur again. It is one of the most effective ways to encourage positive behaviour and develop new habits.

When a person receives a reward, praise, or any positive outcome after performing a behaviour, that behaviour is strengthened.

a) Examples of positive reinforcement

Positive reinforcement is common in everyday life:

- A parent praises a child for completing homework independently.
- A teacher rewards students with free time for good classroom behaviour.

- An employee receives a bonus for good performance.
- A student is given a reward after achieving good academic results.

These rewards motivate individuals to repeat the desired behaviour.

b) Uses of Positive Reinforcement

Positive reinforcement is widely used in different settings:

- At home: Parents encourage good behaviour, such as cleaning or studying, by giving praise or small rewards.
- At school: Teachers use praise, marks, or rewards to encourage participation and discipline.
- At the workplace: Employers use incentives, promotions, or recognition to improve performance.
- In therapy: It is used to help individuals develop positive behaviours and coping skills.

Positive reinforcement is most effective when it is planned, consistent, and applied immediately after the behaviour.

c) Categories of positive reinforcement

Positive reinforcers can be grouped into different types:

- Natural (Direct) Reinforcers: Rewards that naturally follow a behaviour.
Example: Feeling proud after performing well in exams.
- Social reinforcers: Praise, approval, or recognition from others.
Example: A teacher saying "well done" to a student.
- Tangible reinforcers: Physical rewards such as money, gifts, or food.
Example: Receiving a prize for good performance.
- Token reinforcers: Points, stars, or marks that can be exchanged for rewards.
Example: A student collecting points to earn a reward.
- Activity reinforcers: Opportunities to engage in enjoyable activities.
Example: Allowing extra playtime after completing tasks.

d) Tips for using positive reinforcement

For positive reinforcement to be effective, the following should be considered:

- It should be given immediately after the desired behaviour.
- It should be consistent and appropriate.
- It should not encourage undesirable behaviour.

Example: If a child is rewarded for misbehaving, bad behaviour may be strengthened instead of reduced.

e) Reinforcement schedules

A reinforcement schedule refers to how often and when reinforcement is given. It affects how strongly behaviour is learned.

- **Continuous reinforcement:** The behaviour is rewarded every time it occurs. Example: A teacher praises a student each time they answer correctly. This is effective when learning new behaviour.
- **Partial reinforcement:** The behaviour is rewarded only sometimes. This is useful for maintaining behaviour over time.

Types of partial reinforcement include:

- **Fixed Ratio:** Reward after a set number of responses.
- **Variable Ratio:** Reward after an unpredictable number of responses.
- **Fixed Interval:** Reward after a fixed period.
- **Variable Interval:** Reward after varying periods. Example: Receiving a salary after a fixed period of work.

2. Negative Reinforcement

Negative reinforcement is a process of strengthening a behaviour by removing or avoiding an unpleasant stimulus or condition after the desired behaviour occurs. The removal of the unpleasant condition increases the likelihood that the behaviour will be repeated in similar situations. Example: A student regularly completes homework to avoid receiving additional remedial work from the teacher. If completing the homework results in the additional work being removed, the student is more likely to complete homework in the future.

a) Key characteristics of negative reinforcement

- An unpleasant stimulus or condition is present or anticipated.
- The individual performs a particular behaviour.
- The unpleasant stimulus is removed or avoided.
- The behaviour becomes more likely to occur again.

b) Examples

- A student studies hard to avoid the anxiety associated with failing an examination.
- A driver fastens a seat belt to stop the car's warning sound.
- A person takes an umbrella to avoid getting wet when it rains.
- An employee completes assigned work on time to avoid repeated reminders from a supervisor.

3. Positive punishment

Positive punishment refers to the addition of an unpleasant consequence after undesired behaviour to reduce it. It is commonly used to discourage inappropriate actions. Examples of positive punishment:

- A driver receives a fine for speeding.
- A student is scolded for breaking school rules.
- A child is given extra chores for misbehaviour.
- An employee receives a warning about poor performance.

a) Effects of punishment

Positive effects:

- Helps individuals understand that certain behaviours are unacceptable.
- Emphasises that every action has consequences.
- Encourages more appropriate behaviour in the future.

Limitations:

- The behaviour may return when punishment stops.
- It may create fear or negative emotions.
- It does not always teach the correct behaviour to replace the wrong one.

For best results, punishment should be combined with reinforcement of good behaviour.

b) Positive Punishment in Daily Life and Workplace

Positive punishment is used in many areas of life:

- Being reprimanded by a teacher or employer.
- Receiving extra tasks for failing to meet expectations.
- Being given warnings or stricter rules after misbehaviour.

Example: An employee who is late to work receives a warning and becomes more punctual.

Reinforcement and punishment are key processes in operant conditioning. Reinforcement increases the likelihood of behaviour being repeated, while punishment reduces unwanted behaviour. When used appropriately, these methods help shape behaviour in schools, homes, and workplaces, leading to improved learning and performance.

4. Negative Punishment

Negative punishment is a type of operant conditioning in which a desirable stimulus, reward, or privilege is removed following an undesirable behaviour in order to reduce the likelihood of that behaviour occurring again. The term “negative” means removing something, while “punishment” means reducing a behaviour.

a) Characteristics of Negative Punishment

- Removal of a desirable stimulus: A valued reward, privilege, or activity is taken away following an unwanted behaviour.
- Reduction of behaviour: Its main purpose is to decrease the frequency or likelihood of the undesirable behaviour.
- Contingency: The removal occurs because a particular behaviour has taken place.
- Immediate consequence: The consequence is generally more effective when it follows the unwanted behaviour soon after it occurs.

- Individual differences: What is considered desirable varies from one person to another.
- Part of operant conditioning: It is based on the principle that consequences influence the likelihood of future behaviour.

b) Examples of Negative Punishment

- A student loses break-time privileges after repeatedly disturbing the class.
- A teenager loses access to a mobile phone for breaking an agreed family rule.
- A child loses the opportunity to play a favourite game after behaving aggressively toward others.
- A student loses participation in a preferred extracurricular activity after repeatedly failing to follow school rules.
- An employee loses a workplace privilege after repeatedly violating organisational regulations.

c) Effects of Negative Punishment

When appropriately applied, negative punishment may:

- Reduce undesirable behaviour by making the behaviour less likely to occur again.
- Encourage rule-following when individuals understand the relationship between their behaviour and the consequence.
- Promote self-control by helping individuals consider the consequences of their actions.
- Reduce disruptive behaviour in educational, family, and organisational settings.
- Establish behavioural boundaries by making expectations and consequences clear.

However, the effectiveness depends on factors such as the importance of the removed privilege, consistency in applying the consequence, and the individual's understanding of why the privilege was removed.

d) Limitations of Negative Punishment

- It may not address the cause of the behaviour: Removing a privilege may stop a behaviour temporarily without explaining why the person behaved that way.
- It can create frustration or resentment: Frequent or excessive removal of privileges may lead to negative feelings toward parents, teachers, or other authority figures.
- It may be ineffective if the removed stimulus is not valuable: If the person does not value what is removed, the punishment may have little effect.
- It may encourage avoidance rather than learning: A person may focus on avoiding consequences rather than understanding and developing appropriate behaviour.
- Inconsistent application reduces effectiveness: If the consequence is applied inconsistently, the individual may not learn a clear relationship between behaviour and consequence.
- It may damage relationships when misused: Excessive or unfair removal of privileges can undermine trust between individuals.

- Behaviour may return: Once the punishment is no longer applied, the undesirable behaviour may reappear if appropriate alternative behaviours have not been taught.

Application activity 6.4



A teenage girl has recently obtained her driving license, but it clearly states that she is not allowed to drive at night. Despite this rule, she repeatedly drives at night several times a week without facing any consequences, which makes her believe that she will not be caught. One evening, while driving to the mall with a friend, she was stopped by traffic police and given a ticket. A few days later, she received an official notice informing her that her driving privileges had been suspended for 30 days as a penalty for breaking the law. After the suspension ends, she gets her license back but continues to drive at night before she is legally allowed to do so, still believing that she will not be caught again.

Discussion questions

1. In your opinion, which consequence in this situation represents negative punishment, and why do you think it fits that category?
2. Why do you think someone might continue to drive at night even after receiving a punishment in a situation like this?
3. Based on this situation, how would you explain the difference between negative reinforcement and punishment in your own words?

6.5. Operant conditioning: Schedules of reinforcement, application of operant conditioning.

Learning activity 6.5



Think about your daily life at school, home, or in your community. You are rewarded every time you do something well, such as being praised for correct answers at other times. Rewards are given only occasionally, such as surprise marks, gifts, or points in games, after several attempts. Also, consider how your behaviour changes depending on the consequences. For example, you may work harder to achieve good grades, completing homework to avoid punishment, or repeating actions that bring benefits at home or in the workplace.

Discuss

1. Why are some behaviours rewarded every time you perform them, while others are rewarded only occasionally?
2. How do rewards and consequences influence your behaviour at school, home, or in your community?
3. Why do you think people continue to repeat certain behaviours even when they are not rewarded every time?

6.5.1. Types of Reinforcement Schedules

A schedule of reinforcement refers to the pattern or timing of how reinforcement is given after behaviour. It determines how often and when a behaviour is rewarded.

a) Continuous Reinforcement

In this schedule, behaviour is reinforced every time it occurs. It is useful when learning a new behaviour. Example: A teacher praises a student every time they answer a question correctly. This helps learners quickly understand the connection between behaviour and reward.

b) Partial (Intermittent) Reinforcement

In this schedule, behaviour is reinforced only sometimes, not every time it occurs. It is used to maintain behaviour over a long period.

Types of Partial Reinforcement:

- **Fixed Ratio:** Reward after a fixed number of responses. *Example:* A worker is paid after completing 10 tasks.
- **Variable Ratio:** Reward after an unpredictable number of responses. *Example:* Gambling wins after unknown attempts.
- **Fixed Interval:** Reward after a fixed time or period. *Example:* Monthly salary.
- **Variable Interval:** Reward after changing time periods. *Example:* Surprise teacher checks in class.

6.5.2. Importance of Reinforcement Schedules

Reinforcement schedules are important because they affect how quickly behaviour is learned and how long it lasts.

- Continuous reinforcement helps in learning new behaviour quickly.
- Partial reinforcement helps behaviour to be strong and resistant to extinction.

Example in daily life: A child may learn to clean their room quickly when rewarded every time (continuous reinforcement), but over time, rewards may be given occasionally (partial reinforcement) to maintain the habit.

6.5.3. Comparison of Operant and Classical Conditioning

Operant and classical conditioning are both learning theories, but differ in important ways:

Classical Conditioning	Operant Conditioning
Learning by association between stimuli	Learning through the consequences of behaviour
Involuntary behaviour	Voluntary behaviour
Example: Salivating at the sound of the bell	Example: Studying for rewards
Developed by Pavlov	Developed by Skinner

Both explain how learning occurs through interaction with the environment. However, classical conditioning focuses on automatic responses, while operant conditioning focuses on chosen behaviours.

6.5.4. Applications of Operant Conditioning

Operant conditioning is widely used in everyday life, especially in education and parenting.

In Education

Teachers use reinforcement to improve learning and behaviour. *Example:* Praising students for good performance encourages them to study harder. Or giving stars or marks for homework completion.

In Parenting

Parents use rewards and consequences to shape children's behaviour. *Example:* A child is allowed extra playtime after finishing homework. Or a child loses phone privileges for misbehaviour.

In the Workplace

Employers use reinforcement to improve productivity. *Example:* Workers receive bonuses for good performance, and employees receive warnings for being late coming.

In Therapy

Behaviour therapy uses reinforcement to change unwanted behaviours. *Example:* Rewarding patients for reducing phobias or harmful habits.

Operant conditioning and classical conditioning both explain learning, but in different contexts. Classical conditioning links stimuli to automatic responses, while operant conditioning links behaviour to consequences. Together, they help explain how people learn behaviours in school, at home, and in society.

Application activity 6.5



Read the situations below and answer the questions by applying your understanding of operant conditioning:

1. In your school, a teacher praises students every time they answer correctly during the first week of a new topic. Later, the teacher only praises them sometimes. Explain why the teacher changed the way praise is given and how this may affect students' learning behaviour.
2. At home, a parent gives a child pocket money every Saturday if chores are completed, but sometimes gives extra rewards unexpectedly. How does this pattern of reinforcement influence the child's behaviour? Give reasons.
3. In the workplace, employees receive a salary every month, but also get surprise bonuses for excellent performance. How do these two types of reinforcement help improve workers' motivation and productivity?

6.6. Social learning theory/Observational learning theory (Albert Bandura)

Learning activity 6.6



After watching many Marvel superhero movies, Tom, a student in your school, starts imitating his favourite heroes. Each time he jumps off the monkey bars at the playground, he lands in a "superhero pose," just like the characters he has seen in the films.

Discussion questions:

1. In your opinion, why do you think Tom started copying superhero poses after watching Marvel movies? Explain using your own understanding.
2. Have you ever seen or experienced a situation where someone (or you yourself) copied behaviours from movies, friends, or media? Describe it and explain why it happened.
3. What does Tom's behaviour tell you about how people learn from what they see in their environment, especially from media and role models?



6.6.1. Introduction to observational/ social learning theory (Albert Bandura)

Canadian/American Psychologist, Albert Bandura was one of the first Psychologists to recognise the phenomenon of observational learning. His theory, Social Learning Theory, stresses the importance of observation and modelling of behaviours, attitudes, and emotional reactions of others. He found that, as social animals, humans naturally gravitate toward observational learning. Children watch their family members and mimic their behaviours. Even infants, at just 3-weeks old, start imitating the mouth movements and facial expressions of adults around them.

Albert Bandura, a Canadian American Psychologist, was one of the first scholars to recognise the importance of observational learning. His Social Learning Theory emphasises the role of observing and modelling the behaviours, attitudes, and emotional reactions of others.

He discovered that, as social beings, humans naturally tend to learn through observation. Children, for example, watch family members and imitate their behaviours. Even infants, as young as three weeks old, can imitate the mouth movements and facial expressions of adults around them.

Observational learning is the process of acquiring new behaviours by watching others. As the name suggests, this type of learning occurs through observation and imitation. When individuals observe how others behave, they often remember this information and later reproduce similar actions.

This type of learning, also known as modelling or shaping, is especially common during childhood. It plays a key role in socialisation, as children learn how to behave and respond to others by observing parents, teachers, and other important figures in their lives.

Although people sometimes intentionally observe experts to learn new skills, observational learning is often unintentional. For instance, a child may learn to use inappropriate language or adopt harmful habits like smoking simply by watching adults. This shows that individuals are constantly learning from their environment, whether the behaviours observed are positive or negative.

6.6.2. Key concepts of observational learning

a) Modelling

Modelling is the process by which an individual learns behaviour by observing another person (the model). The model demonstrates behaviour, and the observer pays attention, remembers it, and may later reproduce it. Albert Bandura explained that modelling is not just copying actions; it also involves learning attitudes, emotional reactions, and social behaviours.

Example: A student observes a teacher speaking politely and respectfully. Over time, the student begins to use similar polite language when talking to classmates and adults.

b) Imitation

Imitation is the act of copying or reproducing the behaviour observed in a model. It is the visible outcome of observational learning.

Children are especially good at imitation because they are still learning how to behave in society. They copy speech, gestures, habits, and even emotional expressions.

Example: A child sees older students greeting teachers respectfully by saying "Good morning." The child begins to imitate this behaviour and starts greeting teachers in the same way.

Types of imitation:

- **Immediate imitation:** Copying behaviour right after observing it.
- **Delayed imitation:** Reproducing behaviour after some time has passed.
- **Selective imitation:** Choosing which behaviours to imitate.

Imitation is influenced by:

- The attractiveness or popularity of the model.
- The perceived consequences of the behaviour.
- The learner's age and developmental stage.

c) Vicarious reinforcement

Vicarious reinforcement occurs when a person observes another individual being rewarded for a behaviour and is consequently motivated to imitate that behaviour.

Similarly, **vicarious punishment** happens when someone observes a person being punished and becomes less likely to imitate that behaviour.

This means that people do not need to experience rewards or punishments directly; they can learn simply by observing others.

Example: In a classroom, a student sees a classmate being praised by the teacher for completing homework on time. As a result, the student is motivated to also complete homework to receive praise. Or if a student sees a peer being punished for making noise, they are less likely to repeat that behaviour.

d) The Bobo Doll Experiment

This famous experiment by Albert Bandura provides strong evidence for modelling, imitation, and vicarious reinforcement. Children observed an adult model behaving aggressively toward a doll. The outcomes differed depending on whether the model was rewarded, punished, or received no consequence.

Key findings:

- Children imitated aggressive behaviour when the model was rewarded or not punished.
- Children showed less aggression when the model was punished.
- Learning occurred even without direct reinforcement.

6.6.3. Benefits of observational learning

Observational learning happens indirectly, meaning that it does not require formal teaching or a learning process. Here are some of the key benefits of observational learning:

- **Learning new skills:** Observational learning allows both children and adults to acquire new skills by watching others. For example, a child may learn how to paint her nails by observing her mother, while an adult may learn how to lift weights by watching instructional videos.

- **Reinforcing positive behaviour:** Some individuals can learn positive behaviour by observing others. For instance, if a student sees a classmate being rewarded with candy for good academic performance, she/he may be motivated to work harder and imitate that behaviour to receive similar rewards in the future.
- **Decreasing negative behaviour:** Observational learning can also decrease negative behaviour in people. If a co-worker is frequently late for meetings and is reprimanded by a supervisor, this discourages others from arriving late.

6.6.4. Four Processes or Stages of Observational Learning

According to Bandura's research, four processes influence observational learning:

1. Attention

To learn, an observer must pay attention to something in the environment. They must notice the model and the behaviour occurring. Attention levels can vary based on the characteristics of the model and environment, including the model's degree of likeness or the observer's current mood. In humans, the observer will likely pay attention to behaviours of models that are high-status, talented, intelligent, or like the observer in any way. There are a few factors that affect a person's ability to focus and give their full attention:

- **Physical similarities:** Observers sometimes pay more attention to others who are of the same sex or similar age.
- **Prestige:** Observers are more likely to be influenced by people they see as attractive, successful, or distinguished.
- **Health:** If an observer is ill or tired, they may not be able to focus completely on the subject they're observing. This can impact their ability to later imitate the behaviour they learned.

For example, if you want to become a Vice President (VP) at your company, it makes sense that you'd observe the current VP's (or other renowned VPs in your industry) and try to mimic their behaviour.

2. Retention

Simple attention is not enough to learn a new behaviour. An observer must also retain, or remember, the behaviour later. To increase chances of retention, the observer must structure the information in an easy-to-remember format. Maybe they use a mnemonic device. Or form a daily learning habit. The behaviour must be easily remembered so the action can be performed with little or no effort.

Factors that might influence retention include interference, time, interest, and motivation.

- **Interference:** If other things in the environment compete for attention, they may interfere with how well the observed learning is retained.
- **Time:** If a great deal of time lapses between the observation and the performance of the behaviour, memory of the action may also be weak.
- **Motivation:** Interest and motivation can significantly influence how well observational learning is retained. If a person is highly interested or strongly motivated to imitate the observed actions, retention is likely to be much stronger.

Using our VP example, let's say the current VP is giving a company-wide presentation. You notice that they are calm, confident, engaging, and make eye contact. You make a list of these attributes and remember them for the next time you give a presentation.

3. Reproduction

Reproduction is the process by which the observer must be able to physically perform the behaviour in the real world. An individual's capacity to recreate behaviour successfully depends on a few factors:

- **Age or physical attributes:** If a small child watches a baseball player hit a home run, they may try to replicate the action before they are physically able to accomplish it.
- **Ability level:** If a toddler watches their parent folding towels and tries to mimic that behaviour, it is likely going to take them an extended period to complete it properly. This is due to the child's motor skills not having fully developed to that level of ability.
- **Difficulty of learned behaviour:** The time to successfully replicate a behaviour depends on whether it is simple or difficult to model. If the behaviour is simple, such as an adult watching someone wash their car, then the learner is likely to model that behaviour with ease. If it's challenging, it may take more time and effort.

Often, producing new behaviour requires hours of practice to obtain the skills. You can't just watch your VP give a brilliant company-wide presentation, then use only the observed tactics in your own presentation 20 minutes later.

For example, using the VP example again, you've observed and identified four skills that the current VP uses during presentations. To be able to perform these skills yourself, you need to deliberately practice them. Maybe you hold small team meetings to test your skills. Or you ask team members for feedback on your presentation skills. In a few months, you will have sharpened your presentation skills and may be ready to produce behaviour like the current VP.

4. Motivation

All learning requires some degree of personal motivation. For observational learning, the observer must be motivated to produce the desired behaviour. Sometimes this motivation is intrinsic to the observer. Other times, motivation can come in the form of external reinforcement, rewards and punishments. Bandura identified several factors that can influence motivation in observational learning:

- If the observed model was rewarded for their actions.
- When the model is the same age and sex as the observer, or if the observer shares similar interests.
- If the observer is unsure of his/her own knowledge and skill.
- When the observer sees the model as being more knowledgeable and skilled.
- People who have a higher status.
- Authoritative people.
- If the situation is ambiguous and the observer is unsure what to do

By using the VP example again, motivation is intrinsic. You understand that the path to becoming VP at your company requires a certain skill set.

6.6.5. Types of models in social learning

1. Live model

A live model is a real person whose behaviour is observed directly by another person. The observer may learn and copy what the person does. For example, a child watches an older brother wash a car and later tries to wash the family car in the same way.

2. Symbolic model

A symbolic model is a person or character whose behaviour is observed through media such as television, films, books, or social media. For example, someone watches a cooking video and learns how to prepare a new meal by copying the person's steps.

3. Verbal model

A verbal model occurs when a person learns a behaviour from someone who explains or describes what to do. For example, a grandmother explains to a child how to plant beans, and the child follows her verbal instructions when planting them.

6.6.6. A few real-world examples of observational learning.

- A child watches their mother eat dinner with a fork. They observe the behaviour and quickly learn how to use a fork themselves.
- A high-school basketball player watches Stephen Curry shoot free throws. They observe details such as the number of ball dribbles and hand follow-through patterns, then try to mimic the behaviour themselves.
- A child watches their parent folding the laundry. They later pick up some clothing and imitate folding the clothes.
- A young couple goes on a date to an Asian restaurant. They watch others in the restaurant eating with chopsticks and copy their actions to learn how to use these utensils.
- A child watches a classmate get in trouble for hitting another child. They learn from observing this interaction that they should not hit others.
- A group of children plays hide-and-seek. One child joins the group and is not sure what to do. After observing the other children play, they quickly learn the basic rules and join in.
- A newer employee avoids being late to work after seeing a co-worker fired for being late.
- A new customer in a store learns the process for lining up and checking out by watching other customers.
- A customer in a clothing store learns the procedure for trying on clothes by watching others.
- A person in a coffee shop learns where to find cream and sugar by watching other coffee drinkers locate that area.

- A tenant sees a neighbour evicted for late rent payment and, as a result, consistently pays her rent on time.
- An inexperienced salesperson is successful at a sales meeting after observing the behaviours and statements of other salespeople.
- A viewer watches an online video to learn how to contour her foundation makeup, then buys makeup, and later tries the look herself.
- Drivers slow down when they see that another driver has been pulled over by a police officer.
- A bank teller watches their more efficient colleague to learn a better way of counting money quickly.
- A shy party guest watches a more popular person talk to different people in the crowd, and later can do the same thing.
- Adult children begin to act the same way that their parents did when they were young.
- A lost tourist watches a local person navigate the subway system and later can do it on their own.

6.6.7. Positive and Negative Impacts of Observational Learning Theory

Observational learning has the potential to teach and reinforce or decrease certain behaviours based on a variety of factors. Particularly predominant in childhood, observational learning can be a key part of how people learn new skills. And learn to avoid consequences.

However, there has also been concern about how this type of learning can lead to negative outcomes and behaviours. For example, During play with other children in the neighbourhood, a child starts using rude words and speaking disrespectfully to others after observing some of his peers behaving in this way. When the child returns home, an older family member notices the behaviour and calmly explains that speaking respectfully to others is important. The child observes how the adult speaks kindly and respectfully to other people in the family. Over time, the child begins to copy the adult's positive behaviour and reduces the use of rude language.

Observational learning is influenced by several factors that determine whether a person notices, learns, and eventually imitates a behaviour. People do not copy everything they observe. The behaviour of the model, the characteristics of the observer, the social environment, the consequences of behaviour, motivation, and the opportunity to practise can all affect observational learning.

1. Characteristics of the Model

The characteristics of the person being observed can influence whether the observer pays attention to and imitates the behaviour. People are more likely to imitate models whom they respect, admire, trust, or consider successful. Similarity between the model and the observer can also make imitation more likely. For example, a child may copy the respectful way an older sibling speaks to neighbours because the child admires and trusts the sibling.

2. Characteristics of the Observer

The characteristics of the person who is observing also influence learning. Age, interests, previous experiences, abilities, confidence, and personal needs can affect what a person learns from observing others. For example, a child who is interested in cooking may pay more attention when watching a family member prepare food and may be more willing to try it later.

3. Social Environment

People learn from the behaviours they observe in their families, peer groups, neighbourhoods, communities, and media environments. A positive environment can encourage positive behaviours, while a negative environment can expose people to undesirable behaviours. For example, a child may initially copy rude language from peers during play but may later adopt respectful language after observing adults at home communicating respectfully.

4. Consequences of the Behaviour

The consequences that a model receives can influence whether an observer imitates the behaviour. If the observer sees that a behaviour is rewarded, appreciated, or produces a positive result, they may be more likely to copy it. If the behaviour leads to an undesirable consequence, they may avoid it. This is known as vicarious reinforcement and vicarious punishment. For example, if a child sees another child being appreciated for helping an elderly neighbour, the child may also become willing to help others.

5. Motivation

Motivation influences whether a person decides to perform a behaviour after observing it. A person may observe and remember a behaviour but may not perform it unless they have a reason or desire to do so. For example, a young person who wants to become a good cook may be motivated to practise cooking after watching a family member prepare different meals.

6. Opportunity and Ability to Reproduce the Behaviour

Observing a behaviour does not always mean that a person can immediately perform it. The observer needs the necessary skills, physical or mental abilities, resources, and opportunities to practise the behaviour. For example, a child may watch an adult repair a bicycle but may need guidance and practice before being able to repair it independently.

6.6.8. Applications of Observational Learning

Observational learning has many practical applications in everyday life. It plays an important role in shaping behaviour in different settings.

1. Education

Teachers use observational learning by demonstrating behaviours and skills that students can imitate.

- Teachers model good behaviour, discipline, and communication
- Demonstrations help students understand difficult concepts

- Peer learning allows students to learn from each other

Example: A teacher demonstrates how to solve a mathematics problem, and students follow the same steps.

2. Parenting

Children learn many behaviours by observing their parents or caregivers.

- Parents act as primary role models.
- Children imitate language, habits, and emotional responses.
- Positive parenting promotes good behaviour.

Example: A child who sees parents showing kindness and respect is likely to develop similar behaviour.

3. Media Influence

Media (television, social media, movies) provide powerful models that influence behaviour.

- Positive media can promote learning and good values.
- Negative media can encourage harmful behaviours such as violence or risky habits.

Example: Children may imitate characters they see on TV or social media, including both good and bad behaviours.

4. Therapy

Observational learning is used in psychological treatment to help individuals learn new behaviours. Therapists use **modelling techniques** to teach appropriate behaviour. Clients observe and imitate positive behaviours. It is effective in treating fears, phobias, and social difficulties.

Example: A person with a fear of public speaking may observe others speaking confidently and gradually learn to do the same.

Application activity 6.6



Read this scenario and answer the questions:

Clara is 17 years old. Every evening at home, she observes her parents drinking alcohol. Although they often tell her that drinking is bad and she should avoid it, their actions show the opposite behaviour. One weekend, Clara attends a party where her friends are drinking beer. She remembers how her parents behave at home and notices that they do not face any negative consequences for drinking.

Question:

1. What is Clara most likely to do?
2. Explain your answer using the concepts of observational learning.

6.7. Cognitive Learning Theory (Jean Piaget)

Learning activity 6.7



Have you ever learned something new and later realised that your understanding of it had changed as you gained more experience? What made your thinking change?

6.7.1. Description and major concepts

a) Description

Cognitive Learning Theory explains how people learn by focusing on internal mental processes such as thinking, memory, perception, and problem-solving. It emphasises that learners are **active participants** who construct knowledge rather than simply receiving information. Jean Piaget, a Swiss psychologist, proposed that learning occurs through interaction with the environment. According to him, children build understanding step by step as they grow and experience the world around them.

This theory highlights that:

- Learning involves understanding, not just memorisation.
- Learners actively organise and interpret information.
- Knowledge develops through experience and reflection.

b) Major Contributors in Cognitive Learning Theory

1. Jean Piaget

- Developed the theory of cognitive development.
- Explained how children's thinking changes in stages.
- Introduced important concepts such as schema, assimilation, and accommodation.

2. Jerome Bruner

- Emphasised the importance of active learning.
- Developed the concept of discovery learning.
- Believed learners should explore and discover knowledge by themselves.
- Introduced the idea that learning is a continuous process, not limited to stages.
Example: A teacher guides students to discover a scientific principle through experiments instead of simply explaining it.

c) Major concepts in cognitive learning theory

i. Schema

A schema is a mental framework or structure that helps individuals organise and interpret information. **Example:** A child may have a schema of a "bird" as something that flies.

ii. Assimilation

Assimilation occurs when new information is added to an existing schema without changing it. **Example:** A child who knows birds can fly may call an aeroplane a “bird” because it fits their existing schema.

iii. Accommodation

Accommodation happens when a learner modifies an existing schema or creates a new one to fit new information. **Example:** The child later learns that aeroplanes are not birds and creates a new schema for “aeroplane.”

iv. Equilibration

Equilibration is the process of balancing assimilation and accommodation to achieve understanding. When learners encounter new information that does not fit their current schema, they feel confusion (disequilibrium). They then adjust their thinking to restore balance. **Example:** A student becomes confused when a bat is called a mammal, not a bird, and adjusts their understanding.

v. Discovery Learning

Jerome Bruner introduced discovery learning, a teaching approach in which learners actively construct knowledge by exploring and finding things out for themselves. Rather than being directly given answers, learners are guided to uncover concepts through activities, experiments, and problem-solving.

Example (classroom):

Students are provided with materials to investigate how plants grow, allowing them to discover the process through observation and inquiry instead of being explicitly told.

6.7.2. Principles of Cognitive Learning

Cognitive learning focuses on how people think, understand, remember, and solve problems. It emphasises internal mental processes rather than just observable behaviour. According to Jean Piaget, learning occurs when individuals actively construct knowledge based on their experiences.

Key Principles of Cognitive Learning are:

a) Learning is an Active Process

Learners do not simply receive information; they actively engage with it by thinking, questioning, and exploring. **Example:** A student solving a mathematics problem uses reasoning rather than just memorising steps.

b) Prior Knowledge Influences Learning

New information is understood based on what the learner already knows. **Example:** A student who understands basic addition will find it easier to learn multiplication.

c) Learning Involves Mental Processes

Cognitive learning includes processes such as attention, perception, memory, and problem-solving. **Example:** Remembering key points from a lesson requires attention and memory.

d) Learning is Goal-Oriented

Learners are motivated to learn when they have clear goals. **Example:** A student studies harder when preparing for an exam.

e) Learning Involves Organisation of Information

People learn better when information is structured and meaningful. **Example:** Using diagrams, charts, or summaries helps students understand complex topics.

6.7.3. Four stages of cognitive development

Jean Piaget proposed that children pass through four stages of cognitive development as they grow.

a) Sensorimotor stage (0–2 years)

- Learning occurs through senses and physical actions.
- Infants explore the world by touching, seeing, and hearing.
- Development of object permanence (understanding that objects still exist even when not seen). **Example:** A baby looks for a toy even after it is hidden.

b) Preoperational stage (2–7 years)

- Development of language and imagination.
- Thinking is egocentric (difficulty seeing others' perspectives).
- Limited ability to think logically. **Example:** A child believes the moon follows them wherever they go.

c) Concrete Operational Stage (7–11 years)

- Logical thinking develops but is limited to concrete situations.
- Understanding of conservation (quantity remains the same despite changes in shape). **Example:** A child understands that water poured into different containers still has the same amount.

d) Formal Operational Stage (12 years and above)

- Ability to think abstractly and logically.
- Development of problem-solving and critical thinking skills.
- Ability to imagine possibilities and think about the future. **Example:** A student can solve complex algebra problems or discuss moral issues.

6.7.4. Benefits of cognitive learning

Cognitive learning is important because it helps people understand information, think effectively, solve problems, and develop skills that are useful in everyday life. These benefits can influence how people make decisions, remember experiences, and deal with different situations.

a) Promotes deep understanding

Cognitive learning helps people understand why and how things happen, rather than simply memorising information. Example: A person who understands why certain foods need to be stored properly is more likely to remember how to keep food safe.

b) Improves Problem-Solving Skills

Cognitive learning helps people analyse situations, identify problems, and find appropriate solutions. Example: When a person discovers that their bicycle is not working, they think about what might be causing the problem and try different ways to fix it.

c) Enhances Memory Retention

Information learned through understanding is often remembered for longer because people can connect it with their previous experiences. Example: A person may remember how to prepare a meal more easily when they understand the purpose of each ingredient and step.

d) Encourages Independent Learning

Cognitive learning encourages people to ask questions, seek information, and learn from their own experiences. Example: A person who wants to learn how to repair a household item may look for information, observe others, and practise until they understand how to do it.

e) Develops Critical Thinking Skills

Cognitive learning helps people evaluate information, consider different ideas, and make informed decisions. Example: Before believing information shared on social media, a person may compare it with information from other reliable sources before deciding whether it is true.

f) Supports Lifelong Learning

Cognitive learning develops skills such as thinking, problem-solving, decision-making, and adapting to new information, which remain useful throughout life. Example: A person can use these skills to learn a new job, manage household responsibilities, or make better decisions when facing unfamiliar situations.

Application activity 6.7



Clara believes that all fruits are sweet. One day, she tastes a lemon and finds it sour. Later, she explores more fruits and learns which are sweet and which are sour.

1. Identify which concepts of cognitive learning theory are shown in this scenario.
2. Explain how Clara's thinking changed.
3. How could a teacher use discovery learning to help Clara learn about fruits faster?

6.8. Insight Learning (Wolfgang Köhler, Max Wertheimer, Kurt Koffka)

Learning activity 6.8



Imagine you are trying to reach a basket of bananas hanging on a high wall in your kitchen or courtyard. You try stretching your hand, but it's too high. Then suddenly, you notice a stool nearby and realise you can stand on it to reach the basket.

Discuss

1. What new idea helped you finally reach the basket of bananas when stretching your hand didn't work?
2. Did the solution come gradually or suddenly? What does that tell you about how you learned it?

6.8.1. Description of insight learning

Insight learning is a type of learning in which an individual suddenly realises a solution to a problem without trial-and-error experimentation. Instead of learning gradually, the solution comes as an **"aha" or sudden understanding**, often after careful thought or observation.

This theory emphasises that learning is **mental rather than purely behavioural**, meaning the learner processes information internally and reorganises it to solve a problem. **Example:** A student struggling to solve a math problem suddenly understands the correct method after thinking about it differently, without trying multiple random approaches.

6.8.2. Major Contributors

a) Wolfgang Köhler

- Conducted famous experiments with chimpanzees.
- Observed that chimps could solve problems, such as reaching bananas, by suddenly combining ideas (stacking boxes, using sticks) rather than repeated trial and error.
- Demonstrated that animals, like humans, can learn through insight.

b) Max Wertheimer

- One of the founders of Gestalt Psychology, which emphasises holistic perception.
- Believed learning occurs when individuals see the "whole picture" rather than just reacting to separate elements.
- Insight comes from understanding relationships and patterns.

c) Kurt Koffka

- Contributed to the development of Gestalt theory.
- Emphasised that problems are solved when the mind reorganises information and sees connections.

- Insight is achieved when the learner perceives the structure of the problem clearly.

6.8.3. Key Features of Insight Learning

- Learning occurs suddenly, not gradually.
- It involves thinking, understanding, and reorganising mental information.
- Trial and error may still occur, but the “aha” moment is crucial.
- Insight learning often leads to long-lasting understanding.

Example: A student trying to figure out how to divide a task among team members suddenly realises the most efficient way to do it, without trying all possibilities randomly.

6.8.4. Applications of Insight Learning Theory

Insight Learning, as described by Wolfgang Köhler, emphasises sudden problem-solving through understanding rather than trial and error. This approach can be applied in many areas:

1. Education

- Teachers can design activities that encourage students to think deeply and solve problems independently.
- Instead of giving step-by-step instructions, students are guided to discover solutions. **Example:** A science teacher gives students materials to build a simple machine, allowing them to figure out the most efficient way to make it work. The moment a student realises the solution is an example of insight learning.

2. Problem-Solving at Home

- Students and children can use insight learning to solve everyday challenges.
- Encourages creative thinking and mental reasoning rather than repeated trial and error. **Example:** A farmer figures out how to water plants more efficiently by connecting two buckets creatively.

3. Workplace and Skills Training

- Insight learning can be applied in practical skill development, where employees or learners figure out solutions to problems efficiently.

Help learners develop critical thinking and innovation. **Example:** An apprentice electrician solves a wiring problem by applying their understanding of electrical circuits and effectively using the available tools, rather than relying on trial-and-error methods.

4. Therapy and behaviour change

- Therapists use insight learning to help clients understand the root of problems and find solutions through reflection.
- Promote self-awareness and mental restructuring. **Example:** A counsellor helps a student recognise patterns causing anxiety. The student suddenly understands a better way to manage stress, leading to behavioural change.

5. Everyday life and creativity

- Insight learning helps in creative thinking, innovation, and planning.
- Encourages learners to see the “whole picture” and connect ideas. Example: A student suddenly realises a clever way to organise books, design a poster, or complete a project more efficiently.

Application activity 6.8



1. Your teacher gives you a puzzle or riddle, and you suddenly understand the solution without trying every possibility.
 - a) Describe what you did to solve it.
 - b) Which type of learning does this demonstrate?
2. Imagine you want to fix a leaking tap at home but don't know how. After thinking, you realize you can tighten a bolt using a wrench you found nearby.
 - a) Explain how you solved the problem.
 - b) How is this similar to the banana basket example?
3. Your group is asked to build a simple bridge from sticks and glue. After observing and thinking, someone comes up with a new way to place the sticks that works perfectly.
 - a) Who in your group showed insight?
 - b) How did their sudden idea help solve the problem?
4. Think of a time when you suddenly figured out a better way to carry books, organise your desk, or manage chores.
 - a) Describe the situation and your sudden solution.
 - b) How does insight learning explain your experience?

6.9. Constructivism (Jerome Bruner, J. Piaget)

Learning activity 6.9



Have you ever learned something by figuring it out yourself instead of being told the answer? How did it feel?

Constructivism is a learning theory that emphasises that learners actively construct their own knowledge rather than passively receiving information. Knowledge is built through experience, reflection, and interaction with the environment. Learners are not empty vessels; they interpret, explore, and make sense of new information based on what they already know. Example: A student learning about plants observes different plant species in a garden, asks questions, and creates their own conclusions about growth conditions.

Key Principles of Constructivism

- **Active Learning:** Learners actively participate in the process of constructing their own knowledge
- **Prior Knowledge:** New knowledge is linked to what learners already know.
- **Learning by Discovery:** Learners explore and discover information themselves.
- **Social Interaction:** Learning is enhanced through collaboration and discussion.
- **Contextual learning:** Knowledge is better understood when it is related to real-life situations.

1. Jean Piaget's constructivist theory

Jean Piaget (1896–1980) emphasised cognitive development. According to Jean Piaget, children construct knowledge as they grow and interact with their environment. Learning occurs when learners experience, reflect, and adapt.

The main key ideas discussed in the previous lesson are Schema: a mental framework to organise information. Assimilation: Incorporating new knowledge into existing schemas. Accommodation: Modifying existing schemas for new knowledge. And Equilibration: Balancing assimilation and accommodation to achieve understanding.

Example: A child who thinks all birds fly sees a penguin and realises some birds do not fly. They accommodate their schema of "bird" to include flightless birds.

2. Jerome Bruner's Constructivist Theory

Jerome Bruner (1915–2016) emphasised learning by discovery. According to him, Learners construct knowledge actively through exploration, not memorisation. Bruner believed learners should understand the underlying structure of concepts.

Key ideas are:

- **Spiral Curriculum:** Concepts should be revisited in increasing complexity over time.
- **Discovery Learning:** Students learn best by exploring and solving problems themselves.
- **Representation of Knowledge:** Enactive: Learning through action (hands-on), Iconic: Learning through images and visual representation, and Symbolic: Learning through abstract symbols (words, numbers). Example: Students exploring how to balance a scale with different weights learn the principle of equilibrium by experimenting for themselves.

Comparison of Piaget and Bruner

Aspect	Piaget	Bruner
Focus	Cognitive development stages	Discovery learning and curriculum design
Learning	Construct knowledge through interaction with the environment	Construct knowledge actively through exploration
Teaching Approach	Teacher facilitates, child constructs	Teacher guides discovery, encourages exploration

Aspect	Piaget	Bruner
Example	A child adjusts their thinking about birds after seeing penguins	A student experiments with weights to understand balance

Applications of Constructivism

a) Education

- Teachers design hands-on activities, experiments, and projects.
- Students explore, discuss, and reflect rather than just memorise.

Example: Science projects, building models, or role-playing historical events.

b) Parenting

- Parents encourage children to ask questions and find solutions independently.
- Learning by experience is valued over rote instruction. Example: A child learns to cook a simple meal by observing and experimenting under guidance.

c) Workplace

- Employees are encouraged to solve problems creatively, learn through experience, and share knowledge. Example: A shopkeeper reorganises the store layout after observing customer behaviour.

d) Therapy and personal development

- Insight, reflection, and guided problem-solving help learners understand themselves and others better.

Benefits of Constructivism

- Promotes deep understanding instead of memorisation.
- Develops problem-solving and critical thinking skills.
- Encourages independent learning and creativity.
- Connects learning to real-life situations.
- Supports lifelong learning and adaptability.

Application activity 6.9



1. Think about how you learn at school or at home. How does Piaget's idea of learning through stages and experience compare to Bruner's idea of learning by discovering things yourself? Use examples from your own life, like learning to cook, solving a problem, or doing a school project.
2. Give examples of how you or your classmates learn best by doing or discovering things instead of just listening to the teacher. For example, building a model in science, organising your study notes, or helping your parents with chores. How does this show constructivist learning?

3. Think of a time when you figured something out on your own, like understanding a math problem, learning a new cooking method, or fixing something at home. Explain what you already knew (schema), how you adjusted your thinking (assimilation or accommodation), or how you discovered the solution (discovery learning)."

6.10. Social Constructivism (Vygotsky)

Learning activity 6.10



1. Think of a time when you learned a new skill with help from your teacher, a parent, or a friend, for example, learning to solve a difficult math problem, cook a local dish, or use a computer program. How did their guidance help you understand or complete the task?
2. When you work with classmates on a project, like building a model, preparing a class presentation, or solving a science problem, how does discussing ideas and helping each other make it easier to understand compared to working alone?
3. Give an example of something you can now do on your own, which you couldn't do before without help, like riding a bicycle, planting a small garden, or using a smartphone app. How did guidance from someone help you reach this level?

Social Constructivism is a learning theory that emphasises the social context of learning, where knowledge is constructed through interaction with others, such as teachers, peers, and family. Learning occurs when learners collaborate, communicate, and receive guidance, and understanding is developed actively, not passively absorbed. Lev Vygotsky (1896–1934), a Russian psychologist, was a major contributor to this theory. He focused on the social and cultural influences on learning and believed that learning leads to development, meaning that interaction with more knowledgeable individuals helps learners achieve higher levels of thinking.

As Vygotsky emphasised, we learn through interaction with others before we can understand on our own. For example, a student learning a new computer program may discuss the steps with a friend and then apply them independently.

a) Key Concepts of Vygotsky's Theory

1. **Zone of Proximal Development (ZPD):** The ZPD is the gap between what a learner can do alone and what they can do with guidance from a more knowledgeable person. Learning is most effective when instruction is targeted within this zone. Example: A student can solve simple algebra problems independently but needs guidance to solve more complex equations. With teacher support, they gradually learn to solve them alone.

2. **Scaffolding:** Temporary support given by teachers, peers, or parents to help the learner achieve a task within their ZPD. Support is gradually removed as the learner becomes competent. Example: A teacher guides students step by step in writing an essay. As students become confident, the teacher gives less help.
3. **More Knowledgeable Other (MKO):** Someone who has more knowledge or skill than the learner and can provide guidance. MKO can be a teacher, parent, peer, or even a digital tool.
4. **Language and Thought:** Vygotsky emphasised that language is a key tool for thinking and learning. Interaction through discussion, questioning, and dialogue enhances cognitive development.

b) Applications of Social Constructivism

1. In the Classroom

- Encourage group work, peer teaching, and collaborative projects.
- Teachers act as facilitators or guides, not just providers of information.
- Use scaffolding to support learners in the ZPD.

Example: Students work in groups to solve a real-life problem, like designing a small water filter. The teacher guides them through the process, gradually reducing support.

2. At Home

- Parents can guide children in tasks they cannot do alone, like cooking, managing money, or using technology.
- Encourage children to ask questions and discuss ideas, helping them construct understanding.

Example: A parent teaches a child to bake a cake, giving guidance initially, then allowing the child to do it independently next time.

3. Workplace / Real-Life Learning

- Social interaction and mentorship improve skills.
- Learning new work skills is most effective when guided by an experienced colleague (MKO).

Example: An apprentice carpenter learns complex measurements and cutting techniques by observing and being guided by a master carpenter before practising independently.

c) Benefits of Social Constructivism

- Promotes collaborative learning and teamwork.
- Encourages critical thinking and problem-solving.
- Supports personalised learning through scaffolding in the ZPD.
- Enhances communication skills and social interaction.
- Prepares learners for real-life tasks by connecting learning to social and cultural contexts.

Application activity 6.10



1. Think about how you learn new things in school or at home. How does Vygotsky's idea that we learn through interaction with others explain this? Give examples, such as working in a group on a science project, learning from a teacher, or being guided by your parents to do a task at home.
2. Recall a time when you learned something new with help from a teacher, parent, sibling, or friend. Describe what the task was and how they guided you. How does this experience show your Zone of Proximal Development (ZPD) and the support (scaffolding) they gave you?
3. Think about learning alone versus learning with others. How is Vygotsky's idea that social interaction helps learning different from Piaget's idea that we build knowledge mostly by exploring on our own? Give examples from your life, like solving a math problem with a friend or working on a project with classmates.

6.11. End unit assessment



Question 1: Knowledge, Skills & Attitude

Think about a time when you were rewarded for doing something correctly, like helping your parents at home, completing your homework, or behaving well in class. Which behavioural learning theory explains why rewards and consequences make you more likely to repeat a behaviour? Explain your answer and give an example from your own life.

Question 2: Skills & Attitudes

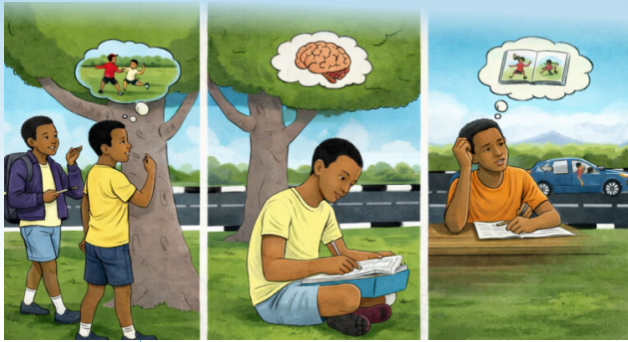
Reflect on different ways you have learned new behaviours or skills, such as learning to ride a bicycle, memorising a song, or helping on the farm. How do these experiences show the different types of behavioural learning, such as learning by association, reinforcement, or sudden understanding? Compare and contrast them using examples from your daily life.

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UNIT 7

MEMORY



Key Unit Competence:

Develop a basic understanding of the nature and characteristics of memory processes and systems.

Learning objectives

By the end of this unit, you should be able to:

1 Knowledge and understanding

- Identify the main components and functions of human memory systems.
- Explain the nature, features, and processes of sensory, short-term, and long-term memory
- Describe theories of forgetting and the factors that influence memory accuracy.
- Illustrate memory processes and retrieval using examples from everyday life.

2 Skills

- Use examples of everyday life to understand how memory works.
- Apply findings from memory research to explain memory retention and forgetting.
- Apply memory strategies to improve recall of information in real-life or academic tasks.
- Analyse real-life situations to identify the type of memory involved.
- Analyse scenarios to explain the information processing.
- Create a visual representation illustrating the processes of encoding, storage, retrieval, and forgetting.

3 Attitudes and values

- Demonstrate a responsible and proactive attitude in supporting the proper functioning of memory.
- Accept people with memory loss (forgetting).

Introduction to the unit

Have you ever walked into a room and forgotten why you went there, yet you can easily remember a song you learned years ago? Or struggled to recall a name during a conversation, only for it to suddenly come back to you later? These everyday experiences show how memory works in fascinating and sometimes surprising ways. Memory is not just about storing information; it is an active process that involves encoding, storing, and retrieving what we learn and experience.

This unit introduces Memory as a fundamental process in human thinking and learning, exploring how we encode, store, and retrieve information in everyday life. You will explore how memory works through key theories such as the information processing model and the levels of processing theory, which explain how information is handled and retained, as well as different types of memory, including implicit memory (unconscious memory) and explicit memory (conscious recall). The unit also explores why we forget, considering explanations such as interference theory, decay theory, retrieval failure, cue-dependent forgetting, repression/motivation, and displacement from short-term memory. In addition, you will learn practical strategies for enhancing memory, enabling you to improve your ability to learn and retain information effectively in everyday situations. By the end of this unit, you will have developed a basic understanding of the nature and characteristics of memory processes and systems, while building essential skills to strengthen your learning and memory.

Introductory activity



Think about your own life for a moment...

Have you ever quickly remembered a friend's phone number, but forgotten what you studied yesterday? Or found it easy to recall a favourite song, yet struggled to remember a teacher's explanation? Your brain is constantly receiving, storing, and retrieving information, but it does not treat all information the same way.

Now,

- In your own words, what does memory mean to you?
- Think about things you remember automatically (like riding a bicycle) and things you consciously recall (like answers in an exam). What different types of memory can you identify from your life?
- Imagine how information moves in your brain from the moment you see or hear something, to the moment you remember it later. How do you think information is processed in memory?
- Reflect on yourself and your classmates: Why do you think some people remember things more easily than others?

7.1. Theory of Memory: Information processing model

Learning activity 7.1



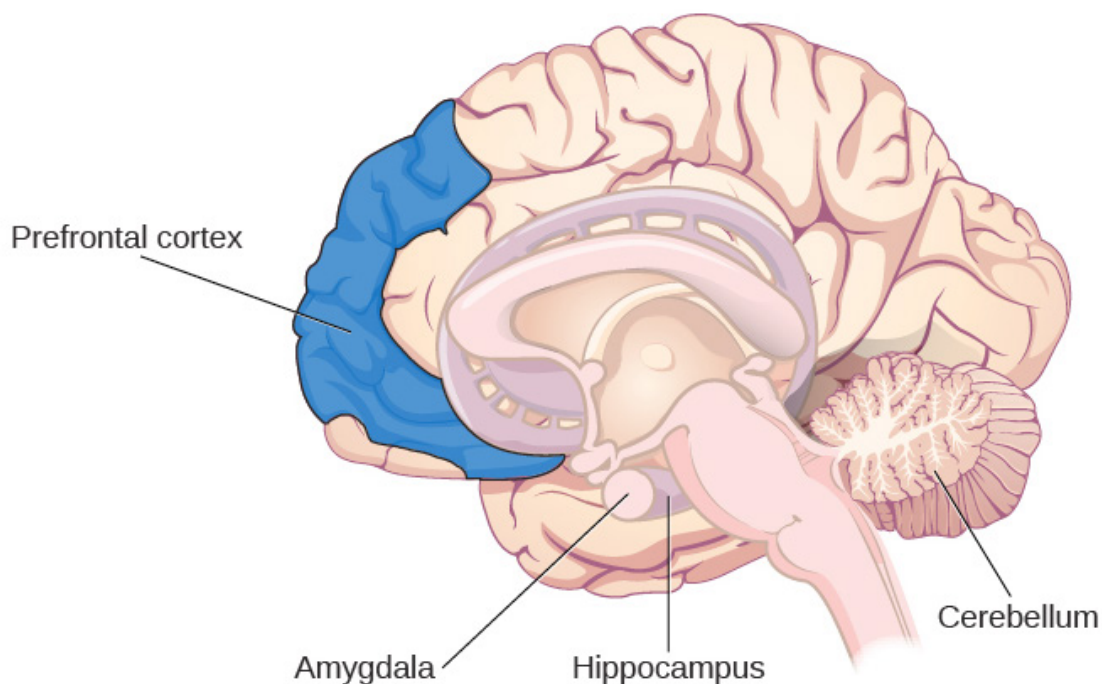
Close your eyes for a moment and imagine this: you are in class, your teacher explains a new concept, you listen, take notes, and later you can recall it during a test. But have you ever wondered what happens inside your mind from the moment you hear that information to the moment you remember it?

Memory is the ability to store, retain, and recall information and experiences. It involves encoding information, storing it in the brain, and retrieving it when needed. However, memory is not perfect; people can forget or misremember things. For example, after studying, you may retrieve stored information to answer questions in an exam.

Key memory structures in the brain

The main parts of the brain involved with memory are the amygdala, hippocampus, cerebellum, neocortex, and basal ganglia.

- The amygdala regulates emotions and influences how memories are stored.
- The hippocampus is important for forming and organising memories, especially recognition and spatial memory, and helps transfer information to long-term memory.
- The prefrontal cortex supports thinking, concentration, and personality.
- The cerebellum and basal ganglia are involved in movement and implicit (unconscious) memory.
- The neocortex helps store long-term explicit memories such as facts and events.



Memory can also be understood as a process that occurs in stages. Information first enters **sensory memory**, then moves to **short-term memory**, and finally to **long-term memory**. Not all information reaches long-term memory; this depends on attention and how the information is processed.

Psychologists describe memory using three main processes:

i. Encoding

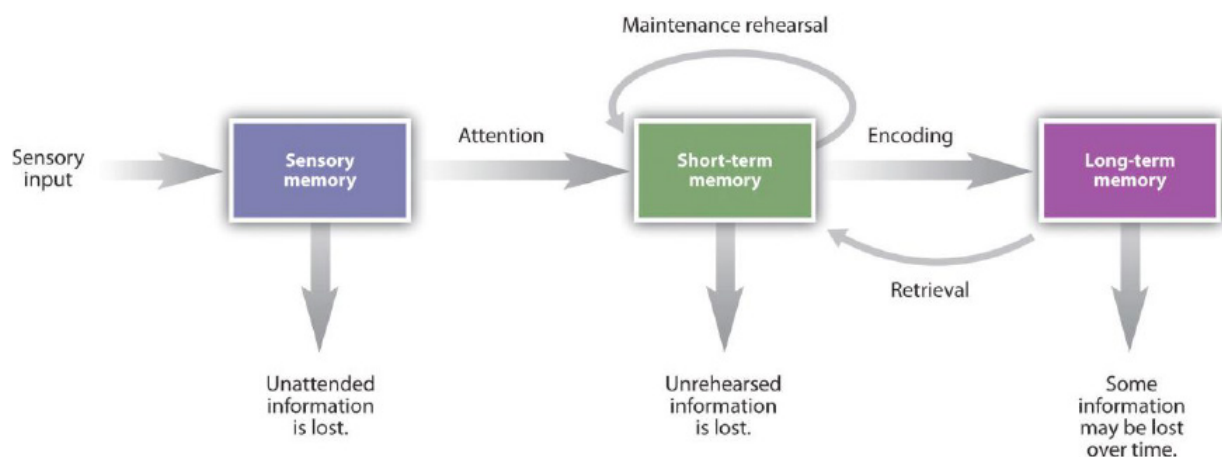
Encoding is the process of converting information into a form that can be stored. It can be visual (images), acoustic (sounds), or semantic (meaning). Attention and understanding improve encoding, while distraction can lead to forgetting. Meaningful learning and making connections (elaborative encoding) help improve memory.

ii. Storage

Storage refers to keeping information over time. Short-term memory has a limited capacity and lasts a short time (about 0–30 seconds), while long-term memory has a large capacity and can last a lifetime. Organising or “chunking” information helps improve storage.

iii. Retrieval

Retrieval is the process of bringing stored information back into awareness. Sometimes retrieval is easy, and sometimes it fails, especially if information was not well encoded. Memory improves when the context or state during retrieval is similar to when the information was learned (context-dependent and state-dependent learning).



Information processing model

Source: Adapted from Atkinson, R. C., & Shiffrin, R. M. (1968). *Human memory: A proposed system and its control*.

Application activity 7.1



1. When you remember something emotional, like a happy or scary moment, which parts of your brain do you think are involved, and how might emotions affect your memory?
2. Imagine how information travels in your mind from the moment you see, hear, or learn something. How do you think it moves through sensory memory, short-term memory, and long-term memory?
3. Have you ever forgotten something because you were distracted? What does this tell you about the role of attention and encoding in memory?
4. What strategies do you personally use to help you remember things? Do you use images, repetition, or meaning? Which one works best for you and why?
5. Think about how you store information: why do you think some memories last only a few seconds while others stay for years?
6. Have you ever known something but failed to recall it at the right moment? Why do you think retrieval failure happens?
7. Do you think you remember things better when you are in the same place or mood as when you learned them? Can you share an example from your own experience?

7.2. Theory of Memory: Levels of processing memory

Learning activity 7.2



Two students in Bwiza Secondary School are preparing for the same exam. Student A reads the note repeatedly, focusing on how the words look and trying to memorise them. Student B tries to understand the meaning, discusses with friends, and connects the topic to real-life situations (like examples from their community or school).

Discuss

1. Which student do you think will remember more? Why?
2. What exactly did Student B do differently?

Have you ever read something repeatedly but still forgotten it, while easily remembering something you truly understood? This shows that memory is influenced not only by how long information is stored, but also by how deeply it is processed. The Levels of Processing Theory, developed by Fergus Craik and Robert Lockhart, explains that the strength of a memory depends on the depth of processing. The deeper and more meaningful the processing, the better the information is remembered.

Information can be processed at three main levels:

1. Shallow Processing (Structural Level)

This is the most basic level, focusing on the physical features of information. Example: Noticing whether a word is written in capital letters or its shape.

It involves little thinking or understanding. Memories formed are weak and easily forgotten.

2. Intermediate Processing (Phonemic Level)

This level focuses on the sound of information. Example: Remembering how a word sounds or rhymes with another word.

It involves more attention than shallow processing. Memory is moderate, but not very strong.

3. Deep Processing (Semantic Level)

This is the deepest level, focusing on the meaning of information. Example: Understanding a concept, relating it to real life, or explaining it in your own words.

It involves analysis, thinking, and making connections. Memories formed are strong, long-lasting, and easy to recall. The deeper the processing, the stronger the memory. Information that is meaningful is remembered better than information processed superficially.

Factors that Promote Deep Processing

Memory can be improved when learners:

- Relate new information to their own experiences.
- Ask questions and think critically.
- Organise ideas into meaningful patterns.
- Use examples and explanations.
- Connect new knowledge with what they already know.

Simply repeating information is less effective than understanding it. Active learning (discussion, explanation, and application) improves memory. Teachers should encourage meaningful learning, not just memorisation.

Application activity 7.2



Three employees are trained in a new procedure. Alice, an office worker, quickly reads the instructions and focuses only on how they are written, without attempting to understand their meaning. Jean, a technician, listens carefully and repeats the instructions aloud to remember them. Grace, a nurse, goes further by asking questions, understanding each step, and relating the procedure to real situations in her work. After a few days, Alice forgets most of the steps, Jean remembers only some parts, while Grace remembers everything clearly and performs the task correctly. This situation shows that people remember information better when they process it deeply by understanding its meaning rather than just seeing or repeating it.

Discuss

1. Imagine you are working like Alice, Jean, or Grace. Which of these three are you most similar to when learning something new, and why?
2. Think about a time when you had to learn something important (at school, an internship, or in daily life). Did you focus on memorising, repeating, or understanding? How did that affect what you remembered later?
3. If you were in Grace's position, what specific actions would you take to understand the procedure deeply and remember it for a long time?

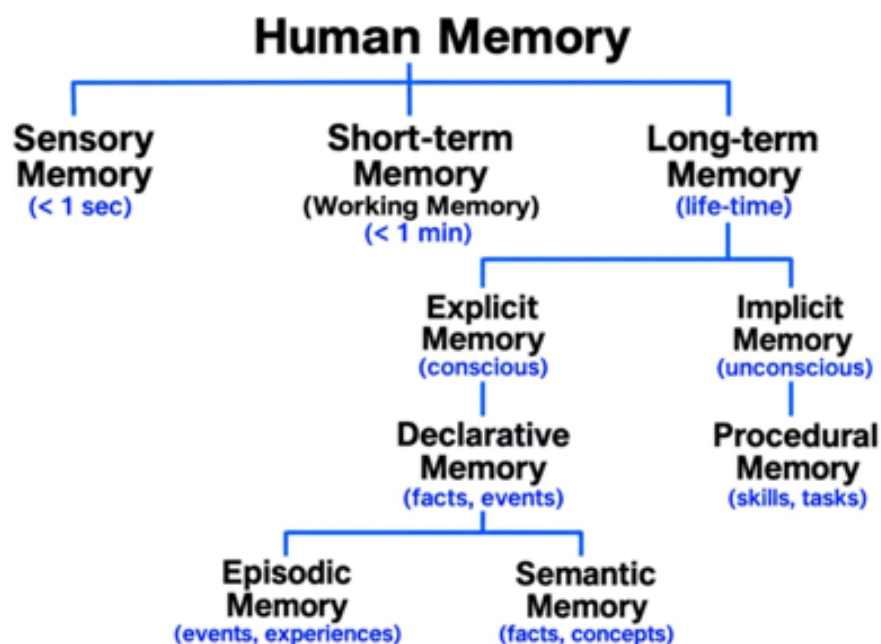
7.3. Types of memory

Learning activity 7.3



1. Think about something you can do automatically, like riding a bicycle or typing on your phone without thinking. Which type of long-term memory are you using that does not need conscious effort?
2. When you perform a skill without actively thinking about it (for example, greeting a friend or tying your shoes), what type of memory are you using?
3. The kind of memory that works automatically without you being aware of it has another name. What is another name for this type of memory?
4. When you practice something many times until it becomes a habit (like writing or playing football), which part of the brain is mainly responsible for controlling this automatic memory?

Memory in psychology is categorized into sensory, short-term, working, and long-term systems, with long-term memory further divided into explicit and implicit types.



7.3.1. Sensory memory

Sensory memory is the briefest form of memory, capturing raw impressions from the senses immediately after a stimulus occurs. It includes iconic memory for visual stimuli, echoic memory for auditory stimuli, and other sense-specific forms like olfactory and haptic memory. Sensory memory acts as a buffer, holding information for a few seconds unless attention is directed toward it, allowing the brain to decide what to process further.

Key Characteristics of Sensory Memory

Sensory memory lasts for a very short time, from a fraction of a second to a few seconds, depending on the sense involved. It has a large capacity and briefly holds detailed information from the environment. It works automatically and without conscious control, serving as a temporary bridge between sensory input and further perception or processing.

Main Types of Sensory Memory

Iconic Memory: Briefly stores visual information, usually for less than a second.

Echoic Memory: Briefly stores auditory information and may last for a few seconds.

Haptic Memory: Briefly stores information received through touch, usually for about a second.

7.3.2. Short-term Memory

Short-term memory is the brain's capacity to hold a small amount of information in an active state for a brief period, typically around 20 to 30 seconds.

Key characteristics of Short-Term Memory

Short-term memory holds information for only a few seconds unless it is actively repeated or rehearsed. Its capacity is limited, allowing a person to hold only a small number of items or chunks of information at a time, often estimated at about 3–5 items. It serves as a temporary mental space for information needed immediately, such as remembering a phone number long enough to enter it.

7.3.3. Long-term memory

Long-term memory is a nearly permanent storage space for learned information and experiences. It can hold memories for years. Long-term memory moves short-term memories to this larger space. Your memories may last a long time, but your ability to retrieve them may be more challenging with age, certain conditions.

Types of long-term memory

Explicit memory (declarative memory): This is the storage of facts, events and locations. You're always adding to this storage space and recalling this information when you need it.

Implicit memory (Nondeclarative memory): This is the storage of learned skills and habits or relationships. You aren't consciously aware of when you're learning this information, but you can recall it at any time.

Application activity 7.3



1. Using example, explain the three types of memory

7.4. Implicit Memory

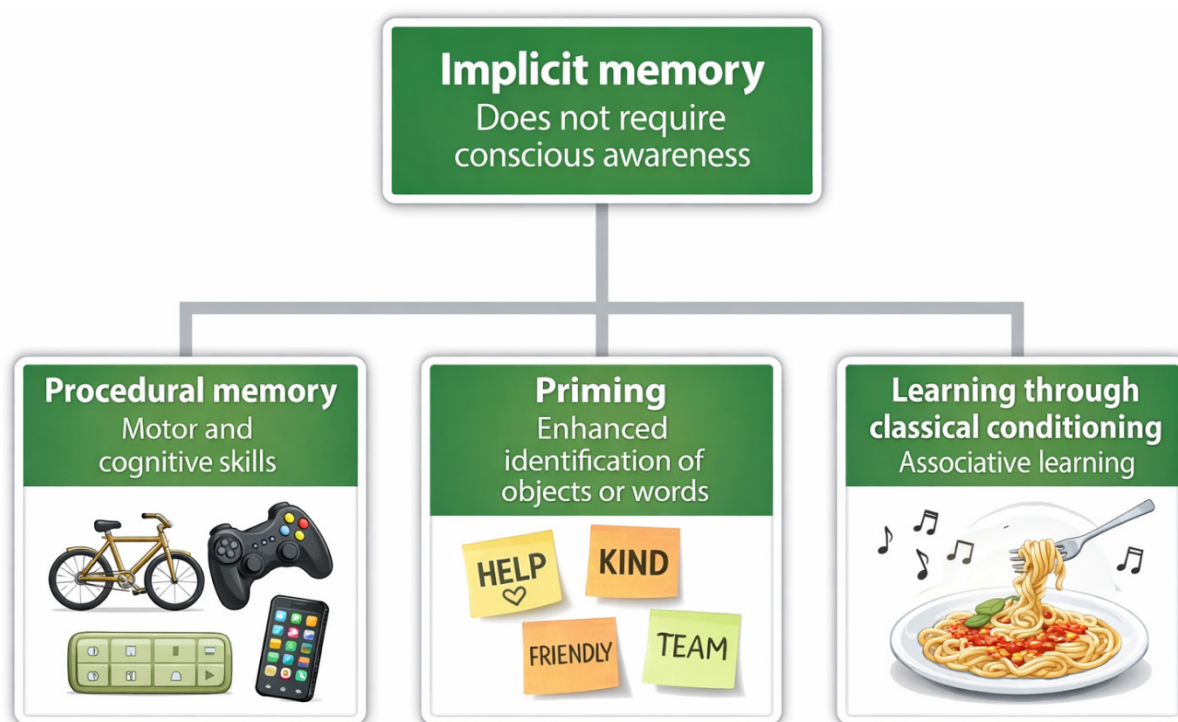
Learning activity 7.4



What do you know about implicit memory?

Implicit (non-declarative) memory refers to knowledge that we cannot consciously access (we are not aware of it). However, it is very important because it directly affects how we behave in everyday life.

There are three main types of implicit memory: procedural memory, classical conditioning effects, and priming.



1. Procedural memory: This is memory for actions or how to do things. It is our knowledge of skills. For example, when we walk, speak English, dial a phone, ride a bicycle, or play a video game, we are using procedural memory.

Procedural memory allows us to do complex activities even if we cannot explain how we do them. For example, it is hard to explain how to ride a bicycle; you learn it by practising, not by explanation.

This also explains how babies learn. Infants learn to crawl, walk, and talk. These are skills (procedures), and they develop them easily. But when we grow up, we do not remember learning these skills.

2. Classical conditioning (associative learning): This type of implicit memory happens when we learn, without effort or awareness, to connect two things. For example, you may hear music in a restaurant and feel happy because you associate the sound (music) with food. At first:

- Food makes you feel happy naturally.
- Later, the sound of the restaurant alone can make you feel happy

So, you learned this connection without thinking about it.

3. Priming: Priming is when our behaviour changes because of something we saw or experienced recently or often. For example, if people see words like "help," "kind," or "friendly," they may behave more kindly afterward. In daily life, priming happens often:

- Seeing your country's flag may make you feel patriotic.
- Seeing a rival school may make you feel competitive.

These effects occur without our conscious awareness, yet they still influence our behaviour.

Application activity 7.4



With examples, explain the three main types of implicit memory.

7.5. Explicit/Declarative Memory

Learning activity 7.5



1. When you try to remember your lesson during a test or recall your last holiday, which type of memory are you using?
2. In your own words, how would you explain this type of memory in a few words?
3. This type of memory also has another name. What is another name for it?
4. When you remember facts (like definitions) or personal experiences (like your birthday), what type of memory is explicit?

When we test memory by asking someone to think and remember information, we are measuring explicit memory. Explicit (or declarative) memory is knowledge or experiences that we can consciously recall. There are two types of explicit memory: episodic memory and semantic memory.

1. Episodic memory

This is a memory of personal experiences. Examples: Remembering your graduation day, recalling your first day at a new job, A teacher remembering a specific lesson that went well, A nurse recalling a particular patient they treated, A driver remembering a difficult trip or accident, and A shop worker remembering a busy day with many customers.

2. Semantic memory

This is a memory of facts, knowledge, and concepts. Examples: Knowing that $2 + 2 = 4$, knowing the meaning of a word, A teacher knowing subject content (e.g., definitions in Psychology), A doctor knowing symptoms of diseases, A mechanic knowing parts of a car, and A bank worker knowing financial rules and procedures.

How explicit memory is tested

Explicit memory is measured when people are asked to actively remember information.

1. Recall (no or few clues)

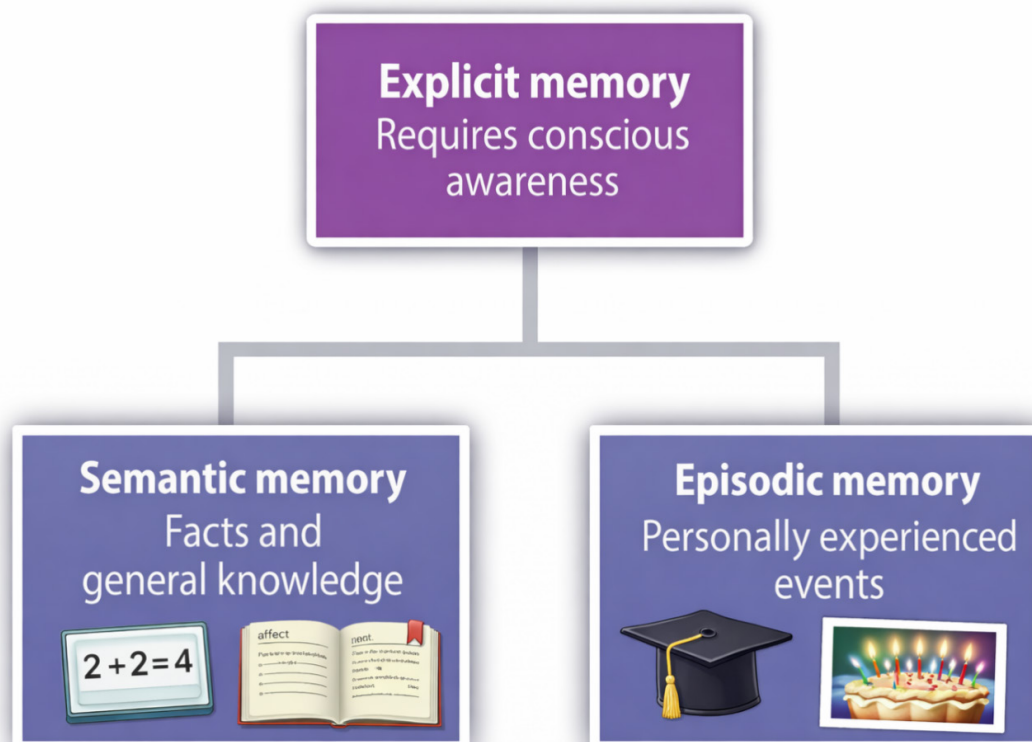
You must bring the information from memory. Example: Essay questions in exams, A teacher explaining a topic without notes, A lawyer recalling laws during a case, A manager remembering instructions during a meeting, Free recall: No help (e.g., "List all you remember"), and cued recall: With hints (e.g., fill-in-the-blank questions)

2. Recognition (choosing the correct answer)

You identify the correct answer from the options. Example: Multiple-choice questions, A security officer recognising a suspect's face, A cashier recognising a product or price, and A student teacher choosing the correct answer in a test. Recognition is usually easier than recall because you only need to recognise the correct answer, not produce it.

3. Relearning

This measures how fast you learn something again after forgetting it. Example: Relearning a language you studied before, A driver quickly relearning routes after a break, A nurse refreshing medical procedures, A teacher relearning content before teaching a new class, and A musician relearning a song faster the second time. Relearning shows that even if we forget something, it is not completely lost.



Application activity 7.5



1. Think about your own life experiences (like your birthday, school events, or things you have learned in class). How would you explain the types of explicit memory (episodic and semantic) using your own examples?
2. When you study, remember, and later recall information in class, how can you explain memory in terms of its types, processes (encoding, storage, retrieval), and stages (sensory, short-term, long-term)?
3. When you perform automatically (like walking or writing) and when you consciously remember facts (like in exams), which parts of the brain are important for implicit memory and which parts play a key role in explicit memory?

7.6. Forgetting: Interference theory and trace decay theories of forgetting

Learning activity 7.6



Have you ever studied for a test but later forgotten some answers, while remembering others very well?

- a) Why do you think this happens?
- b) Is it because the memory fades over time, or because other information confuses it?

7.6.1. The interference theory

Interference occurs when information stored in similar categories becomes mixed up and begins to affect another. There are two types of interference: **proactive interference** and **retroactive interference**. This means that our existing memories can affect new learning, either in a forward or backward way.

Retroactive interference occurs when learning new information makes it harder to remember information learned earlier. For example, if you learn one programming language and later learn another similar one, you may start making mistakes in the first language. Here, new memories affect old memories. Retroactive interference (retro = backward) (RI) is when new learning interferes with previously learned material. It reduces the ability to recall earlier information because of learning new information.

In contrast, proactive interference works in a forward direction. It occurs when earlier learning makes it harder to learn or remember new information. For example, knowing French may make it harder to learn Spanish because they are similar.

Proactive interference (pro = forward) (PI) is when previous learning interferes with remembering new learning. It happens when memories reduce the ability to store new memories.

Proactive and retroactive interference are more likely to occur when the information is similar, for example, confusing old and new telephone numbers. Chandler (1989) stated that students studying similar subjects at the same time often experience interference.

Previous learning can interfere with new learning (e.g., difficulty using foreign currency), and new learning can also interfere with previous learning (e.g., learning French affecting Spanish vocabulary).

In summary, proactive interference occurs when past learning disrupts the acquisition of new information, while retroactive interference occurs when new learning interferes with the recall of previously stored memories or old memories. Example: Confusing an old telephone number with a new one.

7.6.2. The Trace Decay Theory

This theory explains forgetting in short-term memory. It says that memories leave a trace in the brain, which is a physical or chemical change in the nervous system. Forgetting happens when this trace fades over time.

Trace decay theory focuses on time and the short duration of short-term memory. It suggests that short-term memory can hold information for about 15 to 30 seconds unless it is repeated. After that, the memory trace fades away.

Memory traces, also called engrams, are formed in the brain. If the information is not used for a long time, these traces become weak, leading to forgetting. According to this theory, time is the main cause of forgetting.

The trace decay theory was developed by Edward Thorndike in 1914, based on earlier work by Hermann Ebbinghaus. It states that memories fade if they are not used.

When we learn something new, the brain forms memory traces. Remembering requires accessing these traces. The longer the time between learning and recall, the more likely we are to forget.

Although this theory explains forgetting in short-term memory, many researchers do not support it for long-term memory. They believe that memories can still be retrieved with the right cues.

One way to prevent forgetting is rehearsal. Maintenance rehearsal is repeating information mentally or out loud to keep it in memory. This helps keep information in short-term memory and may transfer it to long-term memory. However, short-term memory has a limited capacity.

The trace decay theory also explains that memory traces become weaker if they are not used regularly, leading to forgetting. For example, using a computer password often keeps it strong in memory, while not using it makes it harder to remember.

Scientists have found that memory involves changes in neurons and brain connections (synapses). However, research on complex memory processes is still incomplete.

Application activity 7.6



1. From your understanding, explain what a memory trace is and describe the trace decay theory of forgetting. Also, state how long short-term memory can hold information without rehearsal.
2. Imagine you are studying different subjects. Which type of interference (proactive or retroactive) are you less likely to experience before sleep, and why? Also, explain the difference between proactive and retroactive interference.
3. Think about your own study habits. Suggest one way you can prevent forgetting information from short-term memory, and explain how it helps.

7.7. The retrieval failure and the cue- dependent theory of forgetting

Learning activity 7.7



1. From the example given, which theory of forgetting do you think is being described?
2. Do you know who developed this theory?
3. In your opinion, what reasons might make you forget information according to this theory?

7.7.1. Retrieval failure theory

Some psychologists believe that forgetting happens because we cannot retrieve information, not because it is lost or destroyed. This is called retrieval cue failure, which means we cannot recall information from memory.

- Retrieval failure happens when information is stored in long-term memory but cannot be accessed.
- The information is still available but not accessible.
- This occurs because the needed retrieval cues are missing.
- When we store a memory, we also store details about the situation. These details act as retrieval cues.
- When we experience a similar situation again, these cues can help us remember.

Retrieval cues can be:

- i. External (context cues) from the environment, such as smell or place.
- ii. Internal (state cues) from within us, such as mood, emotions, or physical state.

A common example is the tip-of-the-tongue experience, where you feel you know something but cannot recall it.

The retrieval failure theory was developed by Endel Tulving in 1974. It explains that forgetting happens when we cannot access stored information at a certain moment, even though it is still in memory.

There are two main reasons for retrieval failure:

- **Encoding failure:** the information was never properly stored in long-term memory.
- **Lack of retrieval cues:** the information is stored, but we cannot access it without the right cues.

This theory is also called cue-dependent forgetting, meaning we fail to remember because we do not have the right cue. For example, a student may “know” an answer but fail to recall it during an exam.

Research shows that people remember more when they are given cues. In one study, a group given category hints remembered more words than a group without hints. This shows that we often have more information in memory than we can recall without cues. For example, visiting a childhood place can suddenly bring back old memories.

7.7.2. Retrieval cues

A retrieval cue is anything that helps us remember information. When we form a memory, we also store details about the situation, which later helps us recall it. Without these cues, retrieval may fail.

Examples of retrieval cues:

- **Semantic cues:** links to other memories. For example, remembering a friend may help you recall details of a trip.
- **State-dependent cues:** related to your mood or emotional state. Being in a similar mood can help you remember past events.
- **Context-dependent cues:** related to the environment, such as sounds, sights, or smells. For example, returning to a place can help you recall events that happened there.

Application activity 7.7



1. With examples, explain what you understand by a retrieval cue.
2. Explain the retrieval failure theory of forgetting.
3. Distinguish state-dependent forgetting from context-dependent forgetting.

7.8. Repression /motivation and displacement from short - term memory theories of forgetting

Learning activity 7.8



1. From your own understanding, how would you explain retrieval failure and how it causes you to forget information?
2. Think about your own experiences. What types of retrieval cues help you remember things, and how do they work for you?
3. In your opinion, what are the main reasons you sometimes fail to remember information, even when you feel you have learned it? Give examples.

7.8.1. The Repression / Motivation Theory

Repression is a way the mind protects us. According to Sigmund Freud, painful or unpleasant experiences are pushed into the unconscious mind so that we do not remember them. Even if these memories try to come back, we avoid thinking about them. This is called motivated forgetting.

We often forget things we do not want to remember. For example, a student may forget the name of a teacher who punished them badly.

Freud developed the idea of repression in his psychoanalytic theory. It happens when a memory or feeling is too painful, so the mind hides it without us being aware. Friedrich Nietzsche also suggested that people may forget memories on purpose, especially if they are painful or shameful.

Later, Freud explained that repression happens without intention. It is a defence mechanism. He believed that hidden memories could be brought back, but modern Psychologists do not fully agree with this idea. However, the theory is still important in Psychology.

7.8.2. Displacement Theory

Displacement theory explains forgetting in short-term memory. It says forgetting happens because memory space is limited.

Short-term memory can only hold a small amount of information (about 7 items). When it becomes full, new information pushes out old information. This old information is then forgotten.

The information that has stayed in memory the longest is usually the first to be removed.

This theory shows that forgetting happens because there is no space for new information. Many psychologists have supported this idea through research, especially using recall experiments.

Displacement theory also fits with the Multi-Store Model of Memory, which explains how some information moves to long-term memory while some is lost from short-term memory.

Application activity 7.8



1. From your understanding, how would you describe the capacity of your short-term memory?
2. Based on your experience, how do you think forgetting happens in your short-term memory?
3. In your opinion, what happens during repression, and how might it affect your memories?
4. Think about your own life. What are three ways you forget things, and how does each happen?

7.9. Strategies for enhancing memory

Learning activity 7.9

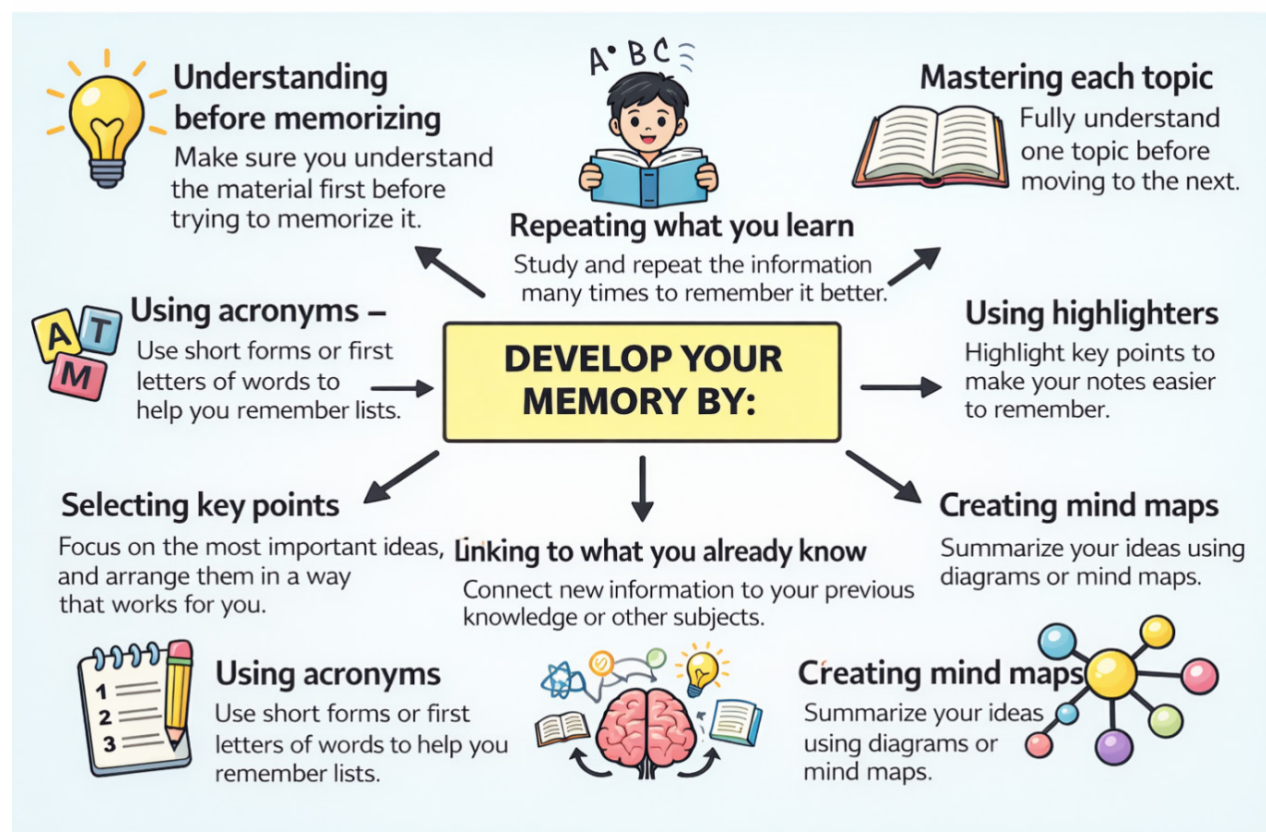


It may be difficult to remember all the new information that you have learned, but there are many techniques out there that can help you improve your memory, enhance recall, and increase retention of information. Suggest some of them.

7.9.1. Useful strategies

- a) Understand the material / deep processing:** Before memorising, first understand the material. Try to connect it to what you already know. The more meaningful it is, the easier it is to remember. Thinking deeply and repeating the material helps you store it in long-term memory.
- b) Organise your learning material:** Group related ideas together. Use diagrams like mind maps or Venn diagrams to show connections. This helps you see how ideas fit together.
- c) Attend carefully:** Pay full attention when studying. Without attention, learning and remembering become difficult.
- d) Have the right study environment:** Study in a quiet place without distractions like noise, TV, or people. A good environment helps you focus better.
- e) Minimise interference:** Avoid studying similar subjects at the same time. For example, study different languages on different days to reduce confusion.
- f) Distributed practice / Spacing effect:** Study in short sessions with breaks in between. Divide your work into parts and learn one part at a time. Do not cram everything at once.
- g) Space out your work:** Spread your study over several days. Focus on small amounts of material each time to remember better.
- h) Using memory aids:** Use cues like images, places, or objects to help you remember information.

- i) Use mnemonic methods:** Create rhymes, songs, or jokes to remember information. This makes learning easier and fun.
- j) Think in pictures and colours:** Use drawings, diagrams, and colours to make notes more interesting and easier to remember.
- k) Repetition:** Repeat information many times by writing, saying, or practising it. This helps store it in long-term memory.
- l) Talk about what you've learned:** Explain what you learned to others or say it out loud. This helps you understand and remember better.
- m) Association:** Link new information to things you already know, such as people, places, or experiences.
- n) Storytelling:** Turn what you learn into a story. This helps you remember information in order.
- o) Acronyms/shorthand code:** Use the first letters of words to create a short word or phrase to help you remember lists.
- p) Overlearning:** Keep studying even after you think you understand. This strengthens your memory.
- q) Vary your study routine:** Change your study place or time to keep learning interesting and improve memory.
- r) Play games:** Use games to learn. Games make studying fun and help you remember through practice.
- s) Chunking:** Break information into small groups. This makes it easier to learn and remember.
- t) Funny sentences:** Create simple or funny sentences to remember difficult words or ideas.
- u) Elaboration:** Add meaning to information by connecting it to personal facts or examples.
- v) Get enough sleep:** Sleep helps your brain store information. Rest well after studying to remember better.



7.9.2. Helpful memory techniques based on psychological research

Technique	Description	Useful example
Use elaborative encoding.	Material is better remembered if it is processed more fully.	Think about how new information relates to prior knowledge.
Make use of self-reference.	Material is better recalled if it is linked to thoughts about the self.	Connect new information about memory strategies to your study habits.
Be aware of the forgetting curve.	Information that we have learned drops off rapidly with time.	Review the material that you have already studied right before the exam to increase the likelihood that it will remain in memory.
Make use of the spacing effect	Information is learned better when it is studied in shorter periods spaced over time.	Study a little bit every day; do not cram at the last minute.
Rely on overlearning.	We can continue to learn even after we think we know the information perfectly.	Keep studying, even if you think you already have it down.
Use context-dependent retrieval.	We have better retrieval when it occurs in the same situation in which we learned the material.	If possible, study under conditions like those in which you will take the exam.

Technique	Description	Useful example
Use state-dependent retrieval.	We have better retrieval when we are in the same psychological state as we were when we learned the material.	Do not study under the influence of drugs or alcohol because they will affect your recall.

Application activity 7.9



1. How do you use different memory strategies, such as studying in small chunks, organising information visually, minimising distractions, spacing out your study sessions, repeated practice, or storytelling, to improve your learning?
2. Why do you think cramming has or has not been effective for you when studying?
3. How do you make new information meaningful to yourself when learning something new?
4. How do your own study habits, such as spacing out learning and testing yourself, compare to cramming or rereading when it comes to helping you remember information long term?

7.10. End unit assessment



Question 1: Knowledge

1. How would you explain declarative memory and how it differs from nondeclarative memory based on your understanding?
2. What types of information do you notice that you process automatically in your everyday life?
3. How do you think your sensory memory works when you experience sights, sounds, or other stimuli around you?
4. How do you understand the capacity of your long-term memory, and where do you think your long-term memories are stored in the brain?

Question 2: Skills

5. What effortful strategies do you personally use or could you use to help remember new information?
6. How would you explain your friend's situation, being able to perform tasks like tying his shoes but struggling to remember conversations, using your understanding of implicit and explicit memory?

Question 3: Attitudes and Values

7. What types of information do you notice that you process automatically in your everyday life?
8. What effortful strategies do you personally use or could you use to help remember new information?

UNIT 8

INTELLIGENCE



Key Unit Competence:

Assess how intelligence impacts human behaviours.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Outline intelligence types.
- Identify different theories and their key views on intelligence.
- Describe the concept of intelligence, including key definitions, primary theories, and main features.
- Describe key research findings on intelligence and explain what they reveal about differences in human behaviour and performance.

Skills

2

- Interpret findings from intelligence research to explain differences in problem-solving, reasoning, and learning among individuals.
- Apply knowledge of intelligence learned to deal with learning challenges, problem-solving tasks, and individual differences in educational or everyday situations.
- Compare and contrast Sternberg's triarchic theory with Piaget's cognitive development stages to examine how various types of intelligence affect problem-solving and decision-making
- Create a personal or group profile mapping the different types of intelligence.

Attitudes and values

3

- Acknowledge and respect individual intelligence differences in all interactions.
- Sympathise with people with low IQ and intellectual disabilities

Introduction to the unit

Have you ever noticed that some learners are good at solving mathematical problems, while others are talented in music, art, or understanding people's feelings? These differences show that intelligence is not only about being good in class, but also about how people think, learn, solve problems, and behave in everyday life.

In this unit, you will learn about intelligence and how psychologists such as Robert Sternberg, Howard Gardner, and Jean Piaget explained it. You will explore different types of intelligence, including linguistic, mathematical, social, self-awareness, creative, and naturalistic intelligence. The unit will also examine the factors that influence intelligence, how intelligence is measured, and intellectual disabilities. By the end of the unit, you will understand how intelligence affects behaviour at school, at home, and in the community, while appreciating individual differences and improving your own learning skills.

Introductory activity



Kalisa and Kamana are students from the same family, sharing the same environment, resources, and learning conditions, yet their academic performance is surprisingly different. Kalisa easily solves complex mathematical problems but struggles to express ideas in writing, while Kamana writes creatively and understands others well but finds calculations difficult; during group work, Kalisa organises tasks logically while Kamana leads discussions and presents confidently.

1. Why do you think these two students perform differently despite having the same opportunities?
2. Could it be that intelligence is made up of different types of abilities?
3. How might these differences influence their behaviour and success in school and daily life?

8.1. Key terms related to intelligence

Learning activity 8.1



Write your name at the centre of a mind map and then add abilities or things you do well in your daily life (at school, at home, or in your community).

After completing your mind map, discuss these questions

1. Are all your abilities the same?
2. Do your classmates have the same strengths as you?
3. What do these differences suggest about intelligence?

8.1.1. Meaning of Intelligence and its importance

In Psychology, intelligence refers to the general ability of a person to understand, think, learn, and solve problems. It helps individuals to adapt to new situations and to deal effectively with challenges in everyday life.

Intelligence is not only used in school, but also in **workplaces and community life**. For example, a person may use intelligence to make decisions, communicate with others, or find solutions to problems.

8.1.2. Key concepts related to Intelligence

Intelligence: The ability to learn, reason, solve problems, and adapt to the environment. Example: A mechanic identifies a problem in a motorcycle and finds the best way to repair it.

Intellect: The ability to think deeply and understand complex ideas. Example: A lawyer analysing a case before presenting it in court.

Ability: The capacity to perform a task, either naturally or through learning. Example: A tailor who can design and sew clothes efficiently.

Aptitude: The potential ability to learn a specific skill. Example: A student who quickly learns computer skills may have aptitude in ICT, which can later help in jobs like data entry or programming.

Achievement: What a person has already learned or accomplished at a given time. Example: Marks obtained in national examinations or completing a training course.

Thinking: A mental process that involves processing information, making decisions, and solving problems. Example: A business owner deciding how to reduce losses during a low-sales period.

Reasoning: The ability to think logically and make sound judgments. Example: A manager decides which employee is best suited for a specific task.

Memory: The ability to store and recall information. Example: A pharmacist remembering different medicines and their uses.

Sensation: The process by which the body receives information through the senses (seeing, hearing, smelling, tasting, touching). Example: A driver hearing a horn and reacting quickly.

Perception: The process of interpreting and giving meaning to sensory information. Example: A security guard sees suspicious behaviour and understands it may be a threat.

Mind: The centre of thoughts, feelings, and memories that controls thinking and behaviour.

Application activity 8.1



You are a student preparing for your future career. Using examples from your daily life at school or in your community in Rwanda, define the following terms: Intelligence, Ability, and Aptitude

8.2. Brief history of intelligence

Learning activity 8.2



Think about your daily life in different places. In a classroom, one student quickly understands a new concept while another needs more time but asks deeper questions. In a health centre, one nurse makes fast decisions in emergencies, while another carefully analyses every detail before acting. In a market or business, some people easily calculate prices in their heads, while others are better at convincing customers.

Why do people think and learn in different ways? Does being “intelligent” mean the same thing for everyone?

As scientific research on intelligence developed, Psychologists proposed different theories to explain how intelligence works and why people differ in their abilities.

The first people to think about intelligence were Philosophers, not Psychologists or teachers. The ancient Greek Philosopher Plato compared intelligence to blocks of wax. He believed people differ depending on the size and quality of their “wax.” If the wax was too hard, too soft, or impure, the person would have learning difficulties. Later, Thomas Aquinas (in the 13th century) said that intelligent people understand things more fully than others, but no human can be as all-knowing as God.

In the 18th century, Immanuel Kant suggested that intelligence has different types or parts, and people differ in how much they have of each. These early ideas helped prepare the way for modern intelligence research. Although people have thought about intelligence for a long time, most scientific knowledge about it has developed since the late 19th century.

In the **late 19th century**, the first pioneers who studied intelligence more scientifically were:

- **Francis Galton:** He is considered one of the first to study intelligence scientifically. He believed intelligence was inherited (passed from parents to children) and tried to measure it using sensory and physical abilities.
- **Wilhelm Wundt:** He established the first psychology laboratory and encouraged the scientific study of the mind, which influenced later intelligence research.
- **James McKeen Cattell:** A student of Wundt, he introduced the idea of “mental tests” to measure intelligence, focusing on reaction time and sensory skills.

These pioneers helped move the study of intelligence from philosophical thinking to scientific measurement and experimentation, which later led to more advanced intelligence testing.

Application activity 8.2



1. In your community or workplace, people often have different ways of thinking and solving problems. Based on the early ideas of philosophers, how did thinkers like Plato and Thomas Aquinas explain these differences in intelligence?

2. Imagine you are explaining to a colleague or community member. How did the ideas of early philosophers differ from later scientific approaches introduced by Francis Galton and James McKeen Cattell?
3. In a real-life setting such as a school or workplace, why is it important to move from philosophical ideas of intelligence to scientific study and measurement?

8.3. Theories of intelligence: Sternberg's triarchic theory and Gardner's theory of multiple intelligences

Learning activity 8.3



1. In your school, people show different abilities; some are good at solving problems quickly, others at communicating or creating ideas. Analyse the statement "intelligence cannot be viewed as a single entity" by using examples from your environment.
2. Based on your observation of people around you, do you agree that intelligence has many forms? Explain your position clearly.
3. Carry out short research to find a Psychological theory that supports your answer, such as the theory proposed by Howard Gardner. Then:
 - Explain the main idea of the theory.
 - Show how it applies to real-life situations in Rwanda (school, market, hospital, etc.).

8.3.1. Sternberg's Triarchic theory of intelligence

Robert Sternberg is an American Psychologist who practically explained intelligence. He said intelligence is the ability to learn from experience, think well, remember important things, and deal with everyday life situations. In simple terms, intelligence helps a person solve problems and adapt to different situations.

Sternberg explained that intelligence is not just one ability. He proposed the triarchic theory of intelligence, which includes three types of intelligence: **(a) analytical, (b) creative, and (c) practical intelligence.**

Analytical (Componential) intelligence

This is the ability to analyse problems and find correct answers. It is mostly used in school learning and exams, so it is often called "book smart."

People with this type of intelligence are good at solving academic problems, think logically, and understand complex ideas. For example, a student who performs well in exams and solves mathematics problems quickly.

Creative (Experiential) intelligence

This is the ability to think in new and different ways and come up with new ideas. People with this type of intelligence are imaginative, find new solutions to problems, and use experience to create new ideas. Example: A young entrepreneur who creates a new way to attract customers or design products.

Practical (Contextual) intelligence

This is the ability to deal with real-life situations. It is often called "street smart" or common sense.

People with this type of intelligence: Adapt to different situations, solve everyday problems, and make good decisions in real life. Example: A farmer who adjusts farming methods based on weather changes or a trader who knows how to negotiate prices.

Ways practical intelligence works

Sternberg explained three ways people use practical intelligence:

- **Adaptation:** Changing yourself to fit the environment. *Example:* Learning new skills to get a job.
- **Shaping:** Changing the environment to suit you. *Example:* Convincing an employer or improving your workplace.
- **Selection:** Moving to a better environment. *Example:* Changing jobs or moving to a place with better opportunities.

This theory shows that intelligence is not only about school performance; it also includes creativity and the ability to handle real-life situations, which are very important in everyday life.

8.3.2. Gardner's theory of multiple intelligences

Howard Gardner said that intelligence is not one single ability. He explained that people have many types of intelligence, and each person has a different combination of them. This means we cannot say one person is more intelligent than another; everyone is intelligent in their own way.

He also explained that having a high IQ (Intelligence Quotient) is not enough. A person must be able to use their abilities in real life. What is considered "intelligent" can also depend on the culture or society. For example, skills valued in a city job may be different from those valued in farming communities in Rwanda.

Types of Intelligence

1. **Linguistic Intelligence (Language skills):** Ability to use words well in speaking and writing. Example: A teacher, journalist, or radio presenter.
2. **Logical-Mathematical Intelligence (Thinking and numbers):** Ability to solve problems, reason, and work with numbers. Example: An engineer, accountant, or student good at mathematics.
3. **Musical intelligence (Music skills):** Ability to create, understand, and enjoy music. Example: A singer, drummer, or music producer.

- 4. Bodily-Kinesthetic intelligence (Body movement):** Ability to use the body skillfully. Example: A football player, dancer, tailor, or surgeon.
- 5. Spatial intelligence (Visual and design skills):** Ability to understand images, shapes, and space. Example: An architect, driver, or artist.
- 6. Interpersonal Intelligence (Understanding others):** Ability to relate well with people and understand their feelings. Example: A teacher, leader, nurse, or counsellor.
- 7. Intrapersonal Intelligence (Understanding oneself):** Ability to understand one's own feelings and goals. Example: A reflective student or a leader who knows their strengths and weaknesses.
- 8. Naturalistic Intelligence (Understanding nature):** Ability to recognise and understand plants, animals, and the environment. Examples: a biologist, a farmer, an environmentalist, or a tourist guide.
- 9. Existential Intelligence (Big life questions):** Ability to think about deep questions like the meaning of life. Example: A philosopher or religious leader.

Gardner's theory teaches that all types of intelligence are important. Schools often focus more on language and mathematics, but other abilities like music, sports, and social skills are also valuable. To help learners succeed, education should support all types of intelligence.

8.3.3. Thurstone's Theory of Intelligence

Louis Leon Thurstone proposed the Theory of Primary Mental Abilities (PMA). He argued that intelligence is made up of several relatively independent mental abilities rather than one general ability. These include verbal comprehension, word fluency, number ability, spatial ability, associative memory, perceptual speed, and reasoning. For example, a person may be very good at understanding words and solving problems but less skilled in numerical tasks.

8.3.4. Emotional Intelligence (Daniel Goleman)

Daniel Goleman emphasized emotional intelligence (EI) as the ability to recognize, understand, and manage one's own emotions and understand those of others. It involves self-awareness, self-management, social awareness, and relationship management. For example, when a person becomes angry during a disagreement, they can recognize the emotion, control their reaction, understand the other person's feelings, and respond calmly.

Application activity 8.3



1. In your school or community, people solve problems in different ways. Using examples from your environment, describe three types of intelligence proposed by Robert Sternberg.
2. In your daily life, you may notice that some people are good at communication, others at calculations, and others at practical tasks. Using real-life examples from Rwanda, differentiate the types of intelligence proposed by Howard Gardner.

8.4. Theories of Intelligence (Piaget's theory of cognitive development)

Learning activity 8.4



Think about what you studied in Senior 4 and relate it to your own learning experience or that of younger students in your school or community. Identify and explain the stages of cognitive development according to Jean Piaget.

In the theory of intelligence developed by Jean Piaget, the stages of intellectual development are very important. He explained that human thinking develops gradually through four main stages.

- a) **The sensory-motor stage**, where a child learns through direct interaction with the environment. At this level, understanding comes from actions such as touching, seeing, and moving objects. For example, a baby learns by shaking or throwing objects. These early experiences later support practical skills used in jobs such as mechanics or carpentry, where understanding comes through handling tools and materials.
- b) **The pre-operational (pre-logical) stage**. At this stage, children begin to use language, symbols, and images, but their thinking is still influenced by what they see. They cannot yet think logically. For example, a child may believe that a taller glass has more juice than a shorter one, even when both contain the same amount. This stage supports the development of communication skills, which are important in workplaces such as teaching, customer care, and business, where explaining and interacting with others is necessary.
- c) **The concrete operational stage**, where the child begins to think logically about real and concrete objects. At this level, thinking is no longer limited to immediate perception, but it still depends on real situations. For example, a child understands that money has the same value whether it is in coins or notes. This kind of thinking is useful in jobs such as trading, farming, and construction, where people deal with real materials, measurements, and calculations.

The fourth stage is the formal operational stage, which is the stage of complete logical thinking. At this level, a person can think abstractly, reason logically, and solve problems without relying on concrete objects. For example, a person can plan future projects or think about solutions to community problems. This level of thinking is important in professions such as leadership, law, medicine, and business management, where decision-making and planning are required.

Piaget believed that intelligence is a mechanism that helps a person adapt to the environment. This adaptation occurs through two important processes: assimilation and accommodation. Assimilation happens when a person uses existing knowledge or experience to understand new information without changing their thinking. For example, a worker may use previous experience to solve a familiar problem at work. Accommodation occurs when a person changes their thinking or creates new ideas to deal with new situations. For example, a farmer may adopt new farming techniques after realising that old methods are not effective.

The concept of a schema is central in Piaget's theory. A schema is a mental structure or pattern of thinking that helps a person organise and understand information. People develop schemata through experience, and these help them respond to different situations. For example, a mechanic develops a schema of how engines work, which helps in repairing different types of vehicles. Even when objects are used in different ways, they remain the same in terms of their basic characteristics such as shape, size, and function.

Piaget also explained that intelligence develops through a balance between assimilation and accommodation, known as equilibration. When a person faces a new or confusing situation, they experience imbalance, which pushes them to learn and adapt. For example, a worker learning to operate a new machine may first struggle but eventually adjust and gain new skills.

In general, intelligence is seen as a process of continuous adaptation. It develops from simple actions in early childhood to complex reasoning in adulthood. This development helps individuals function effectively in different environments, including school, workplaces, and the wider community.

Application activity 8.4



1. Think about children you have seen growing up in your family or in your community. Using clear examples from real-life situations or different workplaces, describe the four stages of intelligence according to Jean Piaget.
2. Using examples from your daily life, define the concept of "schema" as explained by Jean Piaget.

8.5. Types of intelligence: Non-personal Intelligences

Learning activity 8.5



Think about your experience in primary school in Rwanda. List the subjects you studied, and then choose one subject in which you performed well and explain why you succeeded in it, using examples from your learning habits or support from teachers, family, or your environment.

Non-personal Intelligences concern abilities related to language, reasoning, the physical world, and nature. They are :

- **Word Smart / Linguistic Intelligence** is the ability to use language (your own language and others) to express your ideas clearly and understand other people. Example: A teacher explaining a lesson clearly, or a journalist reporting news in Kinyarwanda or English.
- **Math Smart / Mathematical Intelligence** is the ability to use numbers well and think logically. A person with this intelligence can see relationships, identify problems, and solve them quickly. Example: A businessperson calculating profit and loss, or a bank worker handling money accurately.

- **Physically Smart / Physical Intelligence** is the ability to take care of your body and manage your energy, health, and stress so that you can work effectively and live well. Example: A construction worker managing strength and balance while working, or an athlete maintaining fitness and energy.
- **Music Smart / Musical Intelligence** is the ability to understand, recognise, and create sounds and music. A person with this intelligence can differentiate and express musical ideas. Example: A musician or choir member in church composing or singing songs, or a radio presenter working with music programs.
- **People Smart / Interpersonal Intelligence** is the ability to understand others and interact well with them. Example: A leader resolving conflicts in the community, or a customer care worker handling clients politely.
- **Body Smart / Bodily-kinaesthetic Intelligence** is the ability to use your body skillfully to perform tasks and solve problems. Example: A mechanic repairing vehicles using hands and tools, or a tailor sewing clothes carefully.

Application activity 8.5



Think about yourself, your classmates, or people in your community. Give a clear example from school, home, or work, and explain when a person is considered smart in terms of intelligence.

8.6. Other Types of Intelligence

Learning activity 8.6



Think about Mwiza, who can recognise her strengths and weaknesses and appreciates the beauty of nature. Using examples from your own life or school, identify which types of intelligence she demonstrates and explain why.

- **Self-smart, also called intra-personal intelligence**, is knowing yourself well and using that knowledge to make good decisions. This includes knowing your strengths and weaknesses, being aware of your intentions, motivations, feelings, and desires, and having self-discipline, understanding yourself, and confidence. Example: A student who knows they are good at writing but struggles in mathematics can focus on improving math while using writing skills well. A manager who understands their own strengths can assign tasks effectively to their team.
- **Naturalistic Intelligence** is the ability to notice and understand patterns in nature. It includes sensitivity to plants, animals, and the environment. People with this intelligence appreciate the beauty of nature and its details. Examples: Farmers who understand seasons and soil conditions for planting crops, tour guides in Volcanoes National Park explaining wildlife to visitors, gardeners maintaining flowers in hotels, and environmentalists protecting rivers and natural resources. Famous example: Charles Darwin, known for his book *on the Origin of Species*, had high naturalistic intelligence.

- **Existential Intelligence** (added as the ninth intelligence by Gardner in 1998) is the ability to think about deep questions about life and human existence, such as the meaning of life, why we are here, why we die, and what happens after death. It involves thinking about the “big picture” and ultimate truths. Example: Philosophers, religious leaders, or community elders guiding people on moral or life questions. Cosmologists who study the universe also show this intelligence.

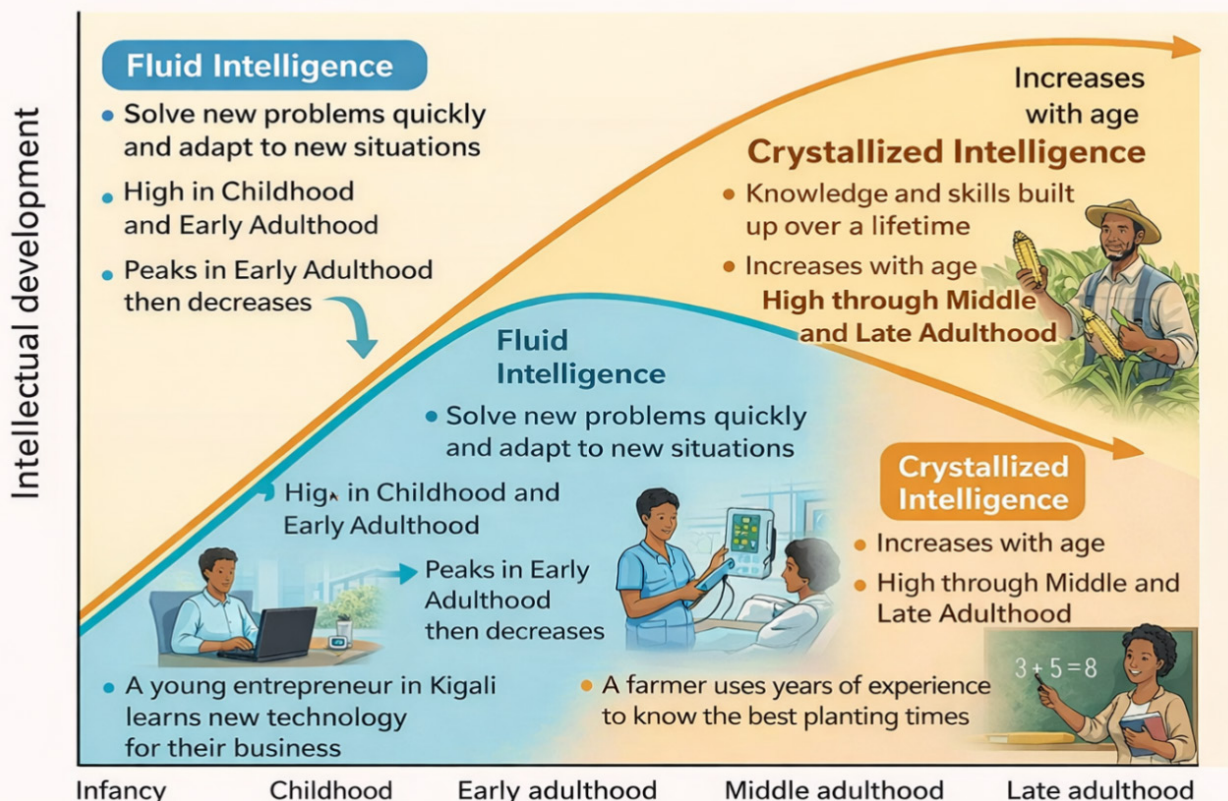
According to Gardner, each of these intelligences has specific abilities that can be observed and measured. The first two, linguistic and logical-mathematical intelligence, are strongly linked to school performance and academic achievement. These are usually emphasised in society.

The next three, bodily-kinaesthetic, musical, and spatial intelligence, are linked to arts and practical skills. People strong in these intelligences include singers, dancers, athletes, architects, designers, and actors.

Our traditional school system often ignores these intelligences, so many children do not get enough chance to develop them. Gardner suggests all types of intelligence should be valued to fully use a person's potential.

- **Visual-Spatial Intelligence** is the ability to understand and use visual and spatial information effectively. People with this intelligence can form mental images, recognise patterns, and see things from different angles. Example: Architects designing buildings, interior designers decorating hotels, surgeons performing operations, pilots navigating aeroplanes, drivers, or sailors navigating rivers and lakes. Besides the types described above, we also have crystallised and fluid intelligence:
- **Fluid Intelligence** is the ability to solve new problems quickly and think abstractly.
- **Crystallised Intelligence** is the knowledge and skills we have learned over time, such as facts, language, and experiences. A businessperson who quickly learns new strategies to improve sales demonstrates fluid intelligence, while a teacher who uses years of knowledge and experience to teach effectively demonstrates crystallised intelligence. Note: Crystallised intelligence usually increases with age, while fluid intelligence tends to decline after middle adulthood.

Understanding the Graph: Fluid and Crystallized Intelligence



Application activity 8.6



Describe the characteristics of a visual-spatial person like you. How do you understand and remember things better? Do you prefer pictures, diagrams, maps, or demonstrations? Think about how you solve problems: do you often imagine ideas in your mind or visualise solutions before acting?

What kind of job would suit you best as a visual-spatial learner? Do you see yourself enjoying careers that involve creativity and design, such as architecture, graphic design, engineering, photography, or art? Explain why these jobs match your strengths and interests.

8.7. Determinants of intelligence

Learning activity 8.7



Nziza and Mulisa are two Primary 6 learners. Nziza's parents are teachers, and they live in a town. They often buy him books to read at home and support his learning. Mulisa's parents are farmers, and they live in a village where access to learning materials may be limited.

Discuss

1. What differences do you notice between Nziza's and Mulisa's home environments, and how is your own home environment similar or different from theirs?
2. How do your parents or guardians support your learning, and how might this affect your performance at school?
3. Apart from school, what things in your home or daily life help you learn better or make learning more difficult?

8.7.1. Heredity

Studies of twins, adopted children, and even animals show that heredity (genes from parents) plays an important role in intelligence. This means that the intelligence you have is partly passed down from your parents, just like eye color or height. For example, scientists have studied identical twins (who share the same genes) raised in different families. Even when they grow up in different places, they often have very similar intelligence levels. This shows how powerful heredity can be. Research also shows that about 40% to 80% of intelligence is influenced by genetic inheritance. This means that genes have a strong effect on how we think and learn.

Another example comes from adopted children. Many studies show that a child's intelligence is often closer to that of their biological parents than their adoptive parents. For instance, a child adopted into a highly educated family may still show intelligence patterns similar to their biological parents.

However, this does not mean the environment is not important. It only shows that heredity has a strong influence on intelligence.

8.7.2. Environment

Although heredity is important, environment also plays a big role in shaping intelligence. In fact, it is difficult to say exactly how much intelligence comes from genes and how much comes from environment. Some experts estimate that about 75% comes from heredity and 25% from environment, but both work together.

The environment includes things like home, school, culture, experiences, and opportunities. These factors can greatly influence how a person thinks and learns.

For example:

- A child who grows up in a home with books, supportive parents, and good schools may develop stronger thinking skills.
- Learners who practice solving problems, reading, and asking questions will improve their intelligence over time.
- Exposure to different cultures, languages, and experiences can also help a person think in more creative and flexible ways.

Studies show that good education and training programs can improve intelligence. For instance, students who attend quality schools and participate in active learning (like discussions, experiments, and projects) often develop better problem-solving skills.

In conclusion, intelligence is shaped by both heredity and environment. While genes give us a starting point, our environment helps us grow, learn, and reach our full potential.

Application activity 8.7



1. Compare heredity (nature) and environment (nurture) as factors that influence intelligence. Which one do you think has affected your own learning more? Explain using your experience.
2. Using examples from your own community or daily life, explain how the environment (home, school, friends, or culture) has influenced how you think and learn.

8.8. Measuring intelligence

Learning activity 8.8



Do you think intelligence can be measured? If yes, what methods or tools do you think can be used to measure it? Give examples from your own learning.

Over the years, the ways of measuring intelligence have greatly improved. Today, there are many trusted and widely used tools for measuring intelligence, making it easier to understand how people think and learn.

8.8.1. Binet and intelligence testing

Modern intelligence testing began with Alfred Binet and his colleagues. They created a test to measure the intelligence of school children in France. Their goal was to identify learners who needed extra help in school.

Binet believed that all children develop intellectually in similar ways, but at different speeds. Some children learn faster, while others take more time. For example, one child may quickly understand mathematics, while another may need more practice and support.

He suggested that a child with lower intelligence might perform like a younger child, while a very bright child might perform like an older child. For instance, a 6-year-old who solves problems like an 8-year-old shows faster intellectual development.

To explain this, Binet introduced the idea of “mental level,” which later became known as Mental Age. Mental age shows the level at which a child is thinking and solving problems.

For example, A child who is 6 years old but performs like an average 8-year-old has a mental age of 8. This means the child is thinking above their actual age level.

Using this idea, Binet and his team created different test activities that measured reasoning, memory, and problem-solving skills. These tests were first introduced in 1905 and became known as the Binet-Simon Test.

Later, another psychologist named Lewis Terman improved this test and introduced a new way of scoring called the Intelligence Quotient (IQ). The idea of IQ was first suggested by William Stern in 1912.

IQ is calculated by comparing a person's mental age with their actual age (chronological age) using this formula:

$$IQ = \frac{\text{Mental Age}}{\text{Chronological Age}} \times 100$$

For example, if a 6-year-old child has a mental age of 9, the IQ will be:

$$IQ = \frac{9}{6} \times 100 = 150$$

This means the child has a very high level of intelligence compared to their age.

Mental age is calculated carefully using test questions. In the Terman-Binet test:

- Each year level has 6 questions
- Each question represents 2 months (because $12 \text{ months} \div 6 = 2 \text{ months}$)

Let us look at an example:

An **8-year-old child** takes the test:

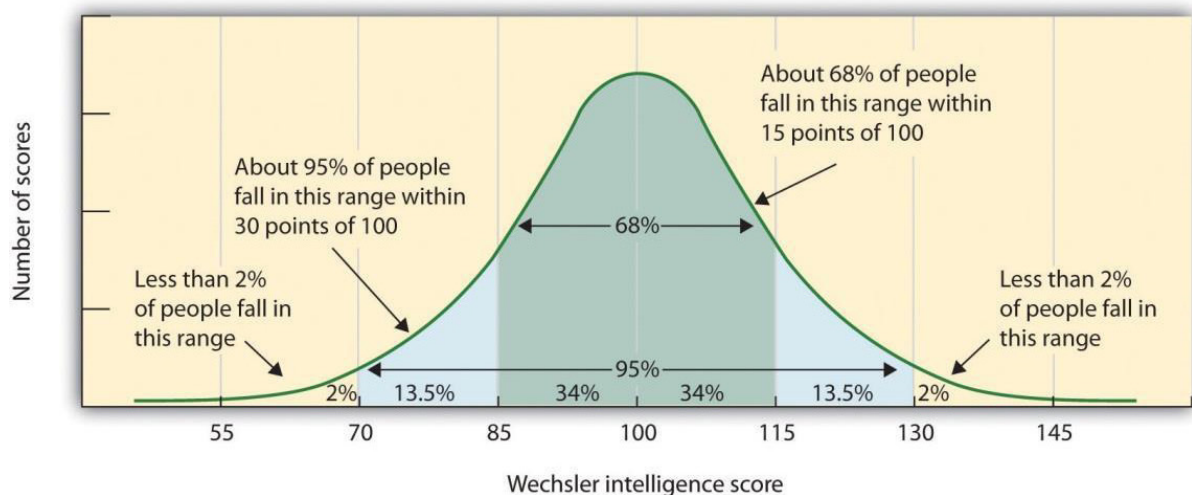
- The child answers all questions for 7 years → gets 7 years
- Answers 2 questions for 8 years → gets 4 months ($2 \times 2 \text{ months}$)
- Answers 1 question for 9 years → gets 2 months

Now add them:

- 7 years + 4 months + 2 months = 7 years and 6 months ($7\frac{1}{2}$ years)

Conclusion:

The child's mental age is 7 years and 6 months, which means the child is 6 months behind their actual age (8 years).



This method helps teachers understand learners better. For example, a teacher can identify which learners need more support and which ones need more challenging work, helping every student improve and succeed.

8.8.2. Factors that affect I.Q scores

❖ Developmental factors

As children grow older, their I.Q. scores become more stable. However, problems that affect physical health or brain development can reduce intellectual abilities and make it harder for children to perform well on intelligence tests.

❖ Emotional factors

Stress, poor adjustment, low motivation, and anxiety can lower a child's performance. Children who experience a lot of stress or anxiety may struggle to concentrate, which can negatively affect their I.Q. scores.

❖ Family and cultural factors

The family environment plays an important role in a child's learning. For example, if parents do not value education, the child may lack motivation to perform well in school. Factors such as family size, parenting values, and socio-economic status (income, parents' education, and work conditions) can influence test performance. Cultural and economic disadvantages, especially in early life, may also lead to lower performance on standardised intelligence tests.

8.8.3. Levels of IQ values

The concept of Intelligence Quotient (IQ) was developed by Lewis Terman in 1916. He proposed a scale used to classify individuals according to their IQ scores.

IQ scores are grouped into different levels as shown below:

IQ Score Range	Level of Intelligence
Above 140	Genius or near genius
120 – 139	Very superior intelligence
110 – 119	Superior intelligence
90 – 109	Average (normal) intelligence
80 – 89	Below average (dullness)
70 – 79	Borderline
50 – 69	Mild intellectual disability
35 – 49	Moderate intellectual disability
20 – 34	Severe intellectual disability
Below 20	Profound intellectual disability

Most people fall within the average range (90–109), while fewer individuals are found at the extreme high or low ends of the scale.

Application activity 8.8



In your school or community, you may notice that learners perform differently in class and on tests. Some concentrate easily, while others struggle due to challenges at home, stress, or health conditions.

Question:

Based on your observations and experiences, describe the factors that can affect I.Q. scores among learners. Give examples to support your answer.

8.9. Intellectual disabilities

Learning activity 8.9



Jane takes longer to learn how to talk compared to other children of her age.

Question:

What could be the possible problem Jane is experiencing?

8.9.1. Meaning

Intellectual disability refers to a condition in which a child has below-average cognitive ability. This affects how the child learns, understands information, solves problems, and adapts to daily life. For example, a child with intellectual disability may: take longer to learn how to speak or read, have difficulty solving simple problems, struggle to perform daily tasks like dressing or communicating with others.

Intellectual disability not only affects the individual, but it can also impact the family and society, as the child may need extra support in school and daily life. In general, intellectual disability means:

- Difficulty in understanding new or complex information.
- Difficulty in learning new skills.
- Difficulty in living independently (social and practical challenges).
- The condition begins during childhood and continues into adulthood.

8.9.2. Criteria for Intellectual Disability

A person is considered to have an intellectual disability if:

1. Sub-average intellectual functioning

This means low ability in areas such as learning, reasoning, and problem-solving. Example: A learner who finds it very difficult to understand lessons even after repeated explanations.

2. Limitations in adaptive skills

These are everyday life skills such as communication, social interaction, and self-care. Example: A child who cannot dress themselves, interact well with others, or follow simple instructions appropriate for their age.

8.9.3. Causes of intellectual disability

The causes of intellectual disability can be biological (organic) or environmental (social).

1. Biological causes

These are related to problems in the body or brain. About 70% of severe cases and 50% of mild cases have biological causes. *Examples:* brain injury, genetic conditions, or complications before or during birth.

Types	Example	Characteristics and considerations
Chromosomal abnormality	Down syndrome	– Most common chromosomal abnormality – Distinctive physical characteristics – Generally mild to moderate intellectual disability
	Fragile X syndrome	– One of the leading inherited causes of intellectual disability – Predominantly affects males – Distinctive physical features – Wide variation in learning characteristics
Metabolic disorders	Phenylketonuria (PKU)	– Inborn error of metabolism, a recessive trait – Dietary intervention initiated shortly after birth prevents the occurrence of intellectual disability
Maternal infections	Rubella (German measles)	– One of the leading causes of multiple impairments in children – Exposure during the first trimester of pregnancy usually results in severe consequences
Environmental Conditions	Fetal alcohol syndrome	– One of the leading causes of intellectual disability – Mild to moderate intellectual disability with concomitant physical deformities
Gestational disorders	Low birth weight/prematurity	– Infant at risk for serious problems at birth – Potential for learning problems as well as sensory and/or major impairments – More common in mothers living in poverty, teenage pregnancy, and women engaged in substance abuse

Types	Example	Characteristics and considerations
Neonatal complications	– Anoxia (oxygen deprivation) – Breech presentation – Prolonged delivery	– Complicating factors surrounding birth may cause intellectual disability and other developmental delays
Infections and intoxications	Meningitis	– Viral infection causing damage to the covering of the brain, the meninges – May result from a typical childhood illness such as chicken pox or mumps – Intellectual disability is a distinct possibility
	Lead poisoning	– Highly toxic substance – Infants/toddlers living in older homes in impoverished areas are at risk for ingesting lead-based paint chips – Potential for causing seizures, central nervous system damage, and brain damage
Environmental factors	Environmental Malnutrition child abuse /neglect	– Correlates, but not necessarily causes, of intellectual disability, especially instances of mild intellectual disability – Best viewed as interacting psychosocial risk factors that heighten the vulnerability of some children to learning difficulties

2. Environmental (psychosocial) causes

These come from the child's surroundings and experiences. Examples: poverty, neglect or lack of care, abuse, lack of stimulation (e.g., no learning materials or interaction at home), poor parent-child relationships

In some cases, a child's low intellectual functioning may result from a combination of both genetic and environmental factors.

8.9.4. Classification of Intellectual Disability

Intellectual disability divides the severity of disability into four categories (mild, moderate, severe, and profound intellectual disability).

Level of Intellectual Disability	IQ range	Approximate mental age in adulthood	% of persons with Intellectual Disability at this level
Mild	55-68	8 years, 3 months to 10 years, 9 months	85
Moderate	36-51	5 years, 7 months to 8 years, 2 months	10

Level of Intellectual Disability	IQ range	Approximate mental age in adulthood	% of persons with Intellectual Disability at this level
Severe	20-35	3 years, 2 months to 5 years, 6 months	3.5
Profound	< 20	< 3 years, 2 months	1.5

8.9.5. Characteristics of people with Intellectual Disability

Characteristics of people with intellectual disabilities that can affect their academic learning, as well as their ability to adapt to home, school, and community environments, are presented under the following points:

- **General Cognition:** People with intellectual disabilities vary physically and emotionally, as well as by personality, disposition, and beliefs. Their apparent slowness in learning may be related to the delayed rate of intellectual development.
- **Learning and Memory:** The learning and memory capabilities of people with intellectual disabilities are significantly below average in comparison to peers without disabilities. People with intellectual disabilities develop learning sets at a slower pace than their peers without disabilities, and they are deficient in relating information to new situations. Children with intellectual disabilities may not spontaneously use appropriate learning or memory retention strategies and may have difficulty in realising the conditions or actions that aid learning and memory.
- **Attention:** To acquire information, children must attend to the learning task for the required length of time and control distractions. Children with intellectual disabilities may have difficulty distinguishing and attending to relevant questions in both learning and social situations.
- **Adaptive Skills:** A child with intellectual disabilities may have difficulty in both learning and applying skills for some reasons, including a higher level of distractibility, inattentiveness, failure to read social cues, and impulsive behaviour.
- **Self-Regulation:** The ability to rehearse a task is related to a broad concept known as self-regulation, or the ability to mediate or regulate one's own behaviour. Information-processing theorists study how a person processes information from sensory stimuli to motoric output. In information-processing theory, the learning differences in people with intellectual disabilities are seen as the underdevelopment of metacognitive processes. The lack or underdevelopment of these skills notably affects memory, rehearsal skills, organisational ability, and being in control of the process of learning.
- **Speech and Language:** People with intellectual disabilities may have delayed speech, language comprehension, and formulation difficulties. Language problems are generally associated with delays in language development rather than with a bizarre use of language.
- **Motivation:** People with intellectual disabilities are often described as lacking motivation or outer-directed behaviour. Past experiences of failure and the anxiety generated by those failures may make them appear to be less goal-directed and lacking in motivation. The result of failure is often learned helplessness.

- **Academic Achievement:** The cognitive inefficiencies of children with mild to moderate intellectual disabilities lead to persistent problems in academic achievement. Children with intellectual disabilities may be able to learn basic computations, but may be unable to apply concepts appropriately in a problem-solving situation.
- **Physical characteristics:** Children with intellectual disabilities with differing biological aetiologies may exhibit coexisting problems, such as physical, motor, orthopaedic, visual, and auditory impairments, and health problems.

Most children with severe and profound intellectual disabilities have multiple disabilities that affect nearly every aspect of intellectual and physical development.

People with intellectual disabilities have strengths, talents, and abilities like all other individuals. Society should provide equal opportunities, respect, support, and inclusive environments that help them participate fully in education and community life

Application activity 8.9



In your community or school, you may have observed learners who experience difficulties in learning, speaking, or understanding instructions compared to their peers.

Questions:

1. Based on your observation and knowledge, determine the possible factors that can lead to intellectual disability in children.
2. Describe the characteristics that may be shown by individuals who have language-related disabilities.

8.10. End unit assessment



Knowledge, skills, values and attitudes

1. From your daily experiences, explain how intelligence influences the way people live, learn, and solve problems in real life. Give examples.

Knowledge and skills

2. In your own words, discuss what intelligence means. Identify and explain the different abilities that show intelligence in a person.

Attitudes and values

3. In your community, people with intellectual challenges may show different levels of difficulty. Explain how intellectual disabilities can be classified according to their severity and describe what each level may look like in real life.

UNIT 9

MOTIVES OF LEARNING



Key Unit Competence:

Justify how the motives of learning contribute to the acquisition of behaviour.

Learning objectives

By the end of this unit, you should be able to:

Knowledge and understanding

1

- Identify the key motives for learning.
- Differentiate motives of learning based on their meanings.
- Describe the different types of physiological, social, and personal motives, including their characteristics and examples.
- Explain how physiological, social, and personal motives influence human Behaviour in everyday life.

Skills

2

- Explain how an individual's motives influence their Behaviour and attitudes.
- Investigate the reasons why people's behaviour patterns and attitudes change in time and in space.
- Analyse real-life scenarios to identify physiological, social, and personal motives influencing human Behaviour.
- Use knowledge of different motives to explain and predict individual actions in various everyday situations.

Attitudes and values

3

- Appreciate how physiological, social, and personal motives influence Behaviour and impact personal and social decision-making.
- Being inspired by the motives of learning to prevent the acquisition of deviant Behaviours.

Introduction to the unit

Have you ever wondered why one learner easily solves mathematical problems, while another expresses an idea beautifully through music or art? Or do some people understand others' feelings quickly, while others prefer working quietly and independently? These everyday differences show that intelligence is not just about being "smart" in class; it shapes how people think, learn, solve problems, and behave in real life.

Learning is not only about ability—it is strongly influenced by motives, but also by the inner drives that push people to act, persist, and change their Behaviour over time. This unit, Motives of Learning, explores how physiological, social, and personal motives influence the acquisition of new behaviours and attitudes, and why Behaviour changes across time and different environments. By the end, you will be able to justify how motives shape Behaviour and apply this understanding in real-life situations at school, in the workplace, and in the community.

Introductory activity



Three young workers are attending a training program to improve their skills. Eric, a construction worker, pays close attention because he knows the new skills will help him earn more money to support his family. Aline, who works at a café, enjoys the training because she likes interacting with others and wants to be appreciated by her customers and colleagues. Samuel, an intern at a technology company, is very curious and asks many questions because he wants to grow his knowledge and become successful in his future career.

Questions

1. Imagine you are part of this training program—who are you most similar to (Eric, Aline, or Samuel) and what motivates you to learn in that situation? Explain why.
2. In your own experience at school or work, what usually drives your effort: basic needs (like success or rewards), social factors (like recognition from others), or personal goals (like self-improvement)? Give a real example.
3. Even though Eric, Aline, and Samuel attend the same training, they behave differently. How would your Behaviour compare to theirs, and how might your motivation influence your future success?

9.1. Key motives to learning: Physiological motives

Learning activity 9.1



Have you ever tried to study when you were very hungry, tired, or sick? What happened?

9.1.1. Meaning of terms

Motives:

- Motives are the results of drives that are closely related to needs, reasons, or things that prompt certain Behaviour.
- A motive is something such as a need or desire that causes a person to act. It is also a specific physiological or psychological state of arousal that directs an organism's energy toward a goal.
- Motive can also be defined as a reason offered as an explanation for or cause of an individual's behaviour.
- Motive can be understood as a reason for doing something.

Needs: These are internal imbalances that occur when we are deprived of something. For example, a lack of food and water can cause problems in the body.

Drives: Are internal energies that create conditions which push an organism to direct its Behaviour usually toward a goal.

9.1.2. Physiological/biological/primary motives

Physiological/biological motives are basic needs that are essential for the survival of the organism. They are basic, inborn, and biological drives that are present in all beings. They are called primary needs because they are vital for life and are necessary for survival. These motives are deeply rooted in biological processes and functioning. Such motives are triggered when there is an imbalance in the body. Primary motives are aimed at satisfying survival needs. Once these physiological needs are not met, an organism dies; hence, it cannot even acquire any Behaviour.

9.1.3. Types/examples of physiological motives

- **Hunger motive:** People eat to live. The food they take is digested, and nutritional substances are absorbed by the body, from which it gets energy to sustain life. When these substances are exhausted, some imbalance exists and hinders learning.
- **Thirst motive:** In our daily life, we regularly take fluids in the form of water. These fluids are essential for our body tissues for normal functioning. When the water level in the body decreases, we develop a motive to drink water.
- **Need for oxygen and respiration:** Our body always needs oxygen, and we get it through continuous respiration. Oxygen is necessary for the purification of blood. Without a regular supply of oxygen, we cannot survive. Lack of oxygen leads to damage to the brain and death.
- **Motive for regulation of body temperature:** Maintenance of normal body temperature is necessary. A rise or fall in body temperature causes many problems in the body.
- **Need for sleep:** Sleep is an essential process for the normal functioning of the body and mind. When our body and mind are tired, they need rest to restore energy.
- **Shelter and clothing:** These protect a person from the cold.

- **Need for avoidance of pain:** We cannot continue to bear pain. Whenever we experience pain, we try to avoid it. One cannot learn well when he/she is in a painful situation/environment.
- **Drive for elimination of waste:** Our body cannot bear anything in excess or any waste. Excess water is sent out in the form of urine or sweat, while digested food, after absorption of nutritional substances, is sent out in the form of stools. Once these wastes are eliminated, we feel comfortable and can learn certain behaviours.

Note that the physiological/biological motives thus have their origin in the physiological state of the body. These motives are triggered by departures from the balanced physiological conditions of the body, as well as by sensory stimuli. This balance is crucial for life.

Application activity 9.1



As a responsible student and citizen of Rwanda, think about your own school and assess the different programs the Government has put in place to meet students' physiological needs (such as food, health, and rest) to support effective learning.

9.2. Key motives to Learning: Social motives

Learning activity 9.2



Have you ever worked harder in class because you wanted your teacher, friends, or parents to be proud of you?

9.2.1. Meaning of social motives

While physiological motives pertain to both animals and human beings, social motives are specific only to human beings. They are called social motives because they are learned in social groups as a result of interaction with the family and society. That is why their strength differs from one individual to another.

9.2.2. Examples of social motives

a) Achievement motive: It refers to a desire to achieve some goal. This motive is developed in the individual who has seen some people in society attaining high success, reaching high positions and standards. He/she develops a concern to do better, to improve performance. Researchers found that high achievers choose and perform better at challenging tasks, prefer personal responsibility, seek and utilise feedback about performance standards, have innovative ideas to improve performance, while low achievers do not accept challenges, set average standards, and accept failure easily. To help children achieve higher, parents must try to inculcate leadership qualities in their children for better achievement in their future lives. They must allow children to make decisions independently and guide them to higher achievement from childhood, so that the children develop high achievement motivation.

- b) Aggressive motive:** It is a motive to react aggressively when faced with frustration. Frustration may occur when a person is obstructed from reaching a goal or when he/she is insulted by others. Even in a fearful and dangerous situation, the individual may resort to aggressive Behaviour. An individual expresses such Behaviour to overcome opposition forcefully, which may be physical or verbal aggression.
- c) Power motive:** People with power motive will be concerned with having an impact on others. They try to influence people through their reputation. They accept those who obey their instructions. People with high power motive choose jobs where they can exert their power. They want people as followers. They expect high prestige and recognition from others. E.g., A person may aspire to go for jobs like a police officer, a politician, etc.
- d) Acquisitive motive:** This motive directs the individual to the acquisition of material property. This motive arises as we come across different people who have earned a lot of money and are leading a good life. It is a human tendency to acquire all those things that appear attractive to him/her.
- e) Curiosity motive:** This is also called the stimulus and exploration motive. Curiosity is a tendency to explore and know new things. We see people engaged in travelling, looking at new places, new things, and new developments taking place outside their environment. These people want to extend their knowledge and experiences by exploring new things. Curiosity motives are very helpful during childhood. This motive facilitates children learning faster.
- f) Gregariousness/affiliation need:** This is a tendency to associate oneself with other members of the group or the same species. The individual will be interested in establishing, maintaining, and repairing friendly relationships and will be interested in participating in group activities. An individual will conform to social norms, mores, and other ethical codes of the groups in which she/he is interested. To a greater extent, gregariousness is developed because many of the needs, like basic needs, safety, and security needs, are fulfilled.
- g)** Apart from the above-described motives, the need for self-esteem, social approval, self-actualisation, autonomy, and mastery is to be included in social motives of learning.

Application activity 9.2



1. How can you work with classmates to encourage teamwork and mutual support during group activities?
2. In what ways can teachers in your school create a sense of belonging and motivation among students?
3. How can school leaders and clubs recognise students' efforts to boost confidence and participation?
4. What practical actions can you personally take to make your classroom a more motivating and supportive place for everyone?

9.3. Key motive to learning: personal motives

Learning activity 9.3



Think about a time when you studied very hard for a test or practised something until you got it right. What pushed you to keep going even when it was difficult?

9.3.1. Personal motives

In addition to physiological and social motives, some other motives are allied with both of the above. These are highly personalised and very much individualised motives.

Personal motives are the internal needs, desires, interests, or goals that drive an individual to think, feel, or behave in a certain way. They arise from within the person and influence decisions and actions.

Examples of personal motives:

- Desire to achieve success
- Need for independence
- Curiosity to learn
- Desire for recognition
- Personal growth and self-improvement

9.3.2. Examples of personal motives

a) Force of habits: We see different people having formed different habits like chewing tobacco, smoking, alcohol consumption, etc. There may be good habits also, like regular exercise, reading newspapers, prayers, meditation, etc. Once these habits are formed, they act as drivers and compel the person to perform the act. The speciality of habits is that they motivate the individual to engage in that act automatically.

b) Goals of life: Every normal person will have some goals in life. They may be related to education, occupation, income, sports, acquisition of property, public service, social service, etc. Once a goal is set, he/she will be motivated to fulfil that goal. The goals people set depend upon various factors like knowledge, information, guidance, support, personality, facilities available, aspirations, family and social background, etc.

c) Levels of aspirations: Aspiration is aspiring to achieve or to get something or a goal. But such achievement depends upon the level of motivation the individual has. Every individual will have a goal in his/her life and strive to reach that goal. But the effort to attain that goal varies from one individual to another. The amount of satisfaction he/she gains depends upon his/her level of aspiration.

Eg, If a student is expecting 80% of marks in the examination, and gets only 75%, he/she may be unhappy. On the other hand, a student expecting failure may feel very happy if he/she gets just 35% passing marks, because the student with a high level of aspiration works hard, whereas the student with a lower level may not.

Note that a higher level of aspiration is advisable. However, it should also correspond with his/her abilities. Because if an individual aspires for higher-level achievement without possessing the required ability, he/she will have to face frustration and disappointment.

d) Attitudes: Our attitudes determine our motivation to learn. Attitudes are specific to an individual. For example, a person within the family may have a positive attitude toward family planning, while others have negative attitudes. Furthermore, attitudes represent a collection of personal beliefs, feelings, and values that influence a person's behaviour in response to specific ideas, objects, persons, or situations. Attitudes represent habits of thought formed over time through exposure to people, events, and ideas over the course of one's life experiences. This storehouse of acquired attitudes acts like a subconscious computer master program for controlling Behaviour. These habits of thought have an impact not only on how we see and react to the world around us, but also on how we see ourselves. Attitudes are often based not on reality but rather simply on how we personally think or feel about something. This then becomes our reality. Because our attitudes reflect what we think, feel, and believe about ourselves and the world around us, they effectively shape our expectations of daily life, and these expectations directly influence the nature of our Behaviour.

e) Interests: Like attitudes, interests also determine people's motivation. And they differ from one person to another. For example, interest in sports, TV, etc., wherever we have positive attitudes, we will be motivated to attain. In a negative attitude, we will be motivated to avoid it. If a person is interested in music, he/she will be motivated to learn it. In this way, our personal motives determine our Behaviour.

f) Initiative: This is the tendency to act, take charge, or move forward before others do.

g) Drive to achieve: The desire to accomplish something versus working for external rewards like money or prestige.

Application activity 9.3



Suppose you are a teacher in one of the schools of Rwanda. What would you do to promote the personal motives of students in learning?

9.4. Acquisition of new behaviours and attitudes

Learning activity 9.4



Think about when you first joined school or learned something new like speaking English, greeting elders respectfully, or working in a group. Were you already behaving that way before, or did you learn it over time?

9.4.1. Attitude

An attitude refers to an expression of the way one feels. It is a general and lasting positive or negative opinion or feeling about some person, object, or issue. It is also a learned tendency to evaluate things in a certain way.

i. Ways by which new attitudes are acquired/formed/founded

- a) **Direct personal experience:** A person's direct experience with the object determines his/her attitude towards it. The personal experience of an individual, whether favourable or unfavourable, will affect his/her attitude deeply, but these attitudes are difficult to change. **For example**, an individual joins a new job, which is recommended to him/her by his/her friend. But when he/she has the job, she/he finds the work repetitive, the supervisor is too tough, and co-workers are not so cooperative, she/he would develop a negative attitude towards the job, because the quality of his/her direct experience with the job is negative.
- b) **Association:** Sometimes an individual comes across a new object that may be associated with an old attitude object. In such a case, the attitude towards the old attitude object may be transferred towards the new attitude object. For example, if a new worker remains most of the time in the company of a worker who is in the supervisor's books, and towards whom the supervisor has a positive attitude, the supervisor is likely to develop a favourable attitude towards the new worker also. Hence, the positive attitude towards the old worker has been transferred to the new worker because of the association between the old and the new worker.
- c) **Family and peer groups:** Attitudes, like values, are acquired from parents, teachers, and peer group members. During our early years, we begin to model our attitudes after those we admire, respect, or even fear. We observe the behaviour of our family members and friends, and we gradually shape our own attitudes and behaviour to align with theirs. This often happens without being directly told what to do and even without having personal experience.
- d) **Neighbourhood:** The neighbourhood in which we live contains different cultural practices, religious groups, and ethnic backgrounds. It is also made up of neighbours with diverse beliefs, values, and ways of life. People from different cultures often display different attitudes and behaviours. Some of these attitudes and behaviours we accept and adopt, while others we may reject or even oppose. Our conformity to certain practices or rebellion against others reflects the attitudes and values we hold.
- e) **Economic status and occupations:** The economic status and occupational position of the individual also affect his/her attitude formation. Our socio-economic background influences our present and future attitudes. Research findings have shown that unemployment disturbs former religious and economic values. Children of the professional class tend to be conservative. Respect for the laws of the country is associated with increased years of higher education.
- f) **Mass communications:** Attitudes are generally less stable than values. Advertising messages, for example, attempt to alter the attitude of the people toward a certain product or service.

Attitudes can be learned in a variety of ways. Consider how advertisers use classical conditioning to influence your attitude toward a particular product. In a television commercial, you see young, beautiful people having fun on a tropical beach while enjoying a sports drink.

g) Operant conditioning: Operant conditioning can also be used to influence how attitudes develop. Imagine a young man who has just started smoking. Whenever he lights up a cigarette, people complain, chastise him, and ask him to leave their vicinity. This negative feedback from those around him eventually causes him to develop an unfavourable opinion of smoking, and he decides to give up the habit.

h) Observation: Finally, people also learn attitudes by observing the people around them. When someone you admire greatly espouses a particular attitude, you are more likely to develop the same beliefs. For example, children spend a great deal of time observing the attitudes of their parents and usually begin to demonstrate similar outlooks.

ii. Attitude change

While attitudes can have a powerful effect on Behaviour, they are not set in stone. The same influences that lead to attitude formation can also create attitude change.

Classical conditioning, operant conditioning, and observational learning can be used to bring about attitude change. Classical conditioning can be used to create positive emotional reactions to an object, person, or event by associating positive feelings with the target object. Operant conditioning can be used to strengthen desirable attitudes and weaken undesirable ones. People can also change their attitudes after observing the Behaviour of others.

Messages that are thought-provoking and that appeal to logic are more likely to lead to permanent changes in attitudes. People can also change their attitudes when they have conflicting beliefs about a topic. To reduce the tension created by these incompatible beliefs, people often shift their attitudes.

iii. Foundations/components of attitudes

a) Affect or emotion/emotional foundations of attitudes

A key part of attitude is the affect or emotion associated with attitude. At a very basic level, we know whether we like or dislike something or find an idea pleasant or unpleasant. For instance, we may say that we know something in our heart or have a gut feeling. In such cases, our attitudes have been formed through our emotions rather than through logic or thinking. This can happen through sensory reactions, values, operant/instrumental conditioning, classical conditioning, mere exposure, semantic generalisation, and evaluative conditioning.

b) Cognition/cognitive foundations of attitudes

What might be called beliefs come from direct experience with the world or through thinking about the world. Thinking about the world includes any kind of active information processing, such as deliberating, wondering, imagining, and reflecting, as well as through activities such as reading, writing, listening, and talking. For instance, if you believe that insects are dirty, then you will probably have the attitude that insects are not food. If you read that locusts and other insects are happily eaten in some cultures, then you may come to believe that locusts may not be so bad. Your attitude here comes from thinking about new facts you read.

c) Behaviour/behavioural foundations of attitudes

Sometimes we form attitudes from our actions. This can happen if we do something before we have an attitude. Example (1): going to an art opening of an unknown artist. When we are unsure of our attitudes. Example (2): going with a friend to a political rally, or when we are not thinking about what we are doing (mindlessly singing along with a random station on the radio). That is, there are times when joking through emotions can cause us to form an attitude consistent with those actions. In the previous examples, people may come to hate the new artist, support free trade, or like classical music because their actions have led them to engage in these behaviours, which then led to the formation of an attitude. Hence, we look at our behaviour and figure out our attitude based on what we have done or are doing. We strive for consistency between our attitudes and our actions, and when the two do not match, we may form a new attitude to coincide with our past actions.

In addition, evidence suggests that attitudes may develop out of psychological needs (motivational foundations), social interactions (social foundations), and genetics (biological foundations).

9.4.2. Meaning of behaviour

Behaviour is the way that someone acts. Although the two are related, they are distinct from one another as attitude focuses on how one feels, and behaviour deals with one's actions. However, one's attitude is often expressed through his/her behaviour.

Techniques for learning new behaviour

Generally, it's important to choose a good, comfortable learning environment and set it up so that it's easy for the animal to learn the intended response. Minimize distractions. Make sure there are no harmful things around.

- a) Practice makes perfect / overlearning:** A technique can be pointed out at this point, which is that practice makes perfect. Any new behaviour you try to learn (or unlearn) will initially be difficult to accomplish. Your body and mind are not used to doing things in the new way and will require repeated practice of these new ways of doing things before they become second nature. With practice, what is initially effortful becomes effortless. The process of practising repeatedly makes the new behaviour more familiar and ultimately something that can be just performed without thought or effort. Repeated practice trains the child's brain to be able to perform the behaviour unconsciously.
- b) Shaping:** New behaviours you want to learn are likely complex in nature, requiring effort and practice to achieve. It is not possible to learn complex behaviours all at once. Instead, you must use the principles of successive approximation and overlearning to gradually master your goal.
- c) Chaining:** Shaping is the technique of choice when you are attempting to learn a single isolated behaviour. A complementary technique, chaining, is useful for situations when you are interested in learning a complex sequence of behaviours. Chaining and shaping approaches may be used concurrently when you are faced with learning a complex behaviour that can be broken into a series of smaller sub-behaviours.

d) Observation: New behaviours are acquired by observing events and by watching other people, and through language, we learn things we have neither experienced nor observed. They also learned by association when our minds connect events that occur in sequence, through reinforcement (reward and punishment), and can be acquired by imitating the Behaviours of others.

9.4.3. Strategies that promote positive attitudes and positive behaviours in children

Positive attitude and behaviour can be encouraged in the classroom through the following:

- **Be consistent with rules:** Establish the rules from the first lesson. It can be helpful to let the students write some of their own class rules, then students use peer pressure to keep them. A rule a student has made is far more likely to be kept than one a teacher introduces. Once the rules are established, follow them consistently with all students.
- **Get the students' full attention before telling them anything:** If students are not focused on you, they will not be listening, so take time of gaining their attention. Better to repeat the instructions to every student again. Make sure you are together before telling them anything.
- **Use positive language and body Language:** Positive language will encourage students to use their own positive language. Eg: don't say stop writing," but say can you all listen please?" Positive body language is just as important. If you have had a bad morning and walk into a classroom in a bad mood, students will pick it up. Try to hide that behaviour; smile as often as you can. Create a warm, welcoming environment for students when they enter your class.
- **Mutual respect:** If you want to be shown respect, you have to role model the behaviour to your students. Listen to students' opinions.
- **Have quality lessons:** Avoid boring lessons. Fast-paced design lessons, and activities should be scattered throughout. Don't expect students to be sitting there for 45 minutes listening to you speak. Allow the students to research and discover knowledge themselves. The more they are engaged, the less their mind will look for trouble.
- **Know the Students:** The teacher should take time to get to know his/her students, their likes and dislikes. If she/he can plan a lesson on what the student is interested in, she/he will find it easier to engage them. Knowing the student also helps if they are going through personal issues.
- **Be able to diagnose learning problems:** Have a basic knowledge of learning problems and the signs of these. Teachers are in the privileged position of being able to diagnose learning problems, and from here, the teacher can help provide tools for the student to succeed.
- **Routines:** Like rules, a routine is important to start from the beginning. Establish a routine that works for you and your students. This could be as simple as one student giving out the books, another giving out the pens. Giving a disruptive student responsibility can minimise their behaviour.

Application activity 9.4



After exploring how you and your classmates develop attitudes and behaviours. Think about how these two are connected in your own life.

- How do your attitudes toward school or a subject influence the way you behave in class?
- Can you share a situation where your behaviour changed because your attitude changed (or the opposite)?
- For example, if you develop a positive attitude toward group work, how does that affect how you participate with others?

9.5. Reasons why people's behaviour patterns and attitudes change in time and in space

Learning activity 9.5



Think about how you behaved when you were in primary school compared to now.

- Do you act the same way today?
- Do you think and feel about things the same way?

9.5.1. Behaviour patterns

Behaviour patterns describe interactions between objects and focus on how objects communicate with each other. Behaviour patterns are recurrent ways of acting by an individual or group toward a given object or in each situation, or a recurrence of two or more responses that occur in a prescribed arrangement or order. It is also a set of dominant Behaviours and emotional reactions that include a high emphasis on competition, impatience, hostility, and aggression.

9.5.2. How to change behaviour patterns?

Behaviour patterns usually give stability and efficiency to our lives. Some examples of Behaviour patterns are performing tasks in a particular way, collaborating on an assignment, working in concert in the laboratory, planning experiments with the team, collaborative decision-making, and managing conflict.

Note that many behaviour patterns develop through reward and punishment and are called learned behaviour patterns (a chain of behaviour that is multifariously linked from simpler, smaller segments).

Behaviour patterns (outside of simple instinct) come from three general areas:

1. A person's particular strengths and weaknesses,
2. Learned reactions from experiences, and
3. Habits.

Behaviours begin mentally; training the body, mind, and emotions to what is wanted is mental exercise. We often think of behaviour in terms of activities, procrastinating, or overworking. When you consider how you behave, you discover that your thoughts, emotions, and beliefs are the true roots of these behaviours. Because all these behaviours are yours alone, you have the right and the power to change them. Some behaviour patterns that team members may want to change are procrastination, over-analysing a problem, being late to meetings, or poor time management.

We may be frustrated by the typical ways we deal with situations and treat other people. We may react all too often with frustration towards others, or be controlling, timid, or waiting to speak up, but not having the self-confidence to act. Changing these types of behaviour patterns can increase confidence in a person's abilities and self-esteem.

9.5.3. Reasons why people's behaviour patterns and attitudes change in time and space

- Attitudes and behaviours are not set in stone and may change over time. Some reasons behind this change are tackled below.
- Behaviour patterns and attitudes may change when people learn new information.
- When they are persuaded by influential people.
- When individuals experience feelings of uneasiness or guilt due to cognitive dissonance.
- Humans have evolved as social, empathetic, collaborating, and altruistic beings in small groups sharing common identities. At the same time, a fear of strangers has been built into our system, which influences the way we perceive events and people, including how we react to influxes of newcomers in our community.

9.5.4. Reasons why people need to change attitudes

- People with a positive attitude tend to enjoy life more and are generally happier.
- Negative attitudes affect others. No one wants to be around a negative person. Negative attitudes can ruin relationships.
- Negative attitudes produce negative circumstances. If we think negatively about a situation or a person, eventually we will give power to those thoughts.
- Maintaining negative attitudes will drain you of physical energy. Becoming angry, upset, or frustrated can release negative emotions that can cause tiredness and a weak immune system. Chronic stress from negative attitudes disrupts the body's hormone balance and depletes the brain chemicals responsible for feel-good emotions.
- A positive attitude helps you to be an overcomer when the adversities of life come.
- A person with a negative attitude blames others for their difficulties.
- A positive attitude will reduce negative emotions and help recover from negative experiences like trauma and loss very quickly.

9.5.5. What can one do to have a better attitude?

- Focus on his/her strengths more than weaknesses.
- Monitor what he/she watches on TV, computer, or what she/he reads.
- Stop negative thoughts immediately. Negative thoughts will come, and in the beginning of change, she/he may even have an increase in negative thoughts.

9.5.6. Importance of behaviour change

- Behaviour change can improve our health. Making healthier choices such as eating nutritious foods, exercising regularly, and getting enough sleep can help us feel our best and enjoy a higher quality of life. Additionally, adopting these healthy habits can help prevent chronic diseases such as heart disease, stroke, cancer, and diabetes.
- Behaviour change can improve our relationships. When we try to change our Behaviour for the better, we often find that our relationships with others improve as well. This is because our new behaviours may be more positive and constructive, and they may help to build trust and communication.
- Behaviour change can lead to success in our work and careers. When we make positive changes in our behaviour, we often find that we are more productive and successful in our jobs. These new behaviours may help us to advance in our careers.
- Behaviour change can improve the world around us. When we make positive changes in our own behaviour, we can set an example for others to follow. In addition, our new behaviours may help to make our communities and the world a better place.
- Behaviour change helps to save money and resources. When we make positive changes in our behaviour, we often find that we save money and help to protect the environment. For example, if we choose to recycle instead of buying new products, we can save money and help to protect the environment. A lot of toxic habits cost money. This is either directly through purchases or indirectly by robbing you of the time, energy, and ability to engage in more fruitful activities.
- Behaviour change helps one adapt. When we make positive changes in our behaviour, we often find that we are better able to adapt to new situations. New behaviours may help to overcome challenges and obstacles.
- Behaviour change will help others through your experience. Sharing your stories and journey to recovery will assist others who are already on the path, and even those who have not yet initiated change. It can also help prevent people from starting the toxic habit in the first place.
- Behaviour change will make you enjoy life. In many instances, the behavioural change you will experience will add much more value to your lives by personal satisfaction and improved relationships with your families and loved ones.

Application activity 9.5



1. Examine how changes in your own behaviour and attitudes can positively affect your everyday life and the people around you.
 - a) Can you think of a time when you changed your attitude (for example, becoming more serious about your studies or more respectful)?
 - b) What difference did it make?
 - c) How does positive behaviour, such as cooperating with classmates or following rules, improve your learning and relationships?
 - d) In what ways can a change in attitude help you handle challenges like stress, conflict, or failure?
2. Use real-life examples from your school or home to explain how positive changes in behaviour and attitudes lead to better outcomes, such as success in learning, improved relationships, and personal growth.

9.6. End unit assessment



Question 1: Knowledge and understanding, Skills & Attitudes and values

Think about your own daily life and learning experiences:

- a) What are some natural needs (biological drives) that push you to learn or act (e.g., hunger, rest, safety)? Give three examples from your life.
- b) What are some social needs (social drives) that motivate you in school (e.g., friendship, recognition, belonging)? Also, give three examples from your experience.
- c) Using your examples, explain clearly how biological drives are different from social drives in influencing your learning.

Question 2: Knowledge, Skills & Attitudes and values

Reflect on children in your community or school:

- a) What do you think happens when children's basic needs (like food, safety, love, or support) are not met?
- b) How might this affect their behaviour, attitudes, and ability to learn? Give real-life examples to support your ideas.

Question 3: Skills & attitudes and Values

Imagine you are a head-teacher at a school in Kigali or any community in Rwanda:

- a) What practical steps would you take to meet students' needs (both biological and social)?
- b) How would your actions help improve students' motivation, behaviour, and learning outcomes? Use realistic examples that could be applied in your school.

10. PROJECT ACTIVITY

❖ Project title:

Influence of school environment on learning

❖ Objectives:

By the end of this project, you will be able to:

- Explain how school environment and emotions influence learning.
- Describe how memory and intelligence relate to learning.
- Identify factors that positively or negatively influence learning.
- Examine how personality traits influence learning.
- Suggest practical ways of creating a school environment that supports effective learning.
- Recognise the importance of creating an inclusive and supportive learning environment.

❖ Project Activity

As a S5 student, investigate how the school environment influences learning. Identify factors within the school environment that may positively or negatively affect students' learning, behaviour, memory, intelligence and emotional experiences. Use observation, a simple questionnaire and/or a safe, simple experiment to collect relevant information. Using the psychological knowledge you have acquired throughout the year, analyse your findings and apply relevant concepts and theories of learning and conditioning, personality traits, personality disorders, Social Learning Theory, memory and intelligence, and emotions.

Based on your findings, suggest practical ways of creating a school environment that promotes effective learning and positive behaviour.

❖ Project outline

After conducting your research and analysing the information collected, write a report. Your report should include:

1. **Introduction:** Briefly introduce the topic and explain the purpose of the research.
2. **Methodology:** Explain how you collected the data, including observation, questionnaire and/or experiment, the participants and the research setting.
3. **Findings:** Present the main findings from the data collected.
4. **Psychological Analysis:** Interpret the findings using relevant psychological concepts, theories and knowledge studied throughout S5.
5. **Conclusion:** Summarise the main findings and what they show about the influence of the school environment on learning.
6. **Recommendations:** Suggest practical ways to address the challenges identified and promote effective learning.
7. **Reflection:** Explain what you learned from conducting the research and how the project helped you apply Psychology to a real-life situation.

❖ Guiding questions

Your research should consider the following questions:

1. Which factors in the school environment appear to have the greatest influence on students' learning?
2. How do these factors influence students' learning and behaviour?
3. Are the effects positive, negative or both? Give examples from your findings.
4. How do rewards, consequences, repetition or other forms of conditioning influence students' behaviour and learning?
5. How might students' personality traits influence their study habits, motivation, participation and responses to learning?
6. How do students learn behaviours, attitudes or study habits by observing and interacting with others?
7. How do emotions influence students' concentration, motivation, learning and memory?
8. What factors appear to help or interfere with students' memory and recall?
9. What do your findings suggest about the role of intelligence in learning, understanding, problem-solving or applying knowledge?
10. Which psychological theory or theories studied in S5 can help explain your findings? How do they apply?
11. How can knowledge of personality help explain differences in students' behaviour without labelling or diagnosing individuals?
12. What conclusion can you draw about the influence of the school environment on learning?
13. What practical measures can be taken by the school, teachers, families and students to create an environment that supports effective learning?

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